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Quantum Random Number Generators and implementations based on Leggett–Garg inequalities

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Abstract

Abstract

Quantum Random Number Generators (QRNGs) are essential tools across various fields, including cryptography, numerical simulations, and metrology, due to their ability to produce truly unpredictable sequences based on the inherently stochastic nature of quantum mechanics. Traditionally, randomness certification in these devices has relied on the violation of Bell inequalities; however, this approach demands complex experimental conditions, such as entanglement and the strict spatial separation of detectors.

This thesis provides a detailed analysis of QRNG operating principles, focusing on alternative implementations that utilize Leggett–Garg inequalities (LGI) as a tool to certify the non-classicality of observed behavior. Originally formulated to test macroscopic realism and the non-invasiveness of measurements in quantum systems, LGIs represent a promising resource for generating certified randomness in temporal scenarios—specifically, those based on repeated measurements over time on a single system.

The primary advantage of LGIs over Bell inequalities lies in their reduced experimental complexity, as they require neither entangled systems nor spatial separation between measurements. This paves the way for more compact, robust implementations that are potentially integrable at the micro- or nano-technological scale. This work explores the theoretical foundations of this approach, examines several recent experimental proposals and realizations, and provides a critical discussion on the future prospects of employing LGIs to develop reliable, certifiable, and scalable QRNGs.

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Summary

The journey toward understanding the true nature of quantum correlations has fundamentally rewritten our perception of physical reality. It began as a profound philosophical debate, sparked by the famous Einstein-Podolsky-Rosen (EPR) paradox, which questioned the completeness of quantum mechanics and sought refuge in local hidden variable theories. Decades later, John Stewart Bell translated this debate into a rigorous mathematical test, proving that nature defies local realism through spatial entanglement. However, in the mid-1980s, Anthony Leggett and Anupam Garg introduced a pivotal paradigm shift: instead of looking at correlations across distant points in space, they turned their attention to time, formulating a framework to test macrorealism and non-invasive measurability within a single evolving system. Today, this rich theoretical lineage serves as the bedrock for a technological revolution, particularly in the realm of Quantum Information.

However, implementing these foundational quantum insights into practical cryptographic protocols introduces a significant technological and architectural challenge. While traditional device-independent quantum random number generators (QRNGs) backed by spatial Bell inequalities offer absolute security guarantees, their experimental overhead remains notoriously high, requiring strict spatial separation to close communication loopholes. This introduces a striking paradox: the demand for certified, high-security randomness is urgent, yet the hardware requirements for Bell-certified architectures remain largely impractical for widespread commercial adoption.

Out of this tension emerges the compelling alternative explored in this work: harnessing temporal quantum correlations and No-Signaling-in-Time (NSIT) principles to drive secure quantum cryptographic applications. By focusing on the temporal violation of Leggett-Garg inequalities (LGI), we can bypass the stringent multi-device requirements of spatial tests, shifting the technological burden toward a single, manageable physical platform while maintaining a mathematically rigorous defense against deterministic classical adversaries.

This Master’s degree thesis delves into the intersection of Quantum Cryptography and Quantum Number Generation, focusing on the quantitative characterization of macrorealism violations and their direct application to QRNG entropy certification. The manuscript begins with an exploration of quantum cryptography and information fundamentals, reviewing the kinematics of two-level systems, the mathematics of mixed states, and the subtle threat that environmental decoherence poses to quantum hardware. The spotlight then shifts to the formal evolution from EPR and Bell’s local realism to the Leggett-Garg framework, examining the structural implications of Fine’s Theorem on temporal joint probability distributions.

An innovative contribution of this thesis lies in the rigorous numerical and statistical analysis of macrorealism violations. By implementing a customized Python simulation framework, we track distinct geometric distance metrics (d_{weak} , d_{int} , and d_{strong}) alongside formal min-entropy formulations. Through this detailed exploration, this work provides a crucial insight: we unveil a profound *decoupling* between non-classicality and statistical randomness, demonstrating that a system can strictly violate macrorealistic bounds even when its certifiable min-entropy drops below classical expectations. Most importantly, this analysis enables the definitive certification and generation of pure, genuine quantum random numbers that remain fundamentally unpredictable and mathematically secure against any adversarial spoofing or classical simulation.

Chapter-by-Chapter Overview

- **In Chapter 1**, we delve into the core concepts of Quantum Cryptography and information processing essential for understanding secure technologies. We review the mathematics of qubits using both state-vector and density matrix formalisms, analyze the properties of the Pauli basis, and investigate how environmental decoherence disrupts quantum systems.
- **As we transition to Chapter 2**, we trace the theoretical evolution from the EPR paradox and Bell’s local realism to the Leggett-Garg framework. We analyze how Leggett-Garg inequalities test the core postulates of Macrorealism Per Se (MRps) and Non-Invasive Measurability (NIM), evaluating the vital role of Fine’s Theorem in establishing macrorealistic boundaries, and finally defining the three distinct degrees of macrorealism.

- **Chapter 3** ventures into the nature of randomness, mapping the profound paradigm shift from classical epistemological unpredictability—arising from incomplete information—to quantum ontological randomness, which is inherently intrinsic to nature. We expound upon the stringent requirements of modern cryptographic applications and outline the device-independent paradigm.
- **In Chapter 4**, we explore the mathematical structures governing temporal correlations. We detail the formal derivation of Leggett-Garg inequalities directly from No-Signaling-in-Time (NSIT) conditions and establish the algebraic and information-theoretic frameworks required for cryptographic entropy estimation.
- **Finally, Chapter 5** defines the violation parameters to evaluate the violation of macrorealism and the min-entropy bounds for randomness certification. By mapping geometric distances against min-entropy bounds, this chapter unveils the intricate trade-offs between quantum non-classicality and secure randomness certification, underscoring the vital necessity of combining both metrics to shield quantum devices from classical simulation and adversarial spoofing.

Without further ado, let's start!

Chapter 1

Foundations of Quantum Mechanics

“Sometimes I’ve believed as many as six impossible things before breakfast.”

— Lewis Carroll, *Alice in Wonderland*

1.1 Introduction

The transition from classical to quantum mechanics represents one of the most profound paradigm shifts in the history of modern science. In the classical framework, the physical world is described by the deterministic evolution of particles and fields, characterized by well-defined trajectories within a canonical phase space. In stark contrast, the quantum regime is governed by a fundamentally probabilistic and algebraic structure, where the state of a system is no longer an objective point in space, but a complex vector in a linear Hilbert space. The emergence of non-classical phenomena, such as **superposition** and **entanglement**, constitutes the primary differentiator between these two descriptions—properties that will be formally developed throughout this chapter.

The purpose of this chapter is to provide a comprehensive overview of the fundamental principles of quantum mechanics. By establishing the mathematical formalism and key conceptual pillars—including the nature of quantum states, the measurement process, and the role of correlations—we define the theoretical bedrock necessary for the subsequent investigation of macrorealism and temporal correlations.

We begin by defining the kinematics of two-level systems, or qubits, utilizing both the state-vector and the more general density matrix formalism. The latter is particularly crucial for addressing the nuances of mixed states and the

algebraic properties of the Pauli basis. Furthermore, we explore the nature of quantum entanglement as the primary differentiator between classical and quantum correlations.

Finally, we address the measurement problem through the lens of the double-slit experiment. By formalizing the transition from coherent superpositions to statistical mixtures, we show how the acquisition of information is intrinsically invasive, leading to the phenomenon of decoherence.

For an exhaustive treatment of these fundamental principles, the reader is referred to standard textbooks such as those by Feynman [1], Sakurai [2], and Nielsen & Chuang [3], which provide the rigorous context for the topics discussed herein.

1.2 Quantum Kinematics: Hilbert Space and Qubits

In the quantum formalism, the state of a physical system is represented by a unit vector in a complex Hilbert space $\mathcal{H}$, which is a complete inner product space. For the purpose of this work, we focus on the simplest non-trivial system: the two-level system, or **qubit**, as it is commonly referred to in the quantum information community.

1.2.1 State Representation and the Computational Basis

A qubit is described by a two-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^2$ [2]. To perform calculations, we define a fixed orthonormal basis $\{|0\rangle, |1\rangle\}$, known as the **computational basis**. From a physical perspective, these basis vectors often correspond to the eigenstates of the **spin operator** $\hat{S}_z$ along a chosen z -axis (e.g., for an electron spin-1/2 system), where $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$.

A fundamental postulate of quantum mechanics is the **principle of superposition**, which states that if $|0\rangle$ and $|1\rangle$ are valid states of the system, then any linear combination of them also describes a valid physical state. Consequently, any state vector $|\psi\rangle$ can be expressed as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}. \quad (1.1)$$

The complex coefficients α and β are defined as **probability amplitudes**. Their physical significance is grounded in the Born rule: the quantities $|\alpha|^2$ and $|\beta|^2$

represent the probabilities of finding the system in state $|0\rangle$ or $|1\rangle$, respectively, upon measurement. This interpretation requires that the total probability must be conserved, leading to the normalization condition:

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1. \quad (1.2)$$

This ensures that $|\psi\rangle$ remains a unit vector in $\mathcal{H}$.

It is important to note that while the focus here is on *pure states* (well-defined vectors in $\mathcal{H}$), the principle of linear combination is a structural property of the Hilbert space. Unlike a classical bit, which is restricted to discrete binary values, a qubit inhabits a continuous landscape of possibilities. In this context, the relative phase between α and β is not a mere mathematical artifact, but a physical observable that plays a critical role in interference phenomena.

1.3 The Density Matrix Formalism

The description of a quantum system via a single state vector $|\psi\rangle \in \mathcal{H}$ is restricted to cases of complete knowledge regarding the system's preparation, referred to as *pure states*. However, in most physically relevant scenarios—such as systems in thermal equilibrium, subsystems of entangled pairs, or systems subject to environmental decoherence—the state is characterized by a statistical mixture. To provide a universal description of quantum states, we employ the **density operator** formalism.

1.3.1 Formal Definition and Fundamental Postulates

Consider an ensemble of pure states $\{|\psi_i\rangle\}$, where each state occurs with a classical probability $p_i \in [0, 1]$ such that $\sum_i p_i = 1$. The state of the system is described by the density operator ρ , defined as the **convex combination** of the corresponding rank-1 projectors:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \quad (1.3)$$

The term *convex* refers to the fact that ρ lies within the state space formed by the weighted sum of pure states, where the weights are non-negative and sum to unity. In the language of functional analysis, the density operator is a linear, self-adjoint operator acting on $\mathcal{H}$. For ρ to represent a physically realizable quantum state, it must satisfy the following three axiomatic properties [4]:

1. **Hermiticity:** $\rho = \rho^\dagger$. This ensures that ρ is diagonalizable and its eigenvalues are real;
2. **Positivity:** $\rho \geq 0$, i.e., $\langle \phi | \rho | \phi \rangle \geq 0$ for all $|\phi\rangle \in \mathcal{H}$. This implies that all eigenvalues λ_k of ρ satisfy $\lambda_k \geq 0$, consistent with their interpretation as probabilities;
3. **Normalization:** $\text{Tr}(\rho) = 1$. This condition guarantees the conservation of total probability.

The distinction between pure and mixed states is formally captured by the **purity** $\gamma(\rho) = \text{Tr}(\rho^2)$. By the spectral theorem, ρ represents a pure state if and only if $\gamma(\rho) = 1$ (where $\rho^2 = \rho$), while for mixed states $\gamma(\rho) < 1$. For a d -dimensional space, the minimum purity is $1/d$, corresponding to the maximally mixed state $\rho = \frac{1}{d}\mathbb{I}$.

1.3.2 Algebraic Structure and the Bloch Representation

For a two-level system ($\mathcal{H} \cong \mathbb{C}^2$), the space of Hermitian operators is spanned by the **Pauli basis** $\{\mathbb{I}, \hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$ [5]. These operators, which form an orthogonal basis for the Lie algebra $\mathfrak{u}(2)$, are defined as:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.4)$$

They satisfy the fundamental commutation and anti-commutation relations:

$$[\hat{\sigma}_j, \hat{\sigma}_k] = 2i\epsilon_{jkl}\hat{\sigma}_l, \quad \{\hat{\sigma}_j, \hat{\sigma}_k\} = 2\delta_{jk}\mathbb{I}. \quad (1.5)$$

where $[A, B] = AB - BA$ and $\{A, B\} = AB + BA$. Any density operator ρ for a qubit can be uniquely decomposed as:

$$\rho = \frac{1}{2}(\mathbb{I} + \vec{a} \cdot \vec{\sigma}), \quad (1.6)$$

where $\vec{a} = (a_x, a_y, a_z) \in \mathbb{R}^3$ is the **Bloch vector**, with $a_i = \text{Tr}(\rho\hat{\sigma}_i)$.

The condition $\rho \geq 0$ restricts the Bloch vector to the **unit ball** (the *Bloch ball*), $|\vec{a}| \leq 1$. Note that while the **Bloch sphere** strictly refers to the surface ($|\vec{a}| = 1$, pure states), the entire ball is required to represent the full state space[3]:

- The **surface** ($|\vec{a}| = 1$) contains pure states, parameterized by $\vec{a} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$.

- The **interior** ($|\vec{a}| < 1$) contains mixed states.
- The **origin** ($\vec{a} = 0$) represents the **maximally mixed state** $\rho = \frac{1}{2}\mathbb{I}$. This point represents the limit of maximal Von Neumann entropy, signifying a total lack of information regarding the system's preparation [3].

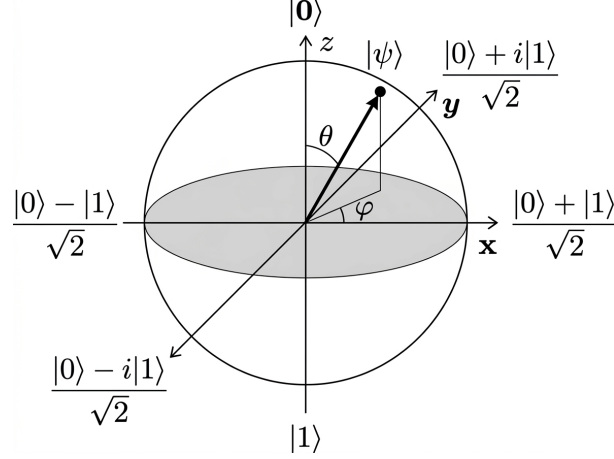


Figure 1.1: Geometric representation of the qubit state space. Pure states reside on the surface of the Bloch sphere, while mixed states lie within the interior of the Bloch ball. The poles correspond to the computational basis states $|0\rangle$ and $|1\rangle$. Note that orthogonal states are represented by antipodal points on the sphere.

1.4 Quantum Entanglement

The most striking feature of quantum mechanics arises in multipartite systems. For a composite system consisting of two qubits, A and B , the state space is defined by the tensor product of the individual Hilbert spaces: $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ [6]. It is fundamental to observe that entanglement is a structural property of the state itself, independent of the dynamical process (interaction) that may have generated it.

1.4.1 Separability and the Definition of Entanglement

A state $|\Psi\rangle_{AB}$ is said to be **separable** (or a product state) if it can be factored into the product of local states: $|\Psi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B$. If such a decomposition is impossible, the state is **entangled**. Entanglement implies that the subsystems do not possess individual identities; their physical properties are inextricably linked through non-local correlations.

The canonical examples are the four **Bell states**, which constitute a **maximally entangled basis** [6]:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \quad (1.7)$$

These states are "maximally entangled" in the sense that they yield the maximum possible value for entanglement measures such as the *Entropy of Entanglement* (where the reduced density matrix is proportional to the identity) and violate Bell inequalities to the algebraic maximum.

1.4.2 Reduced Density Matrix and Partial Trace

To describe a single subsystem within a composite state, we employ the **partial trace** operator, which averages over the degrees of freedom of the other subsystem. The reduced state of subsystem A is given by:

$$\rho_A = \text{Tr}_B(\rho_{AB}), \quad (1.8)$$

and similarly for subsystem B , $\rho_B = \text{Tr}_A(\rho_{AB})$.

A key signature of entanglement is that if the state ρ_{AB} is pure and entangled, the reduced states ρ_A and ρ_B are necessarily **mixed**. This reflects a fundamental quantum principle: local information regarding a subsystem is "lost" or, more precisely, resides in the global correlations of the entangled pair. In the case of the Bell states, the reduced density matrix is the maximally mixed state $\rho_A = \frac{1}{2}\mathbb{I}$, indicating a complete lack of local information.

1.5 Dynamics and Measurement

1.5.1 The Schrödinger Equation and the Time-Evolution Operator

The dynamical evolution of a quantum state is deterministic and, for a closed system—defined as a system that does not exchange information or energy with its environment—is governed by the Schrödinger equation [2]:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (1.9)$$

The idea behind this formulation is that the state at any time t can be uniquely determined from the initial state $|\psi(t_0)\rangle$. For a time-independent Hamiltonian $\hat{H}$, this transition is described by the **unitary evolution operator** $\hat{U}(t, t_0) = \exp[-i\hat{H}(t - t_0)/\hbar]$, such that:

$$|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle. \quad (1.10)$$

In the more general density matrix formalism, the evolution is given by:

$$\rho(t) = \hat{U}(t, t_0)\rho(t_0)\hat{U}^\dagger(t, t_0). \quad (1.11)$$

Unitary evolution is fundamentally reversible and preserves the purity of the state, $\text{Tr}(\rho^2(t)) = \text{Tr}(\rho^2(0))$.

1.5.2 The Measurement Process: Observables and Projective Collapse

Unlike the continuous evolution described by the Schrödinger equation, measurement introduces a non-unitary, stochastic element into the theory.

In quantum mechanics, a physical quantity is represented by an **observable** $\hat{Q}$, which is a Hermitian operator acting on $\mathcal{H}$. According to the spectral theorem, if the eigenvectors $\{|q\rangle\}$ of $\hat{Q}$ form a complete orthonormal basis, the operator can be decomposed as:

$$\hat{Q} = \sum_q q \hat{\Pi}_q, \quad (1.12)$$

where q are the real eigenvalues and $\hat{\Pi}_q = |q\rangle\langle q|$ are the **orthogonal projectors** onto the corresponding eigenspaces. These projectors satisfy the completeness relation $\sum_q \hat{\Pi}_q = \mathbb{I}$ and the idempotency property $\hat{\Pi}_q^2 = \hat{\Pi}_q$.

The probability of obtaining the outcome q when measuring a state ρ is given by the **Born rule**:

$$P(q) = \text{Tr}(\hat{\Pi}_q \rho). \quad (1.13)$$

For a pure state $|\psi\rangle$, this reduces to the familiar expression $P(q) = |\langle q|\psi\rangle|^2$. Immediately following the measurement, the state undergoes an irreversible "collapse" to the eigenspace associated with the observed result:

$$\rho_{after} = \frac{\hat{\Pi}_q \rho \hat{\Pi}_q}{\text{Tr}(\hat{\Pi}_q \rho)}, \quad \text{or for pure states, } |\psi_{after}\rangle = \frac{\hat{\Pi}_q |\psi\rangle}{\sqrt{P(q)}}. \quad (1.14)$$

This process destroys quantum coherences—the off-diagonal elements of the density matrix in the measurement basis—effectively transitioning the system from a coherent superposition to a statistical mixture of outcomes.

1.6 The Double-Slit Experiment: Quantum Interference and Information

The double-slit experiment, famously described by Richard Feynman [1] as a phenomenon which has in it the “only mystery” of quantum mechanics, provides the definitive empirical evidence for the principle of linear superposition and the wave-particle duality. Beyond its historical significance, it serves as the primary conceptual framework for understanding how the acquisition of *which-way* information—defined as the knowledge of which specific path the particle followed—is intrinsically linked to the destruction of quantum coherence. This transition is central to the measurement problem and the emergence of classicality.

1.6.1 Formalism of Quantum Interference

Consider a particle of mass m propagating towards a barrier containing two infinitesimal slits, denoted as L (left) and R (right). In the absence of any intermediate measurement, the state at the slits is described by a **coherent superposition** of the two possible spatial paths:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|L\rangle + |R\rangle), \quad (1.15)$$

where $|L\rangle$ and $|R\rangle$ are orthonormal states representing the particle’s localization at slit L or R , respectively. The term *coherent* implies that a fixed phase relationship exists between the two components, allowing for interference.

To determine the probability distribution on a detection screen, we project the evolved state onto the position basis $|x\rangle$. Let $\psi_L(x) = \langle x|\hat{U}|L\rangle$ and $\psi_R(x) = \langle x|\hat{U}|R\rangle$ be the probability amplitudes, often referred to as **wavefunctions**, associated with each path. According to the Born rule introduced in section (1.5), the total probability density $P(x)$ is:

$$P(x) = |\langle x|\Psi\rangle|^2 = \frac{1}{2} |\psi_L(x) + \psi_R(x)|^2. \quad (1.16)$$

Expanding this expression, we obtain:

$$P(x) = \frac{1}{2} \left[|\psi_L(x)|^2 + |\psi_R(x)|^2 + 2\text{Re}(\psi_L^*(x)\psi_R(x)) \right]. \quad (1.17)$$

The terms $|\psi_L(x)|^2$ and $|\psi_R(x)|^2$ represent the classical intensities as if each slit were isolated. The crucial non-classical component is the **interference term** $2\text{Re}(\psi_L^*(x)\psi_R(x))$, which can be expressed via the relative phase $\Delta\phi(x)$:

$$2\text{Re}(\psi_L^*(x)\psi_R(x)) = 2|\psi_L(x)||\psi_R(x)|\cos(\Delta\phi(x)). \quad (1.18)$$

As illustrated in Fig. 1.2, this cosine term results in the characteristic fringe pattern, where detection probability is modulated by constructive and destructive interference.

1.6.2 Which-Way Information and the Loss of Coherence

The mystery deepens when we introduce a measuring device $\mathcal{D}$ to identify the particle's path. This process is formally described by the interaction and subsequent entanglement between the particle and the detector.

Let the detector be initially in state $|d_0\rangle$. The interaction leads to the global state:

$$|\Psi_{\text{global}}\rangle = \frac{1}{\sqrt{2}} (|L\rangle \otimes |d_L\rangle + |R\rangle \otimes |d_R\rangle), \quad (1.19)$$

where $|d_L\rangle$ and $|d_R\rangle$ are the detector states after interaction [5]. The state of the particle is found by taking the partial trace over the detector's Hilbert space:

$$\rho_{\text{particle}} = \frac{1}{2} (|L\rangle\langle L| + |R\rangle\langle R| + \langle d_R|d_L\rangle |L\rangle\langle R| + \langle d_L|d_R\rangle |R\rangle\langle L|). \quad (1.20)$$

The term $\langle d_L|d_R\rangle$ quantifies the "distinguishability" of the paths. If the detector is perfectly efficient, it means it can distinguish the two paths with certainty, which implies that its final states must be perfectly distinguishable (i.e., orthogonal, $\langle d_L|d_R\rangle = 0$) [4, 5]. In this limit, the off-diagonal *coherences* vanish, and the density matrix becomes purely diagonal:

$$\rho_{\text{particle}} = \frac{1}{2} (|L\rangle\langle L| + |R\rangle\langle R|). \quad (1.21)$$

The interference term in the probability distribution $P(x) = \langle x|\rho_{\text{particle}}|x\rangle$ disappears, and the pattern would revert to the classical sum $P(x) = \frac{1}{2}(|\psi_L(x)|^2 + |\psi_R(x)|^2)$.

This result shows a key principle: it is not necessary for a conscious observer to measure or record the *which-way* information. The mere fact that such information is, in principle, available—encoded in the environment or a detector—is enough to destroy quantum coherence. As a consequence, the system transitions from a coherent superposition to a statistical mixture. This process, known as **decoherence**, plays a central role in the measurement problem and will be important for understanding the boundary between quantum and macrorealistic descriptions of nature.

Young's Double Slit Experiment Diagram

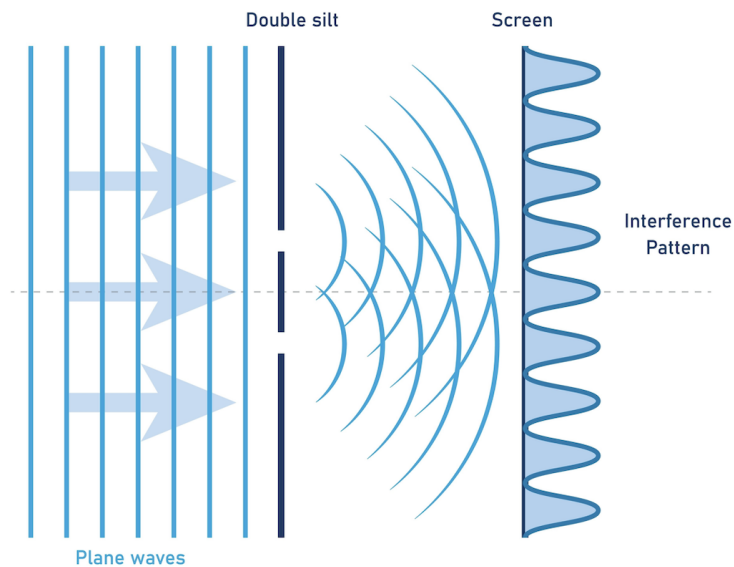


Figure 1.2: Conceptual diagram of the double-slit experiment. A coherent superposition of paths leads to the formation of interference fringes on the detection screen, demonstrating the wave-like nature of the particles.

Chapter 2

From EPR to Macrorealism

“Reality is fabulous. If men would steadily observe realities only, and not allow themselves to be deluded, life, to compare it with such things as we know, would be like a fairy tale and the Arabian Nights’ Entertainments.”

— Henry David Thoreau, *Walden* (1854)

2.1 Historical Context: From Physical Reality to Experimental Testability

The journey from the conceptual foundations of quantum mechanics to the experimental violation of local realism spans over thirty years of intense theoretical debate. It began in **1935**, when Albert Einstein, Boris Podolsky, and Nathan Rosen published their seminal paper [7], challenging the completeness of quantum mechanics. Their argument, known as the **EPR paradox**, was not intended to claim that quantum mechanics was "wrong," but rather that it was an incomplete description of an objective, local reality.

Einstein could not accept what he famously termed "spooky action at a distance" (*spukhafte Fernwirkung*), arguing that if due to a common past two particles were perfectly correlated, their properties must have been predetermined at the moment of their creation. This concept of predeterminism is precisely what is formalized through the hypothesis of **Local Hidden Variables (LHV)**: the idea that there exist underlying parameters, not accounted for by the wave function, which uniquely determine the results of local measurements.

For nearly three decades, the EPR argument remained largely in the realm of philosophy of science, as most physicists followed Niels Bohr’s **Copenhagen interpretation**, which posited that physical properties only become defined upon

the act of measurement. The debate was considered "metaphysical" because, at the time, both the LHV and Copenhagen views predicted the same experimental results for all available setups.

The breakthrough occurred in **1964**, when the Irish physicist **John Stewart Bell** transformed this philosophical stalemate into a mathematically testable hypothesis [8]. Bell showed that *any* LHV theory—regardless of its specific details—is subject to strict statistical constraints on the correlations between distant measurements. These constraints, known as **Bell's Inequalities**, are obeyed by classical-like realistic theories but can be violated by the predictions of quantum mechanics.

Finally, in **1969**, John Clauser, Michael Horne, Abner Shimony, and Richard Holt (CHSH) refined Bell's work [9]. They realized that Bell's original inequality required perfect correlations (such as perfect anti-parallelism), which are difficult to achieve in real laboratories due to detector inefficiencies and environmental noise. The **CHSH inequality** generalized the bound for any measurement settings, opening the door for the first experimental tests.

The subsequent decades saw the implementation of increasingly sophisticated experiments, starting with the pioneering work of Stuart Freedman and John Clauser (1972) [10], followed by the definitive tests using time-varying analyzers by Alain Aspect et al. (1982) [11]. This progression marks one of the most profound shifts in modern physics: the realization that the nature of reality itself is not merely a matter of interpretation, but a property that can be decided in a laboratory.

2.2 EPR Correlations, Local Realism, and Bell Inequalities

In this section, we analyze the transition from the conceptual problem of completeness in quantum mechanics to the formulation of experimentally testable constraints on local realistic theories. We begin by formalizing the EPR argument through the lens of entangled states, and subsequently derive Bell's original inequality and its more versatile generalization, the CHSH form [7, 8, 9, 2].

2.2.1 The EPR Argument: Perfect Correlations

The core of the Einstein-Podolsky-Rosen (EPR) paradox [7] lies in the existence of "elements of reality" inferred from perfect correlations. In other words, if the

outcome of a measurement on one subsystem allows us to predict with certainty the outcome of a corresponding measurement on a distant subsystem—without in any way disturbing it—then that outcome must correspond to a pre-existing property of the system. Such properties, according to EPR, constitute elements of reality and should be accounted for in any complete physical theory.

To see this, consider a bipartite system composed of two subsystems A and B , prepared in a maximally entangled Bell state $|\Phi^+\rangle$ as defined in Eq. (1.7):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle), \quad (2.1)$$

where $\{|0\rangle, |1\rangle\}$ is an orthonormal basis. We define the local observables for each subsystem as the difference between the projectors onto these basis states:

$$A_0 = P_0^{(A)} - P_1^{(A)}, \quad B_0 = P_0^{(B)} - P_1^{(B)}. \quad (2.2)$$

As already discussed in Sec. 1.3, the effect of the projectors on the basis states is to assign a value of $+1$ to $|0\rangle$ and -1 to $|1\rangle$.

When we compute the expectation value of the product of these observables, we are effectively measuring the degree of correlation between the two particles:

$$\langle \Phi^+ | A_0 \otimes B_0 | \Phi^+ \rangle = \frac{1}{2} (\langle 0_A 0_B | + \langle 1_A 1_B |) (A_0 \otimes B_0) (|0_A 0_B\rangle + |1_A 1_B\rangle). \quad (2.3)$$

Since the action of the observables on the basis states is $A_0|0\rangle = +|0\rangle$ and $A_0|1\rangle = -|1\rangle$, the entangled components evolve as:

$$A_0 \otimes B_0 |0_A 0_B\rangle = |0_A 0_B\rangle, \quad (2.4)$$

$$A_0 \otimes B_0 |1_A 1_B\rangle = |1_A 1_B\rangle. \quad (2.5)$$

Substituting these results back into the correlation integral, we obtain:

$$\langle \Phi^+ | A_0 \otimes B_0 | \Phi^+ \rangle = \frac{1}{2} (\langle 0_A 0_B | 0_A 0_B \rangle + \langle 1_A 1_B | 1_A 1_B \rangle) = 1. \quad (2.6)$$

This result, $\langle A_0 B_0 \rangle = 1$, indicates a perfect correlation. Crucially, this property is invariant under a rotation of the **measurement basis**. If we switch to a new basis $\{|\theta\rangle, |\theta_\perp\rangle\}$ defined by the rotation:

$$|\theta\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle, \quad |\theta_\perp\rangle = -\sin \theta |0\rangle + \cos \theta |1\rangle, \quad (2.7)$$

the state $|\Phi^+\rangle$ preserves its form due to its rotational symmetry. Therefore,

$\langle A_\theta B_\theta \rangle = 1$ for any choice of the measurement direction θ .

According to the EPR criteria, since we can predict the outcome of a measurement on B with certainty by measuring A (and vice versa) without in any way disturbing the second particle, the measured value must have been a "pre-determined element of reality". This implies that the quantum state $|\Phi^+\rangle$ cannot be a complete description of the system, necessitating the introduction of a hidden variable model to account for these pre-existing values.

2.2.2 Bell's Inequality (1964)

John Bell moved the discussion from philosophy to physics by asking if any local hidden variable (LHV) model could reproduce these quantum predictions. Assume the existence of hidden variables λ with a probability distribution $\rho(\lambda)$. In this classical-like model, the measurement outcomes are not probabilistic but are deterministic functions of the setting and the hidden variable:

$$A(\hat{a}, \lambda), B(\hat{b}, \lambda) \in \{-1, +1\}. \quad (2.8)$$

The observed correlation function is the average over the distribution of λ :

$$E(\hat{a}, \hat{b}) = \int d\lambda \rho(\lambda) A(\hat{a}, \lambda) B(\hat{b}, \lambda). \quad (2.9)$$

To simplify the derivation, following Bell's original work [8], we consider a **singlet state** $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, which exhibits perfect anti-correlation for identical measurement settings: $A(\hat{a}, \lambda) = -B(\hat{a}, \lambda)$. While this specific anti-correlation is a property of the state and not a general requirement for all LHV theories, it allows for a more direct derivation of the bound without loss of generality for the purpose of demonstrating the conflict with quantum mechanics.

We then compare the correlations between three different measurement directions $\hat{a}, \hat{b}$, and $\hat{c}$. By considering the difference between two correlation functions:

$$E(\hat{a}, \hat{b}) - E(\hat{a}, \hat{c}) = - \int d\lambda \rho(\lambda) A(\hat{a}, \lambda) [A(\hat{b}, \lambda) - A(\hat{c}, \lambda)]. \quad (2.10)$$

By taking the absolute value and noting that $|A(\hat{a}, \lambda)| = 1$, we can bound the difference as:

$$|E(\hat{a}, \hat{b}) - E(\hat{a}, \hat{c})| \leq \int d\lambda \rho(\lambda) |A(\hat{b}, \lambda) - A(\hat{c}, \lambda)|. \quad (2.11)$$

Using the algebraic identity $|x - y| = 1 - xy$, which holds for $x, y \in \pm 1$, we can

rewrite the integrand. Re-introducing the correlation function $E(\hat{b}, \hat{c})$ through the anti-correlation assumption $A(\hat{b}, \lambda)A(\hat{c}, \lambda) = -A(\hat{b}, \lambda)B(\hat{c}, \lambda)$, we arrive at the famous Bell Inequality:

$$|E(\hat{a}, \hat{b}) - E(\hat{a}, \hat{c})| \leq 1 + E(\hat{b}, \hat{c}). \quad (2.12)$$

Quantum Mechanical Violation Quantum mechanics predicts a correlation function of the form $E^{(q)}(\hat{a}, \hat{b}) = -\cos \theta_{ab}$, where θ_{ab} represents the relative angle between the measurement settings $\hat{a}$ and $\hat{b}$.

If we choose three coplanar directions such that the angles are $\theta_{ab} = \theta_{bc} = \pi/3$ and $\theta_{ac} = 2\pi/3$, the local realistic bound is challenged:

$$\text{Left Side: } |\cos(2\pi/3) - \cos(\pi/3)| = |-1/2 - 1/2| = 1, \quad (2.13)$$

$$\text{Right Side: } 1 - \cos(\pi/3) = 1 - 1/2 = 1/2. \quad (2.14)$$

The fact that $1 > 0.5$ represents a definitive failure of local realism to account for the predictions of the singlet state. This result demonstrates that no LHV theory can reproduce the statistical correlations of quantum mechanics.

2.2.3 The CHSH Inequality (1969)

The CHSH inequality [9] generalizes Bell's work by removing the need for perfect anti-correlation, making it suitable for experimental testing. This formulation considers two possible measurement settings for Alice ($\hat{a}$ or $\hat{a}'$) and two for Bob ($\hat{b}$ or $\hat{b}'$). For a single run of the experiment, governed by a hidden variable λ , we define the combination:

$$S(\lambda) = A(\hat{a}, \lambda)B(\hat{b}, \lambda) + A(\hat{a}, \lambda)B(\hat{b}', \lambda) + A(\hat{a}', \lambda)B(\hat{b}, \lambda) - A(\hat{a}', \lambda)B(\hat{b}', \lambda), \quad (2.15)$$

which can be rearranged as:

$$S(\lambda) = A(\hat{a}, \lambda) [B(\hat{b}, \lambda) + B(\hat{b}', \lambda)] + A(\hat{a}', \lambda) [B(\hat{b}, \lambda) - B(\hat{b}', \lambda)]. \quad (2.16)$$

Since each observable $A, B \in \{-1, +1\}$, it follows algebraically that one term in the square brackets must be 0 and the other must be ± 2 . Thus, for any individual run, $S(\lambda) = \pm 2$. Averaging over the probability distribution $\rho(\lambda)$, we obtain the CHSH bound:

$$|S| = \left| \int d\lambda \rho(\lambda) S(\lambda) \right| \leq 2. \quad (2.17)$$

Quantum Mechanical Violation In quantum mechanics, for a maximally entangled state (e.g., the singlet state $|\Psi^-\rangle$), the correlation function is $E(\hat{a}, \hat{b}) = -\cos \theta_{ab}$. By choosing the specific "Bell angles" ($\theta_{ab} = 3\pi/4, \theta_{ab'} = \pi/4, \theta_{a'b} = \pi/4, \theta_{a'b'} = \pi/4$), we obtain:

$$S^{(q)} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}. \quad (2.18)$$

The result $|S^{(q)}| \approx 2.82$ clearly exceeds the local realistic limit of 2. This violation, which reaches its theoretical maximum at the **Tsirelson bound** [12], provides the most robust evidence against the existence of local hidden variables.

The demonstration that quantum correlations between spatially separated systems cannot be explained by local realism sets the stage for a fundamental question: does a similar conflict arise when we consider temporal correlations within a single system? In the following sections, we will explore how this conceptual framework is extended to the time domain through the **Leggett–Garg inequalities**, shifting the focus from local realism to the concept of **macrorealism**.

2.3 Macrorealism and the Leggett–Garg Framework (1985)

The concept of macroscopic realism, or **macrorealism (MR)**, as formulated by Leggett and Garg [13], rests on the premise that a macroscopic system evolving over time possesses well-defined physical properties at any given moment, independent of whether an observation is performed. Historically, MR was developed as a temporal analogue to the concept of local realism in spatially entangled systems. This framework aims to determine whether the classical intuition of an objective reality can be formally tested against the predictions of quantum mechanics.

To provide a rigorous basis for this investigation, the concept of macrorealism is traditionally decomposed into three fundamental postulates:

1. **Macrorealism per se (MR_{ps})**: A macroscopic system capable of occupying two or more macroscopically distinct states is always in exactly one of those states at all times. This represents the ontological core of the theory, asserting that physical properties exist objectively and independently of any measurement process.

2. **Non-Invasive Measurability (NIM):** It is possible, in principle, to determine the state of the system without inducing any change in the state itself or affecting its subsequent dynamical evolution. This assumption ensures that the act of measurement does not perturb the system.
3. **Induction (IND):** Often referred to as the **arrow of time**, this postulate states that the present state of a system is not influenced by measurement choices or events occurring in the future. This causality requirement guarantees that the system's history remains independent of future experimental interventions.

In any empirical test of macrorealism, one has to evaluate the joint validity of these assumptions. Since Induction is generally accepted as a fundamental causal principle in physics, any conflict with experimental data is typically interpreted as a failure of the conjunction of MR_{ps} and NIM.

2.3.1 Ideal Negative Measurements and the Non-Invasive Postulate

The practical verification of macrorealism faces a significant challenge: the NIM postulate is an idealization that is difficult to guarantee in a laboratory setting. To address this, Leggett and Garg proposed the use of **Ideal Negative Measurements (INM)** as a rigorous method to minimize and control measurement-induced disturbances.

An INM is defined by a measurement protocol where the detector is designed to interact strongly with the system only when it occupies a specific state (e.g., $Q(t) = +1$) and to remain entirely passive otherwise. If the detector is activated at time t and no signal is elicited, we can infer that the system is in the state $Q(t) = -1$. Under the assumption of MR_{ps} , in the limit of an arbitrarily short measurement time, such a "null result" implies that no physical interaction occurred between the system and the probe. Consequently, the subsequent dynamical evolution must remain unperturbed, thereby satisfying the NIM requirement.

In an experimental setup designed to evaluate the temporal correlations $C_{ij} = \langle Q(t_i)Q(t_j) \rangle$ —which represent the statistical average of the product of the measurement outcomes at two different times—this protocol can be implemented as follows:

- To determine the joint probability $p_{ij}(-1, s_j)$, an INM is performed at t_i using a detector that only couples to the $Q = +1$ state. We discard all runs

where the detector fires. The remaining "silent" runs provide the value of the observable at t_i without a direct interaction.

- To obtain $p_{ij}(+1, s_j)$, a different experimental run is used where the detector couples exclusively to the state $Q = -1$.
- Since the evolution after the final time t_j is irrelevant for the correlation C_{ij} , a standard measurement can be performed at the final time step.

While this protocol is logically consistent within a macrorealistic framework, it reveals a fundamental conflict with quantum mechanics. In the quantum regime, even a "null result" provides information that restricts the system's possible histories, leading to a collapse of the quantum state. This divergence makes it difficult to pinpoint the exact origin of a violation: it could stem from a fundamental failure of macrorealism or from the fact that the measurement, although "ideal" in a classical sense, still disturbs the quantum system.

This ambiguity is central to the so-called **clumsiness loophole**. Macrorealism does not claim that all measurements are non-invasive; rather, it asserts that it is *possible* to perform a measurement that does not disturb the system. Therefore, an experimental violation could indicate either:

- (i) that the system does not obey the postulates of macrorealism;
- (ii) that the measurement protocol, despite its design, remains invasive and disturbs an otherwise macrorealistic system (the "clumsy" measurement).

It is worth noting that while other loopholes exist in Bell-type tests—such as the *locality loophole* or the *detection loophole*—the Leggett-Garg framework is primarily concerned with the clumsiness loophole. This is because LG tests are performed on a single system evolving in time, where spatial separation is not the main focus, making the invasiveness of the measurement the most critical point of vulnerability in the logical argument.

2.3.2 The Double-Slit Experiment as a Benchmark for NIM

The conceptual tension between macrorealism and quantum mechanics is best illustrated by the double-slit experiment, where the setup follows the formalism introduced in Sec. 1.6. In a classical framework, a particle is assumed to travel through one specific slit, regardless of whether it is observed. However, the emergence of an interference pattern on the detection screen suggests that,

in the absence of measurement, the system exists in a coherent superposition of the form:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle), \quad (2.19)$$

where $|L\rangle$ and $|R\rangle$ represent the states of the particle passing through the left and right slits, respectively.

The "crisis" of realism arises when an observer attempts to determine the specific path taken by the particle (the "which-way" information). This experiment serves as the ultimate benchmark for the Leggett–Garg framework because it perfectly illustrates the inherent fragility of the **Non-Invasive Measurability (NIM)** postulate.

Mathematical Origin of the Collapse To understand why an *ideal negative-result measurement* (where the detector at Slit R remains silent) is invasive, we must look at the mathematical description of the measurement process. As derived in Eq. (1.21), the acquisition of information—even without physical interaction—is formally an entanglement between the particle and the detector $\mathcal{D}$:

$$|\Psi_{global}\rangle = \frac{1}{\sqrt{2}}(|L\rangle \otimes |d_{silent}\rangle + |R\rangle \otimes |d_{fire}\rangle), \quad (2.20)$$

where $|d_{fire}\rangle$ and $|d_{silent}\rangle$ are the orthogonal states of the detector. When the detector at Slit R does not fire, the global state is projected onto the subspace corresponding to $|d_{silent}\rangle$. Following the projection postulate, the state of the particle immediately collapses to:

$$|\Psi_{after}\rangle = \langle d_{silent} | \Psi_{global} \rangle = |L\rangle. \quad (2.21)$$

In this process, the off-diagonal terms of the density matrix (the *coherences*) vanish: $\rho_{LR} = \langle L | \rho | R \rangle \rightarrow 0$. Consequently, the interference term $2\text{Re}(\psi_L^* \psi_R)$ in the probability distribution $P(x)$ disappears, and the pattern on the screen reverts to the classical two-band distribution.

As shown in Figure 2.1, while such a "null" result is non-invasive in a classical sense—since no physical force or energy is exchanged with the particle passing through Slit L—it remains profoundly invasive in a quantum sense. The mere fact that the state $|R\rangle$ is "ruled out" destroys the phase coherence necessary for interference. For Leggett and Garg, the challenge was to translate this spatial mystery into the temporal evolution of a single system, providing a formal way to quantify this "collapse" through observable correlation inequalities.

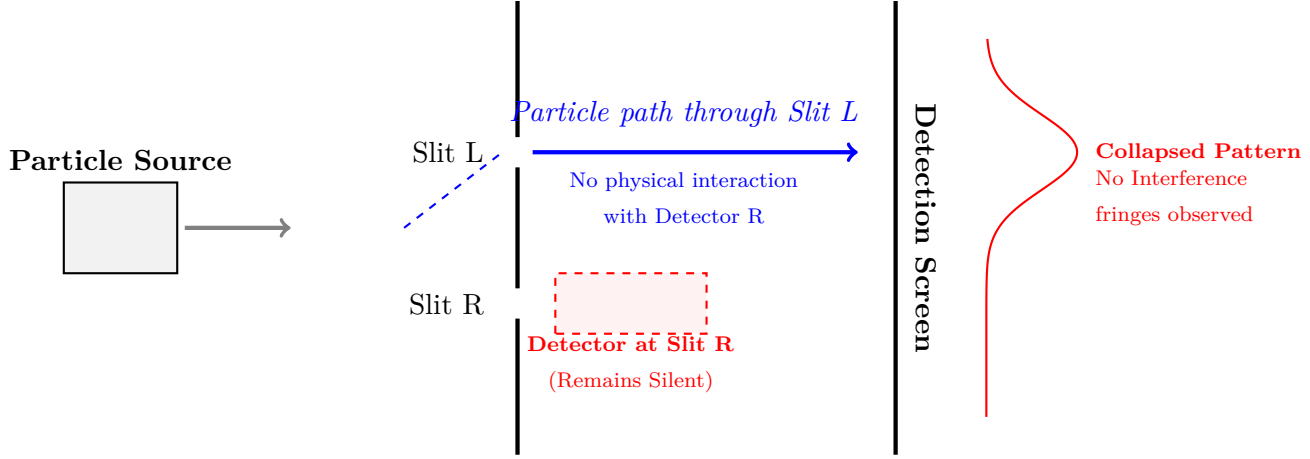


Figure 2.1: Conceptual diagram of an **Ideal Negative-Result Measurement (INRM)**. The detector at Slit R remains silent, implying the particle passed through Slit L. Despite the lack of physical interaction, the acquisition of "which-way" information triggers the projection of the quantum state, eliminating the interference fringes.

2.4 Leggett-Garg Inequalities

Originally designed for macroscopic systems such as Superconducting Quantum Interference Devices (SQUIDs), the Leggett–Garg inequalities (LGIs) [13] analyze temporal correlations between measurements performed at different times on a single quantum system. Unlike Bell inequalities, which probe spatial non-locality, LGIs test the compatibility of a system's evolution with the principles of macroscopic realism.

In its original formulation, the LGI involves a dichotomic observable $Q = \pm 1$ measured at three successive times $t_1 < t_2 < t_3$. The fundamental quantity is the correlation function $C_{ij} = \langle Q_i Q_j \rangle$, defined as:

$$C_{ij} = \sum_{q_i, q_j = \pm 1} q_i q_j P_{ij}(q_i, q_j), \quad (2.22)$$

where $P_{ij}(q_i, q_j)$ is the joint probability of obtaining outcomes q_i and q_j at times t_i and t_j . For a system that satisfies the postulates of Macroscopic Realism (MR_{ps}) and Non-Invasive Measurability (NIM), Leggett and Garg derived a bound on the combination of two-time correlation functions, expressed as:

$$\text{LGI}_{123} \equiv C_{12} + C_{23} - C_{13} \leq 1. \quad (2.23)$$

To clarify how these postulates constrain the statistical properties of a system, we formalize the Leggett–Garg theorem following the approach of Emary et al.

[14].

Proposition 2.4.1 (Leggett–Garg Inequality). *Consider a system described by a dichotomic observable $Q = \pm 1$ measured at three successive times $t_1 < t_2 < t_3$. If the system obeys the postulates of MR_{ps} and NIM, then the two-time correlation functions $C_{ij} = \langle Q_i Q_j \rangle$ must satisfy:*

$$LGI_{123} \equiv C_{12} + C_{23} - C_{13} \leq 1. \quad (2.24)$$

Proof. Under the assumption of MR_{ps} , the observable Q possesses a well-defined value $q_i \in \{\pm 1\}$ at all times. Mathematically, this implies that any two-time joint probability $P_{ij}(q_i, q_j)$ obtained from experiments is necessarily a marginal of a more fundamental, global three-time joint probability distribution $P(q_3, q_2, q_1)$. The postulate of NIM further asserts that a measurement can be performed without altering the system's subsequent dynamics, ensuring that the same unique distribution describes the system regardless of which pairs are measured.

Using the shorthand notation $P(q_3, q_2, q_1)$, where each q_i takes values ± 1 , the correlators defined in Eq. (2.22) can be expanded as:

$$\begin{aligned} C_{12} &= +P(+++) + P(-++) - P(+ - +) - P(- - +) \\ &\quad - P(++-) - P(- + -) + P(+ - -) + P(- - -) \\ C_{23} &= +P(+++) - P(-++) + P(+ - +) - P(- - +) \\ &\quad - P(++-) + P(- + -) - P(+ - -) + P(- - -) \\ C_{13} &= +P(+++) - P(-++) - P(+ - +) + P(- - +) \\ &\quad + P(++-) - P(- + -) - P(+ - -) + P(- - -) \end{aligned}$$

By constructing the linear combination $LGI_{123} \equiv C_{12} + C_{23} - C_{13}$ and invoking the normalization condition $\sum_{q_i} P(q_3, q_2, q_1) = 1$, the expression simplifies to:

$$LGI_{123} = 1 - 4[P(q_3 = +, q_2 = -, q_1 = +) + P(q_3 = -, q_2 = +, q_1 = -)]. \quad (2.25)$$

Since a realist description requires all probabilities to be non-negative, it follows directly that $LGI_{123} \leq 1$. The lower bound $LGI_{123} \geq -3$ is reached when the system is exclusively found in "alternating" trajectories, such as $(+, -, +)$ or $(-, +, -)$ with probability 1. $\square$

2.4.1 Generalizations of the Leggett–Garg Inequalities

While the three-time inequality is the most historically significant, the framework can be extended to various configurations. Note that in a three-time setting, LGI_{123} is only one of a set of four possible inequalities that can be derived by permuting signs (e.g., $-C_{12} - C_{23} - C_{13} \leq 1$), as shown in [15].

Two-Time LGIs A fundamental set of constraints can be derived by considering only two measurement times, t_i and t_j . For a dichotomic observable Q , the two-time LGIs are:

$$LGI_{ij} \equiv 1 + s_i \langle Q_i \rangle + s_j \langle Q_j \rangle + s_i s_j \langle Q_i Q_j \rangle \geq 0, \quad (2.26)$$

where $s_i, s_j = \pm 1$ are fixed parameters and $\langle Q_i \rangle$ denotes the single-time expectation value:

$$\langle Q_i \rangle = \sum_{q_i=\pm 1} q_i P(q_i). \quad (2.27)$$

Under the assumptions of MR_{ps} and NIM, the non-negativity of the joint probability $P(q_i, q_j)$ directly implies Eq. (2.26). A violation ($LGI_{ij} < 0$) is a sufficient condition to exclude a macrorealistic description.

n -Time String Inequalities The protocol can be generalized to a sequence of n measurements at times $t_1, \dots, t_n$ through the string $K_n = C_{12} + C_{23} + \dots + C_{(n-1)n} - C_{1n}$. For a macrorealistic system, the following constraints hold [14]:

$$\begin{aligned} -n \leq K_n \leq n-2 & \quad \text{for odd } n \geq 3, \\ -(n-2) \leq K_n \leq n-2 & \quad \text{for even } n \geq 4. \end{aligned} \quad (2.28)$$

For $n = 3$, K_3 reduces to the original LGI_{123} in Eq. (2.23). In the quantum regime, the maximum value for K_n —the temporal analogue of the Tsirelson bound [12]—is $K_n^{max} = n \cos(\pi/n)$, which exceeds the limit $n-2$ for any $n > 2$.

2.5 Fine’s Theorem

Before diving into the mathematical derivation of the joint probability distributions, it is essential to clarify the motivation behind Fine’s theorem and its role in the debate over the foundations of quantum mechanics.

The Bell and CHSH inequalities were originally derived as a *necessary* consequence of local realism. This means that if a LHV theory exists, then the

inequalities must hold. However, from a logical and experimental standpoint, this leaves an open question: if we find a set of experimental data that satisfies the Bell inequalities, can we be absolutely certain that a local realist model exists for those data?

Arthur Fine's contribution [16] was to close this logical circle. In 1982, he demonstrated that the satisfaction of Bell-type inequalities [8] is not only necessary but also a **sufficient** condition for the existence of a deterministic LHV model. Mathematically speaking, Fine's theorem shifts the problem from physics to probability theory: it proves that the existence of an LHV model is equivalent to the existence of a *single global joint probability distribution* that returns all measured experimental correlations as its marginals.

2.5.1 Marginal Probabilities and Global Distributions

First, let us consider a simple system composed of two variables with measurement outcomes s_1 and s_2 , such that $s_i \in \{\pm 1\}$. There are $2^2 = 4$ possible combinations of measurement outcomes (s_1, s_2) , each associated with a probability $p_{12}(s_1, s_2)$. These four probabilities must satisfy the normalization condition:

$$\sum_{s_1, s_2 = \pm 1} p_{12}(s_1, s_2) = 1. \quad (2.29)$$

Since the system is completely described by these two variables, the union of the four combinations must equal the entire sample space. In other words, for this two-variable system, the pairwise probabilities $p_{12}(s_1, s_2)$ represent the underlying probability distribution of the system.

Given an underlying distribution, we can derive the *marginal probabilities*. In probability theory, a marginal probability is defined as the probability of a subset of variables occurring regardless of the values of the other variables in the distribution. In this case, by summing over one of the variables s_j , we find the marginal probability $p_i(s_i)$ for $i \neq j$:

$$p_1(s_1) = \sum_{s_2 = \pm 1} p_{12}(s_1, s_2). \quad (2.30)$$

This concept extends naturally to systems with n variables [16, 15]. For a system with n measurement outcomes, the underlying joint probability distribution is denoted as $p_{12\dots n}(s_1, s_2, \dots, s_n)$, and the marginal probabilities are obtained by summing over the appropriate variables. As an example, the marginal probability

for the first variable is given by:

$$p_1(s_1) = \sum_{s_2, s_3, \dots, s_n = \pm 1} p_{12\dots n}(s_1, s_2, \dots, s_n). \quad (2.31)$$

In the experimental context of the CHSH inequalities, the system involves four variables s_1, s_2, s_3, s_4 , representing the outcomes for Alice (settings 1 and 2) and Bob (settings 3 and 4). If an LHV theory provides the correct description, there must exist a global joint probability distribution $p_{1234}(s_1, s_2, s_3, s_4)$ such that the four experimentally accessible pairwise probabilities— $p_{13}(s_1, s_3)$, $p_{14}(s_1, s_4)$, $p_{23}(s_2, s_3)$, and $p_{24}(s_2, s_4)$ —are its marginals.

It is important to note that combinations like p_{12} or p_{34} are not considered because they would require measuring two different settings on the same particle simultaneously (e.g., Alice measuring along axis 1 and axis 2 at the same time), which is prohibited by the uncertainty principle and the experimental setup. However, Fine’s theorem ensures that if the *measurable* marginals satisfy the CHSH inequalities, a global distribution p_{1234} exists, even for the non-measurable sectors.

Consequently, violating the CHSH inequalities implies that such a global distribution cannot exist, thereby proving that the experimental data is incompatible with any LHV model.

2.5.2 Proof of Fine’s Theorem

Without loss of generality (WLOG), we focus on the CHSH scenario involving four variables s_1, s_2, s_3, s_4 . While the standard derivation of CHSH inequalities shows they are a *necessary* consequence of local realism, it does not immediately prove that their satisfaction implies the existence of an underlying probability distribution. Fine’s theorem addresses this by proving that satisfying these inequalities is also a *sufficient* condition.

Proposition 2.5.1 (Fine, 1982). *For a correlation experiment, the following statements are equivalent:*

1. *There exists a deterministic Local Hidden Variable (LHV) model for the experiment.*
2. *There exists a factorizable, stochastic model for the experiment.*
3. *There exists a global joint distribution for all observables whose marginals match the experimental probabilities.*

4. The Bell/CHSH inequalities hold for the measured correlations.

Proof. To demonstrate the sufficiency, we follow the derivation by Halliwell [15] for a system with three variables and their pairwise marginals: $p_{12}(s_1, s_2)$, $p_{23}(s_2, s_3)$, and $p_{13}(s_1, s_3)$. The underlying global distribution $p_{123}(s_1, s_2, s_3)$ can be expanded in terms of its moments:

$$p_{123}(s_1, s_2, s_3) = \frac{1}{8} \left(1 + \sum_i B_i s_i + \sum_{i < j} C_{ij} s_i s_j + D s_1 s_2 s_3 \right), \quad (2.32)$$

where $B_i = \langle s_i \rangle$ are the expectations of single variables, $C_{ij} = \langle s_i s_j \rangle$ are the pairwise correlations, and $D = \langle s_1 s_2 s_3 \rangle$ is the triple correlation.

The marginal probabilities are found by summing over the hidden variable. For example, the marginal p_{13} is given by:

$$p_{13}(s_1, s_3) = \sum_{s_2=\pm 1} p_{123}(s_1, s_2, s_3) = \frac{1}{4} (1 + B_1 s_1 + B_3 s_3 + C_{13} s_1 s_3). \quad (2.33)$$

Since there are three pairs (1, 2), (2, 3), (1, 3) and each pair has $2^2 = 4$ possible outcomes, we obtain a total of 12 equations of this form. These equations fix the values of all B_i and C_{ij} coefficients based on the experimentally accessible marginals.

For the model to be physically valid, p_{123} must be non-negative for all $2^3 = 8$ combinations of s_i . Defining $A(s_1, s_2, s_3) = 1 + \sum B_i s_i + \sum_{i < j} C_{ij} s_i s_j$, the condition $p_{123} \geq 0$ translates to:

$$A(s_1, s_2, s_3) + D s_1 s_2 s_3 \geq 0. \quad (2.34)$$

This requirement constrains the possible values of the unobserved triple correlation D :

$$D \geq -A(s_1, s_2, s_3) \quad \text{for } s_1 s_2 s_3 = 1, \quad (2.35)$$

$$D \leq A(s_1, s_2, s_3) \quad \text{for } s_1 s_2 s_3 = -1. \quad (2.36)$$

A consistent value for D exists if and only if every possible lower bound is less than or equal to every possible upper bound (i.e., $\max(\text{lower}) \leq \min(\text{upper})$). This comparison generates 16 inequalities (4 lower $\times$ 4 upper).

Twelve of these are redundant as they reduce to the non-negativity of the original marginals. The remaining four "new" inequalities, however, involve only

the pairwise correlations:

$$1 + C_{12} + C_{13} - C_{23} \geq 0 \quad (2.37)$$

$$1 + C_{12} - C_{13} + C_{23} \geq 0 \quad (2.38)$$

$$1 - C_{12} + C_{13} + C_{23} \geq 0 \quad (2.39)$$

$$1 - C_{12} - C_{13} - C_{23} \geq 0 \quad (2.40)$$

These four expressions are precisely the rearranged forms of **Bell's inequalities**. Thus, the existence of a valid global probability distribution p_{123} is equivalent to the satisfaction of these inequalities. This proves that they are a sufficient condition for the existence of an LHV model. $\square$

2.5.3 Significance and Limitations of Fine's Theorem

The significance of Fine's theorem lies in its ability to reduce a deep ontological question—the existence of a deterministic hidden-variable model—to a purely statistical one. It proves that Bell-type inequalities are not merely consequences of local realism, but are its *defining boundaries*. If these inequalities are satisfied, it is mathematically impossible to find any statistical test that could rule out a local realist description for that specific set of observables.

However, a crucial distinction arises when attempting to apply this logic to the Leggett–Garg framework. While satisfying CHSH inequalities is sufficient for realism in the Bell scenario, the same does not hold for the standard Leggett–Garg inequalities (LGIs) due to two primary factors:

1. **Temporal Compatibility and Invasiveness:** Fine's theorem assumes that pairwise marginals are compatible, meaning they can be treated as parts of a single global distribution. In the LG framework, measurements are performed on the same system at different times. As discussed in the context of NIM, the act of measuring at t_1 can physically alter the state at t_2 . Therefore, the mere mathematical existence of a joint distribution does not automatically guarantee that the measurements performed were non-invasive in a physical sense.
2. **Incompleteness of the LGI Set:** As shown in the proof of Fine's theorem, sufficiency requires the satisfaction of the *complete* set of inequalities (12 for non-negativity of marginals and 4 Bell-like inequalities). Most LG experiments test only a single string, such as $LGI_{123} = C_{12} + C_{23} - C_{13} \leq$

1. While this is a necessary condition for macrorealism, it is not sufficient; a system could satisfy this specific LGI while still failing to admit a global joint distribution if the other three required inequalities (e.g., $-C_{12} - C_{23} - C_{13} \leq 1$) are violated.

In conclusion, Fine’s theorem provides a definitive proof of sufficiency for Bell’s theorem, but the temporal nature of Leggett–Garg tests introduces dynamical complexities that prevent a single LGI from being a standalone sufficient condition for macrorealism. Consequently, it is necessary to consider an extended set of inequalities and compatibility conditions to fully characterize macrorealism in the temporal domain. This leads us to a more nuanced classification—distinguishing between **Weak**, **Intermediate**, and **Strong Macrorealism**—which we will explore in the next section.

2.6 Weak Macrorealism

In this section, we present a framework for establishing necessary and sufficient conditions for macrorealism by utilizing an extended set of Leggett–Garg inequalities. This approach assumes a weaker form of non-invasiveness, where the NIM postulate is only satisfied in a piecewise (pw) manner across separate experiments (NIM_{pw}). We follow the methodology proposed by Halliwell [15], which stays as close as possible to the original LG framework where correlation functions are determined in distinct experimental runs.

Adopting this methodology, we describe a series of experiments designed to extract the expectation values $\langle Q_i \rangle$ and second-order temporal correlations C_{ij} across three time steps. This is achieved by measuring only compatible observables within each individual run, typically through ideal negative-result measurements. A practical approach involves four distinct experiments:

1. Measure $\langle Q_1 \rangle$ and C_{12} ;
2. Determine $\langle Q_2 \rangle$ and C_{23} ;
3. Measure C_{13} ;
4. Measure $\langle Q_3 \rangle$.

(Depending on the setup, the latter two can be combined if C_{13} and $\langle Q_3 \rangle$ are mutually compatible).

Note on Quantum Resources From an experimental perspective, testing MR requires significant resources. For a fixed number of shots per run, the total number of measurements scales with the number of terms in the set of correlation functions. Specifically, while a standard Bell test requires correlation measurements between two particles, the MR_{weak} protocol demands the reconstruction of six different moments ($\langle Q_1 \rangle, \langle Q_2 \rangle, \langle Q_3 \rangle, C_{12}, C_{23}, C_{13}$) across independent trials, effectively quadrupling the required data acquisition time compared to a single-run temporal correlation test.

◇

The core of the macrorealistic inquiry is whether a fundamental joint probability distribution $p(s_1, s_2, s_3)$ exists that is consistent with these experimental moments. We construct this distribution through three stages. First, the single-time probabilities are defined as $p(s_i) = \frac{1}{2}(1 + s_i \langle Q_i \rangle)$, which are inherently non-negative. Second, we define the two-time joint probabilities as in Eq. (2.26):

$$p(s_i, s_j) = \frac{1}{4} (1 + s_i \langle Q_i \rangle + s_j \langle Q_j \rangle + s_i s_j C_{ij}). \quad (2.41)$$

Since the averages $\langle Q_i \rangle$ are assumed to be obtained through non-invasive procedures—such as the ideal negative-result measurements discussed in Sec. 2.3.1—these distributions must satisfy the following compatibility constraints:

$$\sum_{s_i} p(s_i, s_j) = p(s_j) = \sum_{s_k} p(s_j, s_k). \quad (2.42)$$

These relations are mathematically equivalent to the **No-Signaling-in-Time (NSIT)** conditions [17]. In this MR_{weak} context, NSIT ensures that the marginal probability of a measurement at time t_j is independent of whether a (non-invasive) measurement was performed at an earlier time t_i . It is important to clarify that, since the two-time probabilities are reconstructed from separate experimental runs rather than sequential ones, these conditions do not verify the absence of a physical disturbance in a single trial. Instead, they act as a **consistency check** for the piecewise reconstruction: they ensure that the statistics of Q_j remain identical across different experimental setups, justifying the assumption that our measurement protocol (e.g., the negative-result setup) is indeed non-invasive.

The two-time probabilities in Eq. (2.41) must be non-negative, leading to the

twelve two-time LGIs previously introduced in Eq. (2.26):

$$1 + s_i \langle Q_i \rangle + s_j \langle Q_j \rangle + s_i s_j C_{ij} \geq 0. \quad (2.43)$$

These conditions are necessary and sufficient at the two-time level; if they hold, the expressions yield valid probabilities that match the experimental data within statistical margins [15].

Finally, the most general form for the three-time joint probability is:

$$p(s_1, s_2, s_3) = \frac{1}{8} \left(1 + \sum_i s_i \langle Q_i \rangle + \sum_{i < j} s_i s_j C_{ij} + s_1 s_2 s_3 D \right), \quad (2.44)$$

where $D = \langle Q_1 Q_2 Q_3 \rangle$ is the triple correlator. Crucially, D is not directly observed in the piecewise protocol because measuring it would require a sequential measurement of all three times, potentially violating NIM. The problem reduces to determining if there exists *any* value of D such that $p(s_1, s_2, s_3) \geq 0$. According to Fine's theorem [16], such a value exists if and only if the two-time LGIs and the four three-time LGIs (Eq. (2.23)) are satisfied.

In summary, we define MR_{weak} as the joint satisfaction of:

$$MR_{weak} = NIM_{pw} \wedge LGI_{12} \wedge LGI_{23} \wedge LGI_{13} \wedge LGI_{123} \wedge IND. \quad (2.45)$$

A violation of MR_{weak} occurs if at least one of these inequalities is violated, as stated by De Morgan's laws [18]:

$$\neg MR_{weak} = \neg LGI_{12} \vee \neg LGI_{13} \vee \neg LGI_{23} \vee \neg LGI_{123}. \quad (2.46)$$

2.7 Strong Macrorealism

In this section, we examine a fundamentally different set of criteria for macrorealism that relies exclusively on **No-Signaling in Time (NSIT)** conditions, bypassing the use of Leggett-Garg inequalities altogether. This approach is primarily based on the comprehensive framework established by Clemente and Kofler [19].

The central premise of this framework is that the system undergoes sequential measurements at three distinct times within the same experimental run. To ensure that these observations are truly non-invasive, specific constraints are imposed under the principle of NIM_{seq} (sequential non-invasive measurability). This protocol allows for the direct determination of the underlying three-time

probability distribution $p(s_1, s_2, s_3)$. However, the sequential nature of the measurements makes this procedure significantly more restrictive than the augmented LG tests previously discussed.

When accounting for the possibility of measurement-induced disturbance in sequential trials, the most general expression for the three-time probability is:

$$p(s_1, s_2, s_3) = \frac{1}{8} \left(1 + s_1 \langle Q_1 \rangle + s_2 \langle Q_2^{(1)} \rangle + s_3 \langle Q_3^{(12)} \rangle + s_1 s_2 C_{12} + s_1 s_3 C_{13}^{(2)} + s_2 s_3 C_{23}^{(1)} + s_1 s_2 s_3 D \right). \quad (2.47)$$

In this expression, the superscripts indicate that expectation values and correlation functions may depend on the history of prior measurements. For instance, $\langle Q_3^{(12)} \rangle$ represents the average at t_3 given that measurements occurred at both t_1 and t_2 , while $C_{13}^{(2)}$ denotes the correlation between t_1 and t_3 when an intermediate measurement is performed at t_2 . Unlike the MR_{weak} framework where the triple correlator D is an unknown parameter, here D is explicitly determined by the measurement process itself.

Assuming the principle of induction holds and that there is no dependence on future measurements, we verify macrorealism through the NSIT conditions. We consider the marginal measurement probabilities involving fewer time steps (specifically single-time and two-time marginals):

$$p_{13}(s_1, s_3) = \frac{1}{4} \left(1 + s_1 \langle Q_1 \rangle + s_3 \langle Q_3^{(1)} \rangle + s_1 s_3 C_{13} \right) \quad (2.48)$$

$$p_{23}(s_2, s_3) = \frac{1}{4} \left(1 + s_2 \langle Q_2 \rangle + s_3 \langle Q_3^{(2)} \rangle + s_2 s_3 C_{23} \right) \quad (2.49)$$

$$p_{12}(s_1, s_2) = \frac{1}{4} \left(1 + s_1 \langle Q_1 \rangle + s_2 \langle Q_2^{(1)} \rangle + s_1 s_2 C_{12} \right) \quad (2.50)$$

$$p_3(s_3) = \frac{1}{2} (1 + s_3 \langle Q_3 \rangle) \quad (2.51)$$

The first NSIT condition, $NSIT_{(2)3}$, is defined as:

$$\sum_{s_2} p_{23}(s_2, s_3) = p_3(s_3). \quad (2.52)$$

This condition ensures that $\langle Q_3^{(2)} \rangle = \langle Q_3 \rangle$. Subsequently, the condition $NSIT_{(1)23}$ is given by:

$$\sum_{s_1} p_{123}(s_1, s_2, s_3) = p_{23}(s_2, s_3), \quad (2.53)$$

which leads to $C_{23}^{(1)} = C_{23}$, $\langle Q_2^{(1)} \rangle = \langle Q_2 \rangle$, and $\langle Q_3^{(12)} \rangle = \langle Q_3^{(2)} \rangle$ (which conse-

quently equals $\langle Q_3 \rangle$). Finally, the NSIT condition $NSIT_{1(2)3}$ is expressed as:

$$\sum_{s_2} p_{123}(s_1, s_2, s_3) = p_{13}(s_1, s_3). \quad (2.54)$$

This ensures that $C_{13}^{(1)} = C_{13}$ and confirms that $\langle Q_3^{(12)} \rangle = \langle Q_3^{(2)} \rangle = \langle Q_3 \rangle$.

These three constraints collectively guarantee that all statistical moments are invariant with respect to prior measurements. Under these conditions, the joint probability simplifies to:

$$p_{123}(s_1, s_2, s_3) = \frac{1}{8} \left(1 + \sum_i s_i \langle Q_i \rangle + \sum_{i < j} s_i s_j C_{ij} + s_1 s_2 s_3 D \right). \quad (2.55)$$

We define **Strong Macrorealism** as the joint satisfaction of these conditions:

$$MR_{strong} = NIM_{seq} \wedge NSIT_{(2)3} \wedge NSIT_{(1)23} \wedge NSIT_{1(2)3} \wedge IND. \quad (2.56)$$

In this definition, we assume NIM_{seq} and IND are satisfied as foundational postulates of a macrorealistic theory. Therefore, an experimental violation of the NSIT conditions directly implies a failure of strong macrorealism:

$$\neg MR_{strong} = \neg NSIT_{(2)3} \vee \neg NSIT_{(1)23} \vee \neg NSIT_{1(2)3}. \quad (2.57)$$

In the quantum regime, these requirements can only be met if the system's interference terms are completely suppressed, making Strong Macrorealism a significantly more stringent criterion than the weak version.

2.8 Intermediate Macrorealism

A third, intermediate perspective naturally emerges from the previous frameworks, as discussed in the context of persistent macroscopic coherence [15, 19]. This approach involves conducting three separate experiments, each employing sequential measurements at only two distinct times.

For macrorealism to hold in this context, we require that the resulting two-time probability distributions satisfy the two-time NSIT conditions, of the form, Eq. (2.52). Furthermore, these measurements must also satisfy the three-time Leggett–Garg inequalities Eq. (2.23). Consequently, this protocol evaluates the following version of MR:

$$MR_{int} = NIM_{seq} \wedge NSIT_{(1)2} \wedge NSIT_{(1)3} \wedge NSIT_{(2)3} \wedge LGI_{123} \wedge IND. \quad (2.58)$$

Similar to the augmented LG protocol described in Eq. (2.46), this intermediate formulation remains aligned with the fundamental principles of the original LG framework while providing a set of necessary and sufficient conditions for a macrorealistic description.

In line with the logic established in the preceding sections, the primary goal of this framework is to identify any violation of this joint set of assumptions. Since MR_{int} is invalidated if even a single constituent condition is not satisfied, the violation criteria can be formally expressed using De Morgan's laws. Assuming that NIM and IND hold as foundational postulates, any violation of MR_{int} arises solely from the breakdown of either the NSIT conditions or the three-time LGI:

$$\neg MR_{int} = \neg NSIT_{(1)2} \vee \neg NSIT_{(1)3} \vee \neg NSIT_{(2)3} \vee \neg LGI_{123}. \quad (2.59)$$

This specific formulation ensures the detection of non-classical signatures, whether they stem from quantum correlations (LGI) or from measurement-induced disturbances ($NSIT$). In other words, a "non-classical signature" here means that the experiment fails to behave classically either because the data violates the "realist" bounds of $Q = \pm 1$, or because the act of measuring at an earlier time t_i has physically changed the statistics at a later time t_j (mimicking a signaling effect).

By evaluating $\neg MR_{int}$, we address the inherent constraints of Weak Macrorealism, recognizing that a strictly classical system must not only comply with correlation boundaries but also remain statistically unaffected by the act of measurement at the level of its marginal distributions.

2.9 Summary of Macrorealistic Frameworks

To provide a clear overview of the different definitions of macrorealism discussed in this chapter, we summarize their logical structures and violation criteria in Table 2.1.

The classification presented above illustrates a clear progression in the stringency of the requirements imposed on a macrorealistic theory.

The **Weak Macrorealism** framework represents the most permissive test, as it relies on piecewise measurements and focuses on the algebraic bounds of correlation functions. It is primarily concerned with whether the observed correlations could, in principle, be derived from a global joint probability distribution, even if the individual moments are collected in separate experimental runs.

Table 2.1: Comparison of Weak, Intermediate, and Strong Macrorealism.

Framework	Measurement	Core Requirements	Violation ($\overline{MR}$)
Weak	Piecewise	$NIM_{pw} \wedge IND \wedge LGI_{ij} \wedge LGI_{123}$	Eq. (2.46)
Intermediate	2-time Seq.	$NIM_{seq} \wedge IND \wedge NSIT_{ij} \wedge LGI_{123}$	Eq. (2.59)
Strong	3-time Seq.	$NIM_{seq} \wedge IND \wedge NSIT_{ijk}$	Eq. (2.57)

In contrast, **Strong Macrorealism** occupies the opposite end of the spectrum. By demanding that the system satisfies a full set of NSIT conditions during a three-time sequential measurement, it requires that the act of observation leaves the system's subsequent evolution entirely unperturbed. In a quantum mechanical context, this is equivalent to demanding the complete suppression of quantum interference at every step of the process.

The **Intermediate Macrorealism** serves as a hybrid protocol. It adopts the sequential measurement character of the strong version but restricts it to two-time pairs, while still enforcing the three-time Leggett–Garg constraints. This allows for a more nuanced detection of non-classicality, distinguishing between violations caused by "anomalous" correlations and those caused by direct measurement-induced disturbance.

Logically, these definitions satisfy the hierarchy $MR_{strong} \implies MR_{int} \implies MR_{weak}$. This means that any system that fails a Weak MR test is necessarily non-macrorealistic under all other definitions, whereas a system could potentially satisfy Weak MR while being revealed as non-classical by the more rigorous Strong MR criteria.

Chapter 3

Randomness Generation: From Classical to Quantum

“In poker you don’t play against the cards, you play against the person sitting across from you.”
— *Casino Royale* (2006)

3.1 Introduction to Randomness

The history of modern computational science is inextricably linked to the study of stochastic processes. One of the most significant breakthroughs in this field occurred not in a laboratory, but in a hospital room in 1946. While recovering from an illness, the mathematician Stanisław Ulam spent time playing Solitaire. Frustrated by his inability to calculate the probability of a successful game using pure combinatorics, he realized that a more practical approach would be to play several games, record the outcomes, and observe the frequency of success. This anecdote is often cited as the conceptual starting point of *Monte Carlo reasoning* in modern computation [20].

This intuition—that complex, deterministic problems could be solved by simulating random processes—led to the development of the **Monte Carlo method** in collaboration with John von Neumann and Nicholas Metropolis [21]. However, implementing this method on early computers like the ENIAC required a fast way to generate long sequences of numbers that appeared random. In response to this need, John von Neumann proposed the first algorithmic generator, known as the *middle-square method*. The process involved taking an initial n -digit seed, squaring it, and extracting the middle n digits to form the next number in the sequence. For example, if a 4-digit seed like 1111 is squared to get 01234321, the

middle four digits (2343) become the next value.

While revolutionary, these early arithmetical methods exposed a conceptual limitation: outputs generated by fixed rules are deterministic by construction. As already noted in early foundational discussions [22], there is a central tension between algorithmic generation and true unpredictability. These early sequences eventually fell into short loops or converged to zero, exposing the inherent fragility of early *Pseudo-Random Number Generators* (PRNGs). It is important to clarify that the primary limitation of PRNGs—which encompass all classical algorithmic generators—is not merely the potential for short periods, but the fact that they are not truly random; they are deterministic expansions of an initial entropy source.

Subsequently, more sophisticated algorithms were developed to overcome the rapid degradation of early schemes. Modern generators, such as the **Linear Congruential Generator** (LCG) [23] and the **Mersenne Twister** [24], rely on modular arithmetic and linear recurrences over finite fields. Compared with the middle-square method, these approaches substantially improve both the statistical quality of the output and the period length (i.e., the number of steps before the sequence starts repeating). For instance, the Mersenne Twister algorithm has a period length of $2^{19937} - 1$.

Despite these mathematical advances, all such systems remain fundamentally dependent on an initial integer seed. Because the generation rule is deterministic, any observer who knows both the seed and the underlying recurrence can reproduce the entire sequence exactly. This dependence on an initial entropy source creates a security bottleneck: the strength of the output randomness is ultimately limited by the quality of the seed itself. Furthermore, because these are classical processes, they lack intrinsic unpredictability; they are merely complex transformations of initial information. In this framework, randomness remains an architectural simulation rather than a physical reality, leaving open the question of whether a source of entropy exists that is fundamentally decoupled from computational complexity or initial knowledge.

3.2 The Dual Nature of Randomness

In scientific literature and foundational physics, randomness is not a monolithic concept but is divided into two distinct categories that reflect the underlying understanding of physical laws: epistemological and ontological randomness.

3.2.1 Epistemological Randomness: The Classical Limit

Epistemological randomness, often associated with pseudo-randomness, describes systems that appear unpredictable only due to a lack of information or computational power. In a classical Newtonian universe, every event is, in principle, deterministic. This worldview is best exemplified by *Laplace's Demon*¹: an intellect that, if provided with the precise position and momentum of every particle in the universe, could use the laws of classical mechanics to calculate the entire past and future.

Within this framework, the toss of a coin or the roll of a die is not inherently random; it is a complex dynamical process governed by initial conditions. If an observer could measure the exact force, air resistance, and friction, the outcome would be predictable. In the computational domain, this is embodied by **Pseudo-Random Number Generators** (PRNGs), which use deterministic algorithms to produce sequences that satisfy statistical tests for randomness but are ultimately periodic. Here, randomness is a measure of human ignorance—an epistemic limitation—rather than a fundamental property of the system itself.

3.2.2 Ontological Randomness: The Quantum Paradigm

On the other hand, ontological randomness (or intrinsic randomness) refers to events that are fundamentally unpredictable, regardless of the amount of information an observer possesses. This form of randomness is absent in classical physics but constitutes the bedrock of the quantum domain. According to the Copenhagen interpretation of quantum mechanics, the outcome of a single measurement on a system in a state of superposition is not determined by any prior hidden variables.

As Roger Colbeck emphasizes [25], this shift represents a transition from *randomness by complexity* to *randomness by nature*. In this paradigm, the unpredictability of a quantum measurement is not a result of experimental noise or limited knowledge, but an inherent, irreducible property of the physical laws. This distinction is crucial for the development of secure technologies: while a Laplacian observer could eventually "break" a classical source of randomness by gaining more information, quantum randomness is fundamentally *private*. In this context, privacy implies that the measurement outcome is unknown to any exter-

¹In 1814, Pierre-Simon Laplace proposed a thought experiment involving an intellect capable of knowing the position and momentum of every particle in the universe. Using Newton's laws, such an entity could theoretically reconstruct the entire past and predict the future with absolute certainty, effectively eliminating the concept of chance.

nal agent (including the manufacturer of the device or an eavesdropper) until the measurement is performed, remaining unpredictable even against an adversary with infinite classical knowledge and computational power.

3.3 The Requirements of Modern Applications

As noted by Robert R. Coveyou [22], the generation of random numbers is too important to be left to chance. This statement emphasizes that for a sequence to be truly useful for scientific or technical purposes, its generation must be governed by rigorous requirements. The criteria for success vary significantly depending on the final application.

For statistical simulations, such as the Monte Carlo methods pioneered by Stanisław Ulam, the primary concern is **statistical quality**. In this context, a sequence must appear uniformly distributed—meaning that every possible value or bit has an equal probability of occurring—and remain free of any detectable patterns or internal correlations that could bias a numerical result.

However, for cryptographic purposes, statistical quality alone is insufficient. A sequence could be perfectly uniform yet completely useless if it lacks **unpredictability**. For instance, a sequence that simply counts upwards in binary (0000, 0001, 0010, 0011, ...) would eventually satisfy tests for uniformity over a long period, yet it is absolutely predictable; if an observer can determine the next bit in a stream based on past observations, the security of the protocol evaporates.

Furthermore, modern secure communications demand **private availability**. Even a sequence that is fundamentally unpredictable to a general observer provides no protection if it is already known to an adversary. This distinction between public and private randomness is crucial, as the latter is a fundamental prerequisite for achieving *unconditional security*. While the classical methods discussed in the previous sections have served science for decades, the escalating demand for absolute privacy and the increasing power of computational attacks have forced a re-evaluation of these traditional sources.

As previously discussed, **PRNGs** rely on deterministic algorithms to expand a short initial seed. While computationally efficient, they are inherently predictable: if the internal state or the seed is compromised, the entire sequence is exposed, making them vulnerable in high-stakes cryptographic tasks. Similarly, physical classical sources that exploit chaotic processes—such as thermal noise or atmospheric fluctuations—suffer from the latent determinism of classical mechanics. From a fundamental perspective, a sufficiently powerful adversary

with complete knowledge of the initial conditions and the governing physical laws (the aforementioned *Laplace's Demon*) could, in principle, predict the outcome. This limitation on absolute privacy drives the scientific community to look beyond classical physics, towards the intrinsic and ontological randomness offered by quantum systems, where unpredictability is a fundamental feature of the physical laws.

3.4 Quantum Randomness and the Device-Independent Paradigm

Quantum mechanics offers a radical paradigm shift by providing inherent, fundamental indeterminism. While classical randomness is, as previously discussed, an epistemological limitation, quantum randomness is ontological: according to the laws of physics, certain measurement outcomes are not merely "hard to predict", but are fundamentally unpredictable, even to an observer with complete knowledge of the system's preparation.

3.4.1 Beyond Computational Security

The superiority of quantum randomness, as explored by Colbeck [25], lies in its foundational robustness. In classical cryptography, security often relies on **computational assumptions**—the belief that an adversary, often personified as *Eve* or *Snoop*, lacks the computing power to solve specific mathematical problems, such as factoring large integers or computing discrete logarithms. However, a quantum generator provides security guaranteed by the laws of nature rather than the current state of algorithmic complexity.

Even if Snoop possesses a universal quantum computer (that is, a machine that processes information through coherent quantum states and can solve some problems more efficiently than classical computers), the outcome of a measurement performed on a quantum state in superposition remains probabilistic. This property allows for **Private Randomness Expansion**: a protocol where a short, initial seed of trusted randomness is used to produce a much longer string of bits. Unlike classical expansion, these new bits are certified to be private and uniform, as their entropy is sourced from the measurement process itself rather than a deterministic expansion rule.

This property allows for Private Randomness Expansion: a protocol where a short, initial seed of trusted randomness is used to produce a much longer string of

bits. Unlike classical expansion, these bits are certified to be private and uniform. To achieve this, we employ Privacy Amplification: a post-processing technique that uses universal hash functions to "distill" a shorter, secret key from a longer, partially compromised raw string. Even if Snoop has partial information about the raw bits, privacy amplification reduces the adversary's knowledge of the final key to a negligible level, ensuring it is effectively independent of any information held by external agents.

3.4.2 The Challenge of Untrusted Devices (Dr. Snoop)

A significant contribution of Colbeck's work is the formalization of the transition towards Device-Independent (DI) security. In traditional quantum protocols, the user must trust their equipment. Colbeck identifies a critical vulnerability here: a malicious manufacturer, "Dr. Snoop," could distribute devices that appear to be quantum but are actually pre-programmed classical boxes (a "Trojan" device).

To counter this, Colbeck proposes protocols where the devices are treated as black boxes. The internal workings are irrelevant; the user certifies the presence of true quantum randomness through the observation of non-local correlations (such as violations of Bell inequalities). This non-locality acts as a fundamental, empirically verifiable certificate of randomness that no classical pre-programmed system can replicate, ensuring that unpredictability is a logical consequence of the physical data rather than a reliance on the hardware provider's honesty. Furthermore, even if Dr. Snoop retains partial information about the process, Colbeck demonstrates that through techniques like privacy amplification, it is possible to extract a final string of bits that is essentially decoupled from any information held by the adversary. In this framework, unpredictability is no longer an assumption about the device's quality or the provider's honesty, but a rigorous logical consequence of the measured correlations. This ensures that the generated randomness remains private and secure, even if the very tools used to create it were designed by an enemy.

3.5 The Dual Challenge of Certification

The transition from classical to quantum randomness, as framed by the work of Colbeck, redefines the standards of security, yet it introduces a dual challenge that must be carefully addressed. Its utility depends on a rigorous two-step

verification process that distinguishes the nature of the source from the quality of its output.

3.5.1 Quantumness vs. Statistical Randomness

A fundamental distinction must be drawn between *quantum certification* and *statistical randomness*. A quantum system is essential not because it inherently produces perfectly distributed sequences, but because it offers a physical guarantee of unpredictability. While an observer, such as *Dr. Snoop*, could in principle predict a classical source by modeling its deterministic laws, the ontological nature of quantum mechanics ensures that no observer can pre-determine the outcome of a certified quantum measurement.

However, it is crucial to recognize that quantum certification is a necessary condition for privacy, but not a sufficient condition for high-quality randomness in practice. A system can be undeniably quantum and yet function as a poor random number generator.

Consider, for instance, a quantum state prepared in a highly unbalanced superposition, such as one with a 90% probability of being measured as *down* and only 10% as *up*. If a Bell test or a Macrorealism (*MR*) test is performed on this system, we may successfully certify its *quantum nature* and its departure from classical macrorealism. Yet, the resulting bitstream would be heavily biased and statistically predictable, making it unsuitable for most practical applications despite its certified quantum origin.

This highlights that while the violation of *MR* or Bell inequalities provides the *ontological* basis for privacy—ensuring that the bits are not pre-programmed—it does not automatically satisfy the *statistical* requirements of uniformity. Consequently, quantumness should be viewed as the prerequisite for generating entropy that is fundamentally decoupled from an adversary’s knowledge, whereas statistical quality must be ensured through subsequent characterization and post-processing.

3.5.2 The Necessity of Post-Processing and Quality Control

The discrepancy between quantumness and statistical quality necessitates a second level of scrutiny: the continuous monitoring of the output’s statistical properties. To evaluate the degree of randomness in a given string, information theory provides rigorous metrics. In the context of cryptographic security, the most

critical metric is the *min-entropy* [26], defined as:

$$H_{\infty}(X) = -\log_2 \left(\max_{x \in X} P(x) \right) \quad (3.1)$$

Unlike the Shannon entropy [27], which measures the average uncertainty or information content of a source, the min-entropy represents the *worst-case scenario*. In cryptography, this distinction is fundamental: we are not primarily concerned with how difficult it is to guess a bit "on average" across many trials. Instead, security depends on the difficulty an adversary faces when trying to guess the most probable outcome in a single, specific attempt.

If a source possesses high average entropy but features even a single highly probable outcome, the security of the entire protocol is compromised, as an adversary would naturally prioritize that specific outcome. The min-entropy explicitly penalizes this vulnerability by focusing solely on the maximum probability $P(x)$, thereby providing a conservative and reliable lower bound on the system's true unpredictability. Consequently, min-entropy serves as the standard input parameter for *privacy amplification* algorithms, which distill a shorter, perfectly uniform key from a longer, partially predictable raw bitstream.

3.6 Conclusion

The transition from classical to quantum randomness represents a fundamental shift in the certification of trust. Modern security can no longer rely on the assumption that an algorithm is computationally too complex to break or that a hardware provider is inherently honest. Through the *Device-Independent* paradigm, trust is transferred from the specific hardware implementation to the universal laws of physics.

In this framework, the generation of high-quality random numbers is defined by two deeply interconnected requirements:

- **Quantumness and Privacy:** The certification of a source's quantum nature is a necessary condition for privacy. This is achieved by verifying the departure from classical physics through the violation of **Macrorealism** (*MR*) conditions or non-local correlations. Such a violation provides an empirical guarantee that the outcomes are ontologically unpredictable and have not been pre-determined by any external agent, including a malicious manufacturer.
- **Statistical Uniformity:** While *MR* violations certify that the entropy

source is quantum, this is not a sufficient condition for high-quality randomness in practical applications. As previously discussed, a certified quantum process can still produce biased outputs. Therefore, statistical quality must be monitored through metrics such as *min-entropy* and subsequently refined via *privacy amplification* to ensure the final bits are uniform and decoupled from any partial information held by an adversary.

By addressing these two levels of verification, the generation of random numbers ceases to be an unquantifiable process and becomes a certified computational resource. The use of temporal correlations and the rigorous testing of *MR* allow for the production of randomness that is not only statistically robust but also fundamentally secure against any classical or quantum adversary. This dual approach ensures that randomness is treated with the scientific rigor required by modern cryptographic and computational standards.

Chapter 4

Predictability and Randomness

"I have nothing to offer anybody except my own confusion."

— Jack Kerouac

In this chapter, we present the theoretical framework for certifying quantum randomness through a single-system architecture. By "single-system", we refer to a protocol that utilizes a single degree of freedom (such as a single qubit or a photonic mode) evolving in time, rather than relying on entangled pairs distributed across spatially separated subsystems. While the results are mathematically applicable to multi-qubit systems, the significance of this approach lies in its experimental simplicity: it enables randomness certification without the need for entanglement or the enforcement of space-like separation, which are strict requirements for Bell-based protocols.

This methodology, primarily based on the work by Nath *et al.* [28], builds upon the fundamental relationship between the violation of Leggett-Garg inequalities (LGI) (see Eq. (2.23)), the operational assumption of No-Signaling-in-Time (NSIT) (see Eq. (2.42)), and the subsequent logical negation of predictability [29, 30]. Unlike the Bell-based schemes discussed in Chapter 2, which are founded on spatial non-locality, this temporal approach focuses on the non-classicality of correlations across different time instants within a single evolving system [28]. The emphasis on a single-system architecture is crucial from an empirical perspective, as it provides a more accessible and resource-efficient path toward semi-device-independent randomness generation.

4.1 Theoretical Framework: Predictability and NSIT

The conceptual foundation of this work rests on a reinterpretation of the Leggett-Garg Inequality (specifically the three-time formulation LGI_{123}). While traditionally derived from the principles of MR_{ps} and NIM, as discussed in Chapter 2, Mal *et al.* [30] demonstrated that the LGI can be derived from a different set of operationally testable assumptions: *Predictability* and *No-Signaling-in-Time* (NSIT). Indeed, they showed that if a system obeys both predictability and NSIT, then the temporal correlations must satisfy $LGI \leq 1$ [30, 28], providing a rigorous logical basis for certifying randomness when this bound is exceeded.

To establish this framework, let us first define the concept of Predictability:

Definition 4.1.1 (Predictability). *A phenomenon is considered predictable if the observed outcomes of a measurement are uniquely determined by the preparation procedure. In this context, a preparation procedure k encompasses all controllable experimental parameters prior to measurement¹. Formally, for any measurement Q_i, Q_j with outcomes $a_i, a_j \in \{+1, -1\}$, predictability implies that the joint probabilities are deterministic [29, 30]:*

$$p(a_i, a_j | Q_i, Q_j, k) \in \{0, 1\}. \quad (4.1)$$

Under this assumption, given the preparation k , the outcome of any measurement is not merely statistically distributed but is fixed to a specific value.

In Appendix B, we provide concrete examples of how to evaluate the predictability condition in different experimental scenarios, demonstrating cases of maximum unpredictability, perfect predictability under identity evolution, and perfect predictability under a logical NOT operation.

In addition to predictability, we must formally define the three-time No-Signaling-in-Time (NSIT) conditions. While a general definition was provided in Chapter 2 (see Eq. (2.52)), the three-time scenario ($t_1 < t_2 < t_3$) relevant to

¹Operationally, k can be viewed as the set of instructions that specifies the initial state and the subsequent evolution before the measurement occurs.

this work requires the simultaneous satisfaction of the following conditions [28]:

$$p(a_2|Q_2) = \sum_{a_1} p(a_1, a_2|Q_1, Q_2) \quad (4.2)$$

$$p(a_3|Q_3) = \sum_{a_1} p(a_1, a_3|Q_1, Q_3) \quad (4.3)$$

$$p(a_3|Q_3) = \sum_{a_2} p(a_2, a_3|Q_2, Q_3) \quad (4.4)$$

The first condition, $NSIT_{(1)2}$, ensures that the statistics at t_2 are independent of a prior measurement at t_1 . Conditions (4.3) and (4.4) ($NSIT_{(1)3}$ and $NSIT_{(2)3}$ respectively) further stipulate that the marginal probability at t_3 remains invariant regardless of whether an intervention occurred at t_1 or t_2 . It follows from these identities that the right-hand sides of Eq. (4.3) and (4.4) must be equal, ensuring that the measurement history does not disturb the final observed statistics.

4.1.1 Derivation of LGI from NSIT and Predictability

The derivation of the LGI bound from these operational assumptions follows a precise logical sequence proposed by Mal *et al.* [30].

First, the assumption of **Predictability** implies that for a given preparation k , the outcome of any measurement Q_i is functionally determined. As defined by Cavalcanti and Wiseman [29], this allows us to assign a definite value $v_i \in \{+1, -1\}$ to each observable Q_i , such that $p(a_i|Q_i, k) = \delta_{a_i, v_i}$. In this framework, v_i is not necessarily a pre-existing property of the system (as in Macrorealism), but rather a deterministic consequence of the preparation procedure k . If an observer knows k , they can predict v_i with certainty.

Second, the **NSIT** conditions (Eq. (4.2)-(4.4)) act as a consistency constraint. They ensure that the predetermined value v_i assigned to the observable Q_i remains invariant regardless of the measurement context or the system's history [28]. For instance, $NSIT_{(1)2}$ implies that the value v_2 is consistent whether Q_2 is measured alone or following a measurement of Q_1 . Without NSIT, the value of v_2 could fluctuate depending on the context, preventing the use of a single stable variable for different correlation terms.

Under the joint satisfaction of these assumptions, for any single experimental run, we can treat the observables as a triplet of stable values $\{v_1, v_2, v_3\}$. This justifies the application of the classical algebraic identity:

$$v_1 v_2 + v_2 v_3 - v_1 v_3 \leq 1. \quad (4.5)$$

This inequality is a strict property of numbers ± 1 . By taking the expectation value over many experimental runs (averaging over the probability distribution $p(v_1, v_2, v_3)$), we obtain:

$$\langle Q_1 Q_2 \rangle + \langle Q_2 Q_3 \rangle - \langle Q_1 Q_3 \rangle = \sum_{v_1, v_2, v_3} p(v_1, v_2, v_3) (v_1 v_2 + v_2 v_3 - v_1 v_3). \quad (4.6)$$

Since the term in the parentheses is bounded by 1 for all v_i , and given the normalization condition $\sum p(v_1, v_2, v_3) = 1$, it follows that:

$$LGI_{123} = \langle Q_1 Q_2 \rangle + \langle Q_2 Q_3 \rangle - \langle Q_1 Q_3 \rangle \leq 1. \quad (4.7)$$

This demonstrates that the joint satisfaction of NSIT and Predictability strictly implies the Leggett-Garg Inequality: $(NSIT \wedge Predictability) \implies LGI_{123} \leq 1$. Consequently, any observed violation $LGI_{123} > 1$ must stem from the failure of at least one of these operational premises: either the system's outcomes are not predictable from the preparation, or the measurement process violates no-signaling-in-time [30, 28].

4.1.2 Randomness Certification Logic

The significance of the derivation presented in Section 4.1.1 lies in its contrapositive. Let $\neg$ denote the logical negation operator; the implication $(NSIT \wedge Predictability) \implies LGI_{123} \leq 1$ can be rewritten as:

$$LGI_{123} > 1 \implies \neg NSIT \vee \neg Predictability. \quad (4.8)$$

In a characterizable experimental setup where *NSIT* is verified to hold—meaning that the experimental deviations from the conditions in Eq. (4.2)-(4.4) are null within statistical error—an observed violation of the LGI ($LGI_{123} > 1$) leads to the inescapable conclusion:

$$(LGI_{123} > 1) \wedge (NSIT) \implies \neg Predictability. \quad (4.9)$$

The rejection of predictability is operationally equivalent to the certification of genuine randomness [30, 28]. If the measurement outcomes cannot be predicted even with full knowledge of the preparation procedure and the measurement settings, they must be regarded as intrinsically stochastic. As argued by Cavalcanti and Wiseman [29], this form of unpredictability is a fundamental consequence of the violation of temporal inequalities, regardless of the underlying ontological

nature of the system.

Our objective in the following sections is to utilize this logical bridge between temporal non-classicality and indeterminism to quantify the amount of randomness—measured by the min-entropy—that can be extracted when the LGIs are violated. This approach ensures that the certification is semi-device-independent: it does not rely on a detailed description of the system’s internal dynamics, but rather on the observed temporal correlations and the verified absence of signaling-in-time.

4.2 System Characterization and Optimization

In order to quantify the amount of certified randomness, we characterize the single-system setup as a qubit (two-level system) that undergoes unitary evolution U between measurement stages (see Section 1.5 for a detailed discussion on the unitary dynamics). We denote the dichotomic observables at times $t_1 < t_2 < t_3$ as Q_1, Q_2, Q_3 , where each measurement yields an outcome $a_i \in \{+1, -1\}$.

The measurements at t_1 and t_2 are performed using selective filters or "stoppers" (as described in the experimental photonic architecture of [28]), which are well-modeled by ideal projective measurements $P_{\pm} = |\pm\rangle\langle\pm|$. However, to maintain a semi-device-independent approach, we do not assume an ideal projective nature for the final measurement at t_3 . Since the final detection stage may involve complex internal dynamics or environmental noise, the measurement at t_3 is more generally described by a Positive Operator-Valued Measure (POVM) [3].

A POVM is defined as a set of positive semi-definite operators $\{M_m\}$ (i.e., $M_m \geq 0$) that satisfy the completeness relation $\sum_m M_m = \mathbb{I}$. In our case, the dichotomic POVM $\{M_+, M_-\}$ at t_3 can be parameterized in the Bloch representation as [28]:

$$M_{\pm} = \frac{1}{2} \left((1 \pm a) \mathbb{I} \pm \vec{b} \cdot \vec{\sigma} \right), \quad (4.10)$$

where $a \in \mathbb{R}$ represents a bias (which may account for systematic offsets or noise) and $\vec{b} \in \mathbb{R}^3$ represents the measurement direction and effective strength. The requirement that $M_{\pm}$ constitute a valid POVM (specifically, that they are positive semi-definite) implies the constraint $|a| + |\vec{b}| \leq 1$.

4.2.1 Quantifying Randomness via Min-Entropy

The fundamental objective of this framework is to quantify the portion of observed unpredictability that is inherently quantum and, therefore, inaccessible

to any external adversary (often referred to as Eve) [30, 28]. To achieve this in a statistically robust manner, we employ the *min-entropy* as our measure of randomness. The min-entropy is the most conservative measure in the Renyi family and is directly related to the maximum probability that an adversary can correctly predict the measurement outcomes.

For a pair of measurement outcomes (a_i, a_j) obtained at times (t_i, t_j) , the min-entropy conditioned on the measurement settings (Q_i, Q_j) is defined as:

$$H_\infty(a_i, a_j|Q_i, Q_j) = -\log_2(p^*(a_i, a_j|Q_i, Q_j)), \quad (4.11)$$

where p^* represents the maximum guessing probability, defined as the maximum value in the joint probability distribution:

$$p^*(a_i, a_j|Q_i, Q_j) = \max_{a_i, a_j} p(a_i, a_j|Q_i, Q_j). \quad (4.12)$$

To certify randomness in a semi-device-independent way, we seek the *minimum* possible value of H_∞ (which corresponds to the worst-case scenario of a maximum p^*) that is compatible with the observed level of LGI violation. This optimization is performed under the strict constraint that the NSIT conditions (as defined in Section 4.1.1) are satisfied, ensuring that the observed violation is not due to signaling or experimental loopholes but stems from the intrinsic indeterminism of the system.

4.2.2 The Optimization Problem

In the work of Nath *et al.* [28], the certification of randomness is achieved by solving a constrained optimization problem, where the objective is to maximize the guessing probability p^* under the physical constraints imposed by the experimental observations. Specifically, we denote the measured value of the Leggett-Garg parameter as $LGI_{123} = 1 + \alpha$, where $\alpha \in (0, 0.5]$ ² represents the amount of violation beyond the classical limit.

The system is initially prepared in a general qubit state ρ , represented in the Bloch vector formalism as:

$$\rho = \frac{1}{2}(\mathbb{I} + \vec{n} \cdot \vec{\sigma}), \quad \vec{n} = (n_x, n_y, n_z) \in \mathbb{R}^3, \quad (4.13)$$

²The upper bound $\alpha = 0.5$ (corresponding to $LGI_{123} = 1.5$) is not arbitrary but represents the maximum violation attainable by a quantum mechanical two-level system, a limit known as the *temporal Tsirelson bound* [31, 32]. For a standard qubit evolution where the correlation functions scale as $C_{ij} = \cos(\theta_{ij})$, the parameter LGI_{123} is maximized when the relative measurement angles are set to $\pi/3$, yielding $2\cos(\pi/3) - \cos(2\pi/3) = 1.5$.

with the purity constraint $|\vec{n}|^2 \leq 1$. The dynamics between measurement stages are described by the most general unitary transformation for a single qubit (up to a global phase):

$$U_i = \begin{bmatrix} e^{ix_i} \cos(z_i) & e^{iy_i} \sin(z_i) \\ -e^{-iy_i} \sin(z_i) & e^{-ix_i} \cos(z_i) \end{bmatrix} \quad \text{for } i = 1, 2, \quad (4.14)$$

where $x_i, y_i, z_i \in \mathbb{R}$ are the parameters defining the rotation and phase shifts.

The optimization problem seeks to determine the maximum joint probability p^* , which defines the worst-case scenario for randomness extraction. As we shall detail, the problem is subject to the observed LGI violation and the strict satisfaction of the NSIT conditions:

$$\begin{aligned} p^* &= \max_{a_i, a_j} p(a_i, a_j | Q_i, Q_j) \\ &\text{subject to:} \\ \langle Q_1 Q_2 \rangle + \langle Q_2 Q_3 \rangle - \langle Q_1 Q_3 \rangle &= 1 + \alpha \\ p(a_2 | Q_2) &= \sum_{a_1} p(a_1, a_2 | Q_1, Q_2) \\ p(a_3 | Q_3) &= \sum_{a_1} p(a_1, a_3 | Q_1, Q_3) \\ p(a_3 | Q_3) &= \sum_{a_2} p(a_2, a_3 | Q_2, Q_3) \end{aligned} \quad (4.15)$$

By solving this optimization³, one obtains a rigorous analytical expression for the maximum guessing probability, which in turn defines the certified min-entropy as a function of the violation α .

Proposition 4.2.1 (Analytical Bound on Certified Randomness). *Given a three-time LGI scenario where the NSIT conditions (4.2)-(4.4) are strictly satisfied and the LGI parameter yields a value $1 + \alpha$, the maximum guessing probability p^* for any pair of measurements is uniquely determined by [28]:*

$$p^* = \frac{1}{4}(1 + \alpha + \sqrt{1 - 2\alpha}). \quad (4.16)$$

Consequently, the certified amount of random bits (min-entropy) generated per measurement pair is:

$$H_\infty(\alpha) = -\log_2 \left(\frac{1}{4}(1 + \alpha + \sqrt{1 - 2\alpha}) \right). \quad (4.17)$$

³For a comprehensive derivation and the analytical steps of this resolution, see [28].

Impact of NSIT on System Parameters

The requirement that $\text{NSIT}_{ij} = 0$ as in Eq. (4.2)-(4.4) and the three times LGI (2.23) is violated, in a quantum mechanical framework imposes severe constraints on the mathematical description of the system [30]. By analyzing the joint probabilities in terms of the state ρ , the unitaries U_1, U_2 , and the POVM $M_{\pm}$, it is found that to satisfy the no-signaling conditions for any LGI violation α [28]:

1. The bias and the direction of the POVM at t_3 must satisfy specific relations with the unitary evolutions to ensure the marginal at t_3 is independent of the previous measurement choices.
2. The unitary phases must satisfy the relation $x_1 + x_2 = y_1 - y_2$ to prevent signaling via the interference terms in the temporal correlations.

This leads to a significant simplification of the model:

- **Initial State:** The state must be maximally mixed, $\rho = \frac{1}{2}\mathbb{I}$, meaning that no information is encoded in the initial preparation ($n_x = n_y = n_z = 0$).
- **Measurements:** The general POVM $M_{\pm}$ at t_3 reduces to a standard projective measurement ($a = 0$ and $|\vec{b}| = 1$), identical in nature to those at t_1 and t_2 .
- **Evolution:** The unitary operators U_1 and U_2 are constrained by the violation parameter α . For simplicity, and without loss of generality for achieving the bound, we consider the symmetric case $U_1 = U_2$. These operators are effectively represented as a rotation of the state vector around the y -axis of the Bloch sphere. where the rotation angle is fixed by the relation: $\cos(2z_1) = \cos(2z_2) = \frac{1-\sqrt{1-2\alpha}}{2}$.

Analysis of the Certified Min-Entropy

The relationship between the LGI violation and the certified randomness, as expressed in Eq. (4.17), is illustrated in Figure 4.1. This analytical bound demonstrates that the certified min-entropy is a strictly increasing function of the violation parameter α . Specifically, as the temporal correlations deviate further from the classical limit ($\alpha > 0$), the maximum guessing probability for an adversary decreases, thereby increasing the amount of extractable randomness.

We have verified through independent numerical simulations that these analytical results are consistent with the findings presented in [28]. Our implementation successfully reproduced the min-entropy curve, confirming that the optimized

configuration—characterized by a maximally mixed initial state and symmetric unitary evolutions—provides a robust benchmark for randomness generation in single-system architectures.

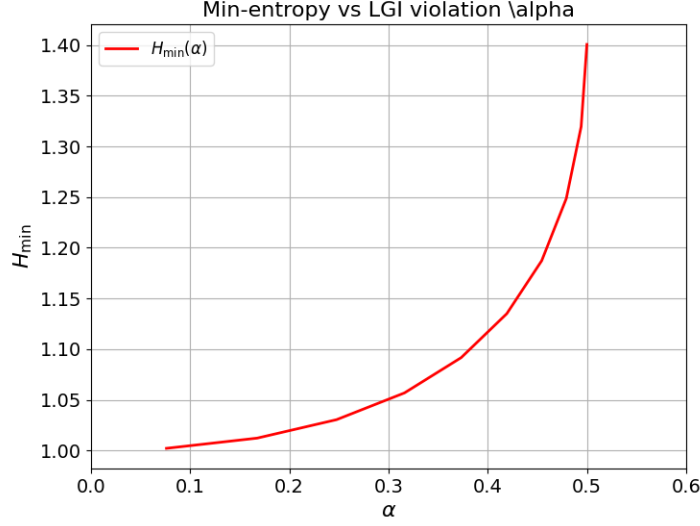


Figure 4.1: Certified min-entropy H_{∞} as a function of the LGI violation parameter $\alpha \in (0, 0.5]$. The curve represents the analytical bound derived from the optimization of the guessing probability. Higher violations of temporal non-classicality directly correspond to an increase in certified randomness, reaching a maximum of $H_{\infty} \approx 1.415$ bits at the algebraic limit of $\alpha = 0.5$.

Chapter 5

Quantitative Characterization of Macrorealism Violations and Min-Entropy Analysis

“Information is the resolution of uncertainty.”

— Claude Shannon

This chapter contains the original results of the thesis.

In this chapter, we establish the formal mathematical framework and the primary physical quantities that constitute the core of our investigation. While the foundational debate surrounding Macrorealism (MR) has historically been framed in terms of boolean conditions—where a system either satisfies or violates a specific set of postulates—the central objective of this work is to extend this binary classification toward a *quantitative characterization*.

Building upon the hierarchical framework proposed by Halliwell [33] (see Chapter 2), we go beyond the mere detection of non-classicality and instead introduce a quantitative measure for violations at each level of macrorealism: weak, intermediate, and strong. By interpreting these violations as a *distance from the set of macrorealistic conditions*, we obtain a refined characterization of the transition between the classical and quantum regimes.

This quantitative approach is essential for identifying the regions of the system’s parameter space where quantum effects are most prominent. Furthermore, it provides the necessary analytical tools to correlate the breakdown of macrorealism with the generation of certified randomness, a relationship that serves as a cornerstone for the device-independent protocols explored in the subsequent sections of this chapter.

5.1 Characterization of Macrorealism via New Quantitative Indicators

5.1.1 Distance from Weak Macrorealism

The weakest level of macrorealism, denoted as MR_{weak} , provides the first quantitative threshold for non-classicality. As established in Sec. 2.6, it is defined by the simultaneous satisfaction of several independent conditions:

$$MR_{\text{weak}} = NIM_{\text{pw}} \wedge LGI_{12} \wedge LGI_{23} \wedge LGI_{13} \wedge LGI_{123} \wedge \text{IND}, \quad (5.1)$$

As discussed in the previous sections, Eq. (5.1) defines a Boolean condition. To move from a qualitative Boolean assessment to a quantitative characterization, we introduce a set of parameters that quantify the magnitude of violation associated with each constituent condition in Eq. (5.1). In our analysis, we assume that the induction principle (IND) is a universal feature of the physical setting, and that the NIM_{pw} conditions are satisfied by the experimental protocol. Consequently, violations of MR_{weak} are entirely determined by violations of the Leggett–Garg inequalities, and therefore the distance from MR_{weak} depends solely on the violation of the LGIs.

We define four non-negative violation parameters— α, β, γ , and δ —which act as coordinates in a violation space. For the three-time LGI, as defined in Eq. (2.23) which are violated for values of $LGI_{123} > 1$, we formally define the parameter α seen in Chapter 4 as:

$$\alpha = \max(0, LGI_{123} - 1). \quad (5.2)$$

This definition ensures that $\alpha = 0$ whenever the inequality is satisfied, while it takes positive values in the event of a violation. As formally derived in the literature [28], for a standard qubit system, this parameter is bounded such that $\alpha \in [0, 0.5]$.

Analogously, for the three two-time LGIs introduced in Eq. (2.26), which are violated for values of $LGI_{1,2}, LGI_{1,3}, LGI_{2,3} < 0$, we define the violation parameters β, γ , and δ . Since these inequalities are formulated to be non-negative in a macrorealistic regime, we invert the sign to ensure that any violation results in a

positive parameter value:

$$\beta = \max(0, -LGI_{12}), \quad (5.3)$$

$$\gamma = \max(0, -LGI_{13}), \quad (5.4)$$

$$\delta = \max(0, -LGI_{23}). \quad (5.5)$$

Numerical simulations, which we performed to validate the theoretical framework, indicate that the physically attainable domains for these violation coordinates are $\beta, \gamma \in [0, \sqrt{2} - 1]$ and $\delta \in [0, 0.25]$. These results are omitted for brevity as they served primarily as a consistency check.

By mapping the system's state onto a four-dimensional parameter space $\mathcal{V} = (\alpha, \beta, \gamma, \delta)$, we can geometrically represent the "classicality" of the system. In this space, the state of perfect macrorealism—where all inequalities are satisfied—corresponds uniquely to the origin $P_{MR} = (0, 0, 0, 0)$. Any departure from the origin signifies a breakdown of the macrorealistic description. We define the *distance from weak macrorealism* as the Euclidean norm of this violation vector:

$$d_{\text{weak}} = \sqrt{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}. \quad (5.6)$$

This scalar quantity d_{weak} provides a unified measure of how strongly the system's temporal correlations deviate from the classical bounds, serving as a certificate for the presence of quantum effects at the correlation level.

5.1.2 Distance from Intermediate Macrorealism

The transition from a correlation-based test to a probability-based framework is formalized through the concept of intermediate macrorealism, MR_{int} . As established in Sec. 2.8, this level of hierarchy is defined by the conjunction of the No-Signaling-in-Time (NSIT) conditions and the three-time Leggett–Garg inequality:

$$MR_{\text{int}} = NSIT_{(1)2} \wedge NSIT_{(1)3} \wedge NSIT_{(2)3} \wedge LGI_{123} \wedge \text{Ind}. \quad (5.7)$$

The boolean $NSIT_{(i)j}$ condition provides a strict test of non-invasiveness: it is satisfied only if the measurement statistics at time t_j remain invariant regardless of whether a prior measurement was performed at time t_i . Any deviation from this identity indicates that the act of observation has "signaled" its presence into the future evolution of the system, a hallmark of quantum invasiveness.

To quantify the degree to which a system violates these conditions, we introduce the parameters η_1, η_2 , and η_3 . These variables represent the numerical discrepancy between the single-time probability distribution (the "unperturbed" state) and the marginalized joint probability distribution (the state "perturbed" by a prior measurement). Without loss of generality, we evaluate these differences for the outcome $s_j = +1$:

$$\eta_1 = P(s_j = +1 \mid Q_2) - \sum_{s_1=\pm 1} P(s_1, s_j = +1 \mid Q_1, Q_2), \quad (5.8)$$

$$\eta_2 = P(s_j = +1 \mid Q_3) - \sum_{s_1=\pm 1} P(s_1, s_j = +1 \mid Q_1, Q_3), \quad (5.9)$$

$$\eta_3 = P(s_j = +1 \mid Q_3) - \sum_{s_2=\pm 1} P(s_2, s_j = +1 \mid Q_2, Q_3). \quad (5.10)$$

As in the previous case, in a perfectly macrorealistic scenario, each η_i must vanish, reflecting the fact that the system possesses definite properties that are not influenced by the measurement process.

Analogously to the weak case, we define the *distance from intermediate macrorealism* as the Euclidean distance from the origin in a violation space that now includes both signaling and correlation terms. This distance is given by:

$$d_{\text{int}} = \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2 + \alpha^2}, \quad (5.11)$$

where $\alpha = \max(0, LGI_{123} - 1)$ is the LGI violation parameter defined in Eq. (5.2). By including α , d_{int} effectively "subsumes" the core non-classicality of the three-time correlations while adding the sensitive detection of measurement-induced disturbances provided by the NSIT parameters. In this sense, d_{int} can be regarded as a "stronger" metric than d_{weak} , as it is capable of identifying quantum regimes where the correlations might appear classical while the underlying probability distributions remain inherently non-classical.

5.1.3 Distance from Strong Macrorealism

The most stringent formulation of the hierarchy, denoted as MR_{strong} defined in Sec. 2.7, requires that the system's dynamics be compatible with a well-defined global joint probability distribution $p(s_1, s_2, s_3)$. This condition is satisfied only if the statistics remain consistent under the marginalization of any measurement, implying that the system evolves along a definite trajectory regardless of the observation sequence. As established in Sec. 2.7, this is expressed through the

conjunction of high-order No-Signaling-in-Time (NSIT) conditions:

$$MR_{\text{strong}} = NSIT_{(2)3} \wedge NSIT_{(1)23} \wedge NSIT_{1(2)3} \wedge \text{Ind}, \quad (5.12)$$

where the two-time and three-time NSIT conditions follow the formal definitions provided in Eq. (2.52), Eq. (2.53), and Eq. (2.54).

To quantify the degree of non-classicality at this level, we introduce specific violation parameters for the three-time NSIT conditions, denoted as η_4 and η_5 . These parameters represent the numerical deviation between the marginalized three-time distribution and the corresponding two-time joint distributions:

$$\eta_4(s_2, s_3) = \sum_{s_1=\pm 1} p_{123}(s_1, s_2, s_3) - p_{23}(s_2, s_3), \quad (5.13)$$

$$\eta_5(s_1, s_3) = \sum_{s_2=\pm 1} p_{123}(s_1, s_2, s_3) - p_{13}(s_1, s_3). \quad (5.14)$$

WLOG we evaluate these parameters for the outcome pair $(s_2, s_3) = (+1, +1)$ and $(s_1, s_3) = (+1, +1)$, respectively.

The calculation of the three-time joint probability distribution $p(s_1, s_2, s_3)$ within a quantum framework must account for the sequential nature of the measurements and the associated state collapse. It is computed via the trace of the evolved density matrix as:

$$p(s_1, s_2, s_3) = \text{Tr} \left(P_{s_3} U_2 P_{s_2} U_1 P_{s_1} U_0 \rho_0 U_0^\dagger P_{s_1} U_1^\dagger P_{s_2} U_2^\dagger P_{s_3} \right), \quad (5.15)$$

where P_{s_i} are the projection operators onto the σ_z eigenstates at time t_i , ρ_0 represents the initial density matrix of the system, and U_i denotes the unitary evolution operators acting between consecutive measurement events.

Following our established geometric methodology, we define the *distance from strong macrorealism* as the Euclidean norm of the violation vector in the relevant parameter subspace:

$$d_{\text{strong}} = \sqrt{\eta_3^2 + \eta_4^2 + \eta_5^2}, \quad (5.16)$$

where η_3 is the two-time violation parameter previously defined in Eq. (5.10). A non-zero value of d_{strong} implies a failure of the global consistency required for a macrorealistic description, indicating that the system's temporal evolution is inherently non-classical.

By employing d_{strong} , we can detect subtle quantum effects that might be invisible to both d_{weak} and d_{int} , such as cases where the measurement at an intermediate time t_2 disturbs the non-local correlations between the past and the future

of the system, even if the individual single-time statistics remain unchanged.

5.2 Min-Entropy Formulations

In this section, we refine the application of min-entropy to temporal measurements, establishing the statistical framework necessary for evaluating the quality of generated randomness. While standard entropic measures, such as the Shannon entropy [27], characterize the average uncertainty of a distribution, our approach focuses on the *min-entropy*, which quantifies the degree of randomness of the most likely outcome. This is particularly relevant in cryptographic contexts, where security is determined by the "worst-case" scenario—the probability that an adversary could correctly guess the measurement sequence.

From a cryptographic perspective, the min-entropy H_∞ serves as the ultimate measure of security against an adversary attempting to predict the measurement outcomes. While a raw sequence may consist of N bits, they are not necessarily all "secure" or perfectly random; any predictability in the system provides a potential advantage to Dr. Snoop. The min-entropy quantifies the "guaranteed randomness" that remains even in the worst-case scenario: the probability of the most likely outcome.

To illustrate this, consider a generic source that produces a raw bitstring of 100 bits, produced in couples, where the maximum randomness is obtained when the four possible outcomes $(s_i, s_j) = \{(+1, +1), (+1, -1), (-1, +1), (-1, -1)\}$ are equally likely, each with a probability of 0.25 and so the min entropy is $H_\infty = 2$. If the calculated min-entropy of this source is $H_\infty = 1.4$ bits per event [28], it means that each event does not contribute a full 2 bits of uncertainty (as would happen in a perfectly random four-outcome scenario), but only 1.4. Consequently, out of the 100 bits generated, only 70 bits (i.e., $100 \times \frac{1.4}{2}$) can be considered truly independent and unpredictable.

In a randomness extraction protocol, this implies that we can safely "expose" or utilize a final string of at most 70 bits. If we were to use all 100 bits, the remaining 30 bits would represent a security leak, as they contain correlations or biases that an adversary with knowledge of the system's dynamics could exploit to guess the sequence with a probability significantly higher than 2^{-H_∞} . Thus, H_∞ dictates the maximum length of the cryptographic key that can be securely distilled from the physical process.

By evaluating the maximum probability of measurement outcomes at different temporal scales, we can quantify uncertainty and predictability from the single-

event level to the full dynamical evolution. We begin by recalling the two-time min-entropy, as defined in [28]:

$$H_{\infty}^{(2)} = -\log_2 \left(\max_{s_i, s_j} p(s_i, s_j \mid Q_i, Q_j) \right). \quad (5.17)$$

This quantity characterizes the predictability of pairs of measurements, providing a bound on the information shared between two distinct points in time t_i and t_j . In a system with dichotomic outcomes, $H_{\infty}^{(2)}$ ranges from 0 (perfect predictability) to 2 (maximum uncertainty for four possible outcomes).

Building upon this framework, we extend the definition to two additional scenarios that allow for a more granular (or, conversely, more comprehensive) analysis of the system's stochasticity. First, we define the *single-time min-entropy* as:

$$H_{\infty}^{(1)} = -\log_2 \left(\max_{s_i} p(s_i \mid Q_i) \right). \quad (5.18)$$

This measure captures the maximum uncertainty associated with a measurement at a specific time step t_i , independent of the system's previous or subsequent history. It serves as a local probe of the randomness produced at each individual stage of the experimental protocol.

Finally, to account for the full temporal evolution across the entire experimental sequence, we introduce the *three-time min-entropy*:

$$H_{\infty}^{(3)} = -\log_2 \left(\max_{s_1, s_2, s_3} p(s_1, s_2, s_3 \mid Q_1, Q_2, Q_3) \right), \quad (5.19)$$

where $p(s_1, s_2, s_3)$ represents the joint probability of observing a specific sequence of outcomes at t_1, t_2 , and t_3 , calculated as in Eq. (5.15). The three-time min-entropy represents the total entropy available in a single "run" of the three-measurement sequence.

The interplay between these measures— $H_{\infty}^{(1)}$, $H_{\infty}^{(2)}$, and $H_{\infty}^{(3)}$ —allows us to distinguish between randomness that is purely local and randomness that is distributed across temporal correlations. In the following sections, these quantities will be compared against the distances from macrorealism to certify that the observed uncertainty is of genuine quantum origin.

5.3 Numerical Results and Discussion

In this section, we present the numerical results obtained by simulating the dynamics of the quantum systems described in the previous chapters. The aim is to

quantify the departure from macrorealism through the distances d_{weak} , d_{int} , and d_{strong} , and to correlate these findings with the certified randomness expressed by the min-entropy $H_{\infty}^{(1)}$, $H_{\infty}^{(2)}$ and $H_{\infty}^{(3)}$.

5.3.1 Numerical Methodology

The results presented in the following sections have been obtained through a numerical framework developed in Python, utilizing the *NumPy* [34] and *SciPy* [35] libraries for linear algebra and matrix exponentiation.

To evaluate the metrics, the simulation samples the dimensionless parameter space $\omega\tau \in [0, 2\pi]$. Here, we consider the specific case of a two-level system governed by the Hamiltonian $H = \frac{\hbar\omega}{2}\sigma_n$, where $\sigma_n = \vec{\sigma} \cdot \hat{n}$ is the projection of the Pauli matrices along the rotation axis $\hat{n}$. Under these specifications, the simulation reconstructs the multi-time statistics through the following algorithmic protocol:

1. **System Initialization:** The system is prepared in a selected initial density matrix ρ_0 . Different initial configurations (e.g., pure states, maximally mixed states, and coherent superpositions) are tested.
2. **First Measurement and State Collapse:** A projective measurement is performed on the initial state along the σ_z basis using the projectors $\Pi_{s_1} = |s_1\rangle\langle s_1|$ (with $s_1 = \pm 1$). The probability of obtaining the outcome s_1 at time t_1 is $p(s_1) = \text{Tr}(\Pi_{s_1}\rho_0\Pi_{s_1})$, and the post-measurement state collapses to:

$$\rho_1(s_1) = \frac{\Pi_{s_1}\rho_0\Pi_{s_1}}{\text{Tr}(\Pi_{s_1}\rho_0\Pi_{s_1})}. \quad (5.20)$$

3. **Unitary Evolution:** The collapsed state $\rho_1(s_1)$ is evolved forward in time over an interval τ . Given our choice of Hamiltonian, the unitary evolution operator depends explicitly on the dimensionless parameter $\omega\tau$ and reads:

$$U(\omega\tau) = \exp\left(-i\frac{H\tau}{\hbar}\right) = \exp\left(-i\frac{\omega\tau}{2}\sigma_n\right), \quad (5.21)$$

yielding the evolved state $\rho'_1 = U(\omega\tau)\rho_1(s_1)U^\dagger(\omega\tau)$.

4. **Sequential Measurement and Joint Probabilities:** A second projective measurement Π_{s_2} is performed at time t_2 . The conditional probability of obtaining s_2 given s_1 is computed, allowing the reconstruction of the

multi-time joint probability distribution via the post-selection rule:

$$p(s_1, s_2) = \text{Tr} \left(\Pi_{s_2} U(\omega\tau) \Pi_{s_1} \rho_0 \Pi_{s_1} U^\dagger(\omega\tau) \Pi_{s_2} \right). \quad (5.22)$$

This procedure is generalized iteratively for any subsequent measurement steps $t_3, \dots, t_n$.

5. **Metric Extraction:** From the complete set of joint probabilities $p(s_i, s_j)$, the code evaluates the macrorealistic observables. Specifically, the Leggett-Garg correlators are computed as:

$$C_{ij} = \sum_{s_i, s_j = \pm 1} s_i s_j p(s_i, s_j). \quad (5.23)$$

These correlators, alongside the joint probabilities, are then used to calculate the Leggett-Garg inequalities, the No-Signaling-in-Time (NSIT) violations and the Euclidean distances from the macrorealistic hierarchies. Concurrently, the min-entropy is extracted as $H_\infty = -\log_2 (\max_{s_1, s_2, \dots} p(s_1, s_2, \dots))$.

This step-by-step numerical reconstruction allows for a precise mapping of the non-classical regions of the Bloch sphere, directly linking the geometric trajectory of the state to the violation of macrorealism.

5.3.2 Application to the Maximally Mixed State under σ_y Evolution

In this section, we apply the previously defined metrics to the quantum system introduced in Chapter 4, in Appendix D we provide a simulation based on bits analysis instead of analytical methods. To perform a consistent numerical analysis, we first establish the physical framework and the specific parameterization used for the simulations.

We consider the initial state of the system as described by the density matrix:

$$\rho_0 = |\Psi\rangle\langle\Psi| = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (5.24)$$

The dynamical evolution is governed by the unitary operator $U(\omega\tau)$ defined in Eq. (5.25), where ω represents the characteristic frequency of the system and τ denotes the time interval between consecutive measurement stages.

To simplify the analysis and focus on the symmetry of the temporal correlations, we consider a specific evolution which, in the formalism introduced in

Sec. 4.2.2, corresponds to

$$U = \begin{bmatrix} \cos(\omega\tau/2) & \sin(\omega\tau/2) \\ -\sin(\omega\tau/2) & \cos(\omega\tau/2) \end{bmatrix}, \quad (5.25)$$

equivalent to setting $z_1 = z_2 = \frac{\omega\tau}{2}$ in Eq. (4.14). This choice allows us to explore the violation of macrorealism and the consequent generation of randomness as a function of the dimensionless parameter $\omega\tau$, which effectively scales the measurement time relative to the system's internal dynamics.

5.3.3 Analysis of the Weak Distance d_{weak}

We first examine the results concerning the distance from weak macrorealism, d_{weak} , as defined in Eq. (5.6), while its full analytical expression can be found in Appendix A (see Eq. (A.4)). As illustrated in Fig. 5.1, d_{weak} is non-zero only within specific intervals of the evolution parameter, namely $\omega\tau \in [0, \pi/2]$ and $[3/2\pi, 2\pi]$. In these regions, the non-classicality of the system is captured exclusively by the violation of the three-time Leggett-Garg inequality (LGI), as expressed in Eq. (2.23). Outside of these ranges, the system remains within the weak macrorealistic bounds, suggesting that this metric provides a relatively restrictive condition for certifying quantumness.

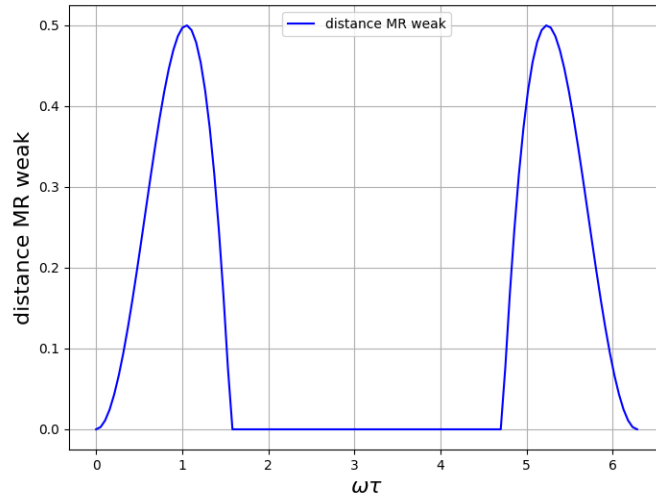


Figure 5.1: Distance from weak macrorealism d_{weak} for the first system, assuming the initial state ρ_0 from Eq. (5.24) and the unitary evolution from Eq. (5.25).

5.3.4 Analysis of the Intermediate Distance d_{int}

The results for the distance from intermediate macrorealism, d_{int} (Eq. (5.11)), are presented in Fig. 5.2, while its full analytical expression can be found in Appendix A (see Eq. (A.5)). Interestingly, the profile of d_{int} matches that of the weak distance shown in Fig. 5.1. This behavior stems from the fact that by choosing the dynamical parameters $z_1 = z_2 = \frac{\omega\tau}{2}$, $x_1 = x_2 = y_1 = y_2 = 0$ in Eq. (4.14) (see Sec. 4.2.2) — which leads to the unitary transformation in Eq. (5.25) — the No-Signaling-in-Time (NSIT) conditions in Eq. (5.11) are identically satisfied for this specific system configuration. Consequently, the only non-vanishing contribution to the violation distance arises from the three-time LGI parameter α [28], rendering the weak and intermediate criteria equivalent in this scenario.

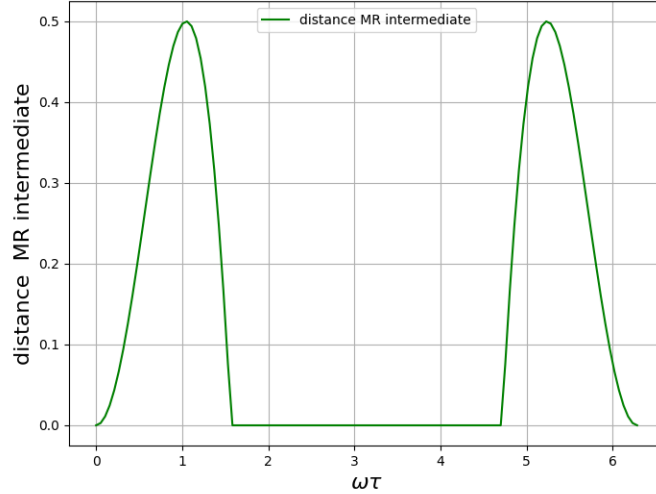


Figure 5.2: Distance from intermediate macrorealism d_{int} for the first system, assuming the initial state ρ_0 from Eq. (5.24) and the unitary evolution from Eq. (5.25). The numerical results confirm the overlap with the weak distance due to the satisfaction of NSIT conditions.

5.3.5 Analysis of the Strong Distance d_{strong}

Finally, we report the results for the distance from strong macrorealism, d_{strong} , defined in Eq. (5.16), while its full analytical expression can be found in Appendix A (see Eq. (A.6)). As shown in Fig. 5.3, d_{strong} exhibits a significantly broader violation range compared to the weak and intermediate cases. Specifically, the distance is non-zero for the entire interval $\omega\tau \in [0, 2\pi]$, vanishing only at the trivial points $\omega\tau = 0, \pi, 2\pi$, where the evolution does not generate temporal correlations.

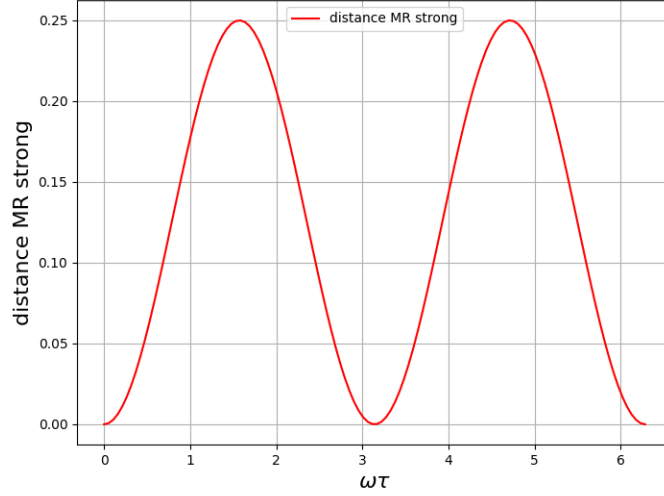


Figure 5.3: Distance from strong macrorealism d_{strong} for the first system, assuming the initial state ρ_0 from Eq. (5.24) and the unitary evolution from Eq. (5.25). Note the significantly wider range of certified non-classicality.

The most remarkable result emerging from Fig. 5.3 is the ability of the strong distance to certify the quantum nature of the system in the region $\omega\tau \in [\pi/2, 3\pi/2]$, where both d_{weak} and d_{int} fail to detect any violation. This discrepancy provides strong evidence for a hierarchical structure among the macrorealistic levels: the strong distance acts as a more sensitive probe, revealing non-classical temporal correlations that remain "hidden" to coarser metrics.

The most remarkable result emerging from Fig. 5.3 is the ability of the strong distance to certify the quantum nature of the system in the region $\omega\tau \in [\pi/2, 3\pi/2]$, where both d_{weak} and d_{int} fail to detect any violation. This discrepancy provides strong evidence for a hierarchical structure among the macrorealistic levels. Specifically, while one subset of No-Signaling-in-Time conditions Eq. (4.2), (4.3) (2.52) are identically satisfied due to the specific choice of the initial state and dynamics, other independent NSIT conditions—such as the one defined in Eq. (2.53) and Eq. (2.54)—are explicitly violated. Consequently, the strong distance acts as a more sensitive probe, capable of isolating these specific violations and revealing non-classical temporal correlations that remain "hidden" to coarser metrics.

5.3.6 Randomness Evaluation: Min-Entropy Analysis

As a subsequent step, we evaluate the genuine randomness produced by the system by analyzing the behavior of the min-entropy, in Appendix A we provide

the full analytical expression for the two-time min-entropy $H_\infty^{(2)}$ (see Eq. (A.7)). In Fig. 5.4, we report the two-time min-entropy $H_\infty^{(2)}$ (black line), which remains consistently above the classical threshold of 1, reaching a maximum value of approximately 1.4. This result is in full agreement with the theoretical predictions established in [28].

The most significant observation arises from comparing the entropic bounds with the distance metrics. Specifically, in the regions where both d_{weak} and d_{int} vanish, $H_\infty^{(2)}$ still maintains values significantly higher than 1. By invoking the strong distance d_{strong} (red line), we are now able to provide a rigorous certification of the system's non-classicality across the entire range of the evolution parameter $\omega\tau$. This confirms that the observed stochasticity is of genuine quantum origin, even in regimes where the macrorealistic tests based on d_{weak} and d_{int} fail to detect any violation.

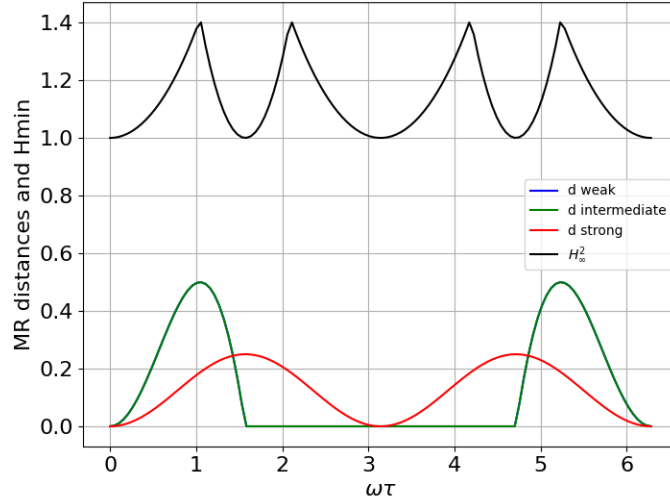


Figure 5.4: Comparison between the two-time min-entropy $H_\infty^{(2)}$ (black line) and the macrorealism distances: d_{weak} (blue), d_{int} (green), and d_{strong} (red) for the first system, assuming the initial state ρ_0 from Eq. (5.24) and the unitary evolution from Eq. (5.25). The strong distance provides a certification of quantum randomness over the full dynamical range.

To provide a comprehensive characterization of the system's entropy, we conclude by comparing the three different min-entropy formulations defined in Eqs. (5.18), (5.17), and (5.19). As shown in Fig. 5.5, the single-time min-entropy $H_\infty^{(1)}$ remains constant at 1, reflecting the maximum uncertainty of a single dichotomic measurement.

As we account for temporal correlations, the available entropy increases: the

two-time min-entropy $H_{\infty}^{(2)}$ fluctuates between 1 and 1.4, while the three-time min-entropy $H_{\infty}^{(3)}$ exhibits the highest values, reaching a peak of approximately 2.8.

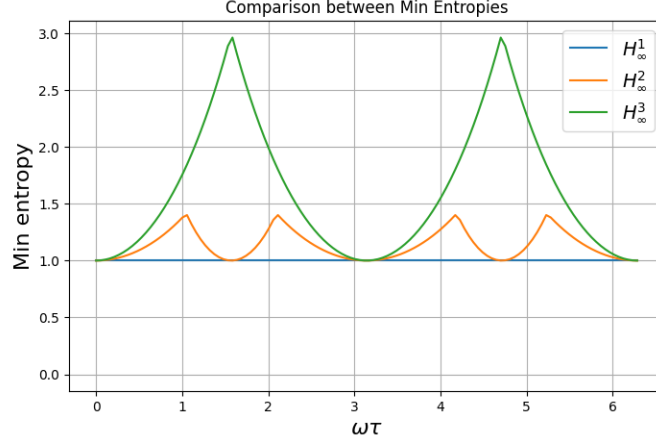


Figure 5.5: Full entropic profile for the first system, assuming the initial state ρ_0 from Eq. (5.24) and the unitary evolution from Eq. (5.25): $H_{\infty}^{(1)}$ (blue line), $H_{\infty}^{(2)}$ (orange line), and $H_{\infty}^{(3)}$ (green line). The increase in entropy with the number of measurement steps highlights the contribution of multi-time temporal correlations.

5.4 Application to the σ_y Autostate under σ_x Evolution

As a second system, we consider a two-level quantum system initially prepared in the pure state

$$|\Psi_0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle), \quad (5.26)$$

where $|\uparrow\rangle$ and $|\downarrow\rangle$ denote the eigenstates of the Pauli operator σ_z , as defined in Ch. 1.

The system's dynamics is governed by the Hamiltonian:

$$H = \frac{\omega}{2} \sigma_x, \quad (5.27)$$

and we consider projective measurements of the observable $A = \sigma_z$ performed at sequential times

$$t_0 = 0, \quad t_1 = \tau, \quad t_2 = 2\tau. \quad (5.28)$$

In the computational basis $\{|\uparrow\rangle, |\downarrow\rangle\}$ (the eigenstates of the observable A),

the Hamiltonian and the observable take the matrix representation

$$H = \frac{\hbar\omega}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

In this basis, the initial state in Eq. (5.26) can be written as

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix},$$

leading to the density matrix

$$\rho_0 = |\Psi\rangle\langle\Psi| = \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}. \quad (5.29)$$

The time-evolution operator is given by

$$U(\tau) = e^{-i(\omega/2)\sigma_x\tau},$$

which in matrix form explicitly is expressed as

$$U(\tau) = \begin{bmatrix} \cos \frac{\omega\tau}{2} & -i \sin \frac{\omega\tau}{2} \\ -i \sin \frac{\omega\tau}{2} & \cos \frac{\omega\tau}{2} \end{bmatrix}, \quad U^\dagger(\tau) = \begin{bmatrix} \cos \frac{\omega\tau}{2} & i \sin \frac{\omega\tau}{2} \\ i \sin \frac{\omega\tau}{2} & \cos \frac{\omega\tau}{2} \end{bmatrix}. \quad (5.30)$$

Measurements at times t_1 , t_2 , and t_3 are assumed to be projective in the σ_z basis, with projectors

$$P_+ = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad P_- = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}. \quad (5.31)$$

5.4.1 Analysis of the LGI violation

We begin the analysis of the second system by evaluating the violation of the three-time Leggett-Garg inequality (LGI), as expressed in Eq. (2.23), using the initial state ρ_0 from Eq. (5.29) and the unitary evolution U defined in Eq. (5.30), while its full analytical expression can be found in Appendix A (see Eq. (A.8)).

As illustrated in Fig. 5.6, the three-time LGI is violated within the intervals $\omega\tau \in [0, \pi/2]$ and $[3/2\pi, 2\pi]$. In the complementary region, the inequality is satisfied, meaning that a standard LGI test would fail to detect any non-classical behavior in that specific temporal range.

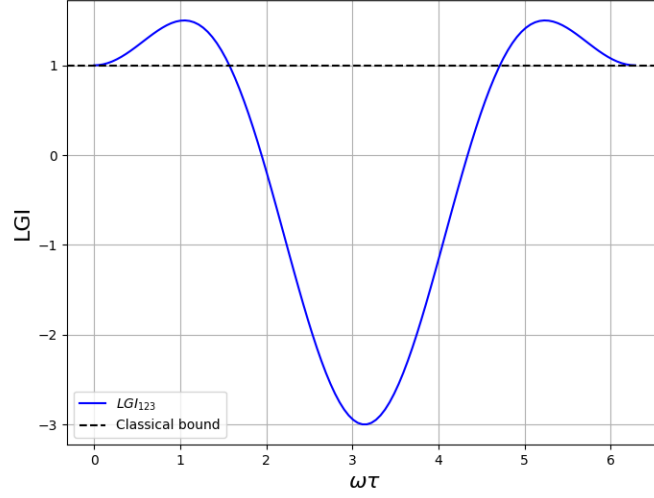


Figure 5.6: Leggett-Garg inequality violation for the second system (ρ_0 from Eq. (5.29), U from Eq. (5.30)).

5.4.2 Analysis of the Weak Distance d_{weak}

We then evaluate the distance from weak macrorealism, d_{weak} , for this second configuration. While its full analytical expression can be found in Appendix A (see Eq. (A.13)). In contrast to the first system analyzed in Sec. 5.3.2 (cf. Fig. 5.2), Fig. 5.7 shows that d_{weak} remains non-zero across almost the entire evolution range, vanishing only at a few discrete points.

This stark contrast lies precisely in the choice of the initial state. For the state defined in Eq. (5.26), the No-Signaling-in-Time (NSIT) conditions are explicitly violated during the dynamics. This ongoing violation prevents the distance from vanishing, leading to $d_{\text{weak}} > 0$ even within the region $\omega\tau \in [\pi/2, 3\pi/2]$, where the previous system configuration instead yielded a null distance. Consequently, these results highlight how the temporal correlations—and the resulting deviations from weak macrorealism—are highly sensitive to the initial state preparation.

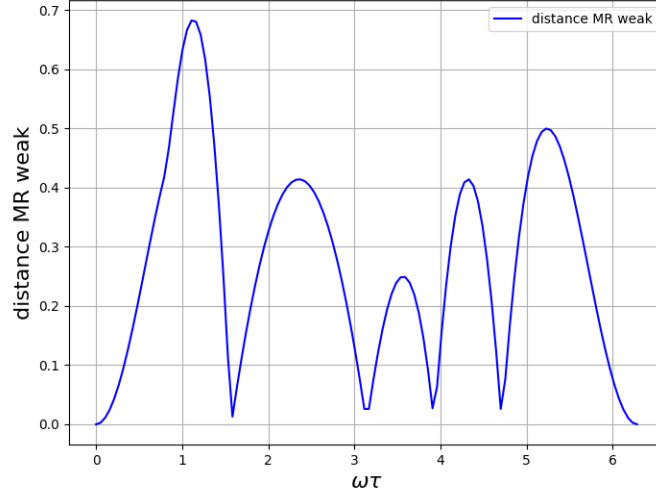


Figure 5.7: Distance from weak macrorealism d_{weak} for the second system, ρ_0 from Eq. (5.29) and U from Eq. (5.30). The non-zero values across most of the range indicate a persistent departure from the weak macrorealistic limit.

5.4.3 Analysis of the Intermediate Distance d_{int}

The analysis of the distance from intermediate macrorealism, d_{int} , is presented in Fig. 5.8, while its full analytical expression can be found in Appendix A (see Eq. (A.14)). While d_{int} is non-zero in the same regions as the weak distance, its profile exhibits significant qualitative differences. Specifically, it displays a more structured behavior with localized peaks and a different distribution of its minima. Since the intermediate distance incorporates the No-Signaling-in-Time (NSIT) conditions, these features suggest that the second system significantly violates the non-invasiveness of measurements, providing additional "geometric" contributions to the distance that were absent in the first system.

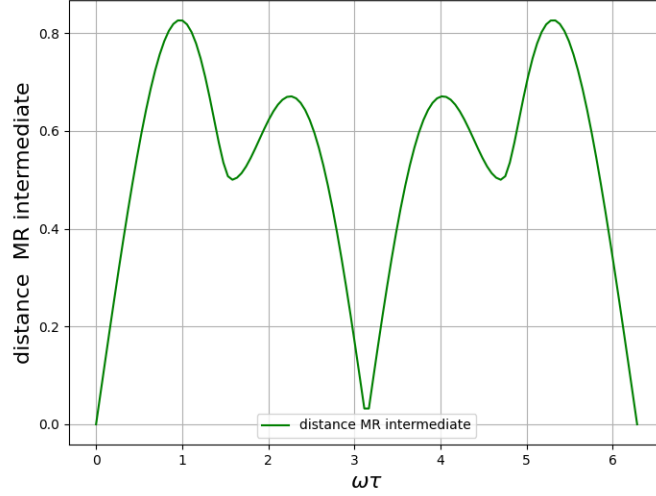


Figure 5.8: Intermediate distance d_{int} for the second system, ρ_0 from Eq. (5.29) and U from Eq. (5.30). The presence of distinct peaks highlights the contribution of NSIT violations to the overall non-classicality.

5.4.4 Analysis of the Strong Distance d_{strong}

Finally, we examine the distance from strong macrorealism, d_{strong} , as shown in Fig. 5.9, while its full analytical expression can be found in Appendix A (see Eq. (A.15)). The results confirm that d_{strong} provides the most robust certification of non-classicality, remaining non-zero for the entire interval $\omega\tau \in [0, 2\pi]$ (except for trivial evolution points).

Comparing the profiles, we observe a striking correlation between the metrics: the peaks in d_{strong} coincide with the local minima (or "wells") observed in d_{int} . This complementarity suggests that the strong distance captures a more fundamental and persistent form of non-classicality, effectively integrating both the violation of temporal inequalities and the signaling effects induced by the measurements. This further reinforces the hierarchy of macrorealism levels, where d_{strong} emerges as the most sensitive tool for quantumness certification in complex unitary evolutions.

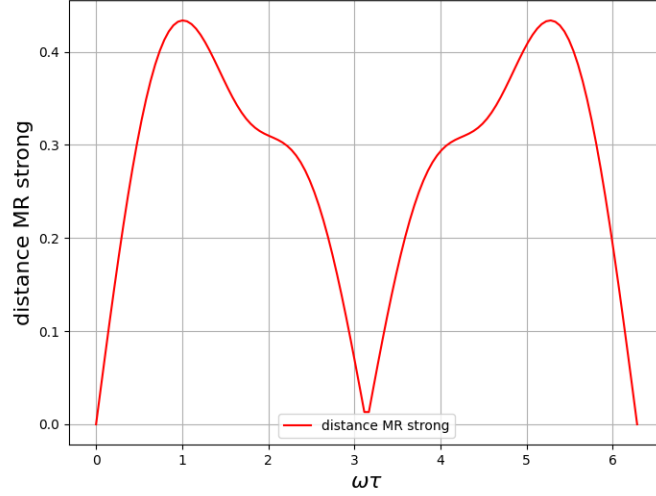


Figure 5.9: Strong distance d_{strong} for the second system, ρ_0 from Eq. (5.29) and U from Eq. (5.30). The comparison with d_{int} reveals a complementary structure in the detection of non-classical temporal correlations.

5.4.5 Min-Entropy Analysis: The Decoupling of Non-Classicality and Randomness

As a final step, we evaluate the min-entropy for the second system to assess the quality of the generated randomness. In Fig. 5.10 (for the analytical expression see Appendix A, Eq. (A.17)), we compare the two-time min-entropy $H_{\infty}^{(2)}$ (black line) with the three distance metrics previously discussed.

The results reveal a striking and somewhat counterintuitive behavior: despite the system exhibiting a persistent violation of all macrorealistic levels—as indicated by d_{weak} (blue line), d_{int} (green line), and d_{strong} (red line) being strictly non-zero—the min-entropy $H_{\infty}^{(2)}$ remains consistently below the classical threshold of 1 for a significant portion of the evolution. This suggests that the presence of non-classical temporal correlations, although sufficient to rule out macrorealism, does not necessarily translate into the production of genuine quantum randomness. In this specific configuration, the system’s dynamics lead to highly predictable outcomes, even while operating in a regime that is formally non-classical.

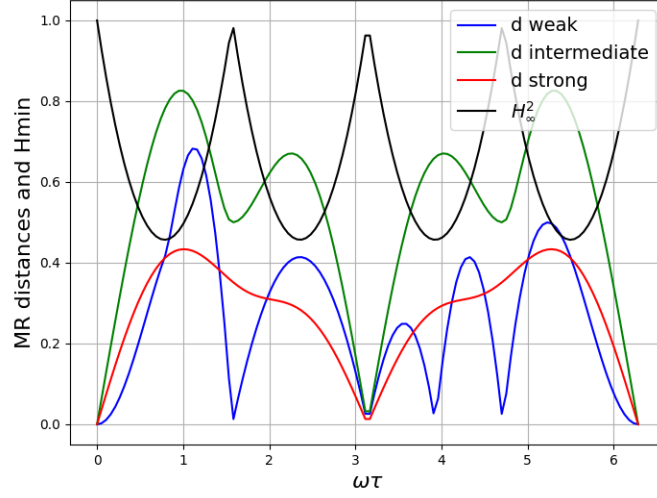


Figure 5.10: Min entropy $H_\infty^{(2)}$ (black line) compared with the distances from macrorealism: d_{weak} (blue), d_{int} (green), and d_{strong} (red) for the second system, ρ_0 from Eq. (5.29) and U from Eq. (5.30). The min-entropy remains below 1 in the regions of non-classicality, indicating a lack of genuine quantum randomness.

Consistent with the analysis of the first system, we examine the hierarchy of the three min-entropy formulations ($H_\infty^{(1)}$, $H_\infty^{(2)}$, and $H_\infty^{(3)}$) for this second case. As shown in Fig. 5.11, both the single-time $H_\infty^{(1)}$ and two-time $H_\infty^{(2)}$ min-entropies fail to exceed the value of 1 in the non-classical regions, confirming a high degree of predictability for short measurement sequences.

However, a different behavior emerges for the three-time min-entropy $H_\infty^{(3)}$ (red line), which manages to exceed the threshold and reaches a maximum of approximately 2.8. This indicates that while local or pairwise measurements do not provide sufficient uncertainty for cryptographic purposes, the full three-event temporal sequence eventually accumulates enough quantum entropy to become useful for randomness generation. This result highlights that in complex unitary evolutions, randomness certification may require higher-order temporal correlations than those captured by standard two-time metrics.

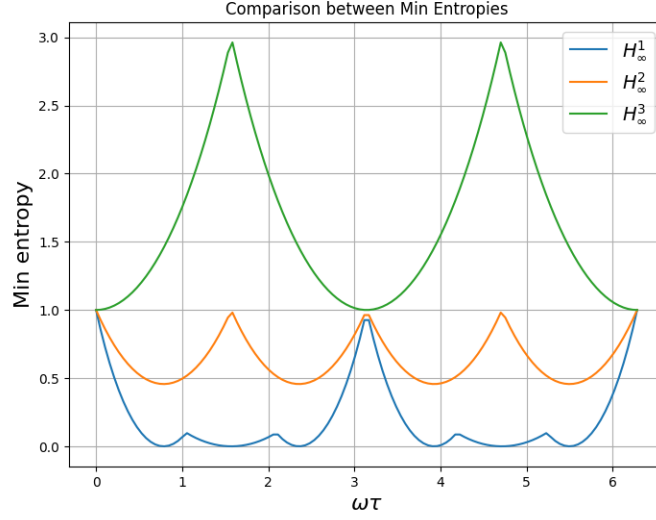


Figure 5.11: Min entropy $H_\infty^{(1)}$ (blue line), $H_\infty^{(2)}$ (green line), and $H_\infty^{(3)}$ (red line) for the second system, ρ_0 from Eq. (5.29) and U from Eq. (5.30). The min-entropy remains below 1 in the regions of non-classicality, indicating a lack of genuine quantum randomness.

5.5 How dephasing influence the system

In this section, we analyze the min-entropy and the violation of macrorealism in the presence of a dephasing channel to establish a clear boundary for these metrics when moving toward the classical regime. Generally, *dephasing* represents a decoherence process wherein the off-diagonal elements of a density matrix decay toward zero in a given reference basis, effectively suppressing quantum superpositions while preserving the diagonal populations.

To model this transition, we consider a system initialized in a state ρ_0 and subjected to an evolution interspersed with dephasing effects. Rather than limiting the analysis to a complete projection, we employ a generalized dephasing procedure at each time step, implemented as a quantum channel $\mathcal{E}(\rho)$. Under this framework, the sequential dynamics and measurement protocol alter the state as follows:

$$\begin{array}{ccccccc}
 \rho_0 & \xrightarrow{\text{dephasing}} & \rho'_0 = \mathcal{E}(\rho_0) & \xrightarrow{\text{evolution}} & U\rho'_0 U^\dagger & \xrightarrow{\text{dephasing}} & \rho_1 = \mathcal{E}(U\rho'_0 U^\dagger) \quad \dots \\
 & & \downarrow \text{measure} & & & & \downarrow \text{measure} \\
 & & M_0 & & & & M_1
 \end{array}$$

where the action of the channel on any evolved density matrix ρ in the computa-

tional basis is defined by:

$$\mathcal{E}(\rho) = \begin{bmatrix} \rho_{00} & \rho_{01}e^{-\gamma\tau} \\ \rho_{10}e^{-\gamma\tau} & \rho_{11} \end{bmatrix}. \quad (5.32)$$

Here, $\gamma \geq 0$ represents the dephasing rate and τ is the time interval. This formulation interpolates between purely coherent quantum evolution ($\gamma = 0$) and a fully classical regime ($\gamma \rightarrow \infty$), where the off-diagonal coherence is completely eliminated.

In the fully classical limit ($\gamma \rightarrow \infty$), the state at each step is strictly reduced to a classical mixture. For instance, an initial state of the form:

$$\rho_0 = \begin{bmatrix} \cos^2 \theta_0 & 0 \\ 0 & \sin^2 \theta_0 \end{bmatrix}, \quad (5.33)$$

where $\theta_0 \in (0, 2\pi)$, represents a configuration characterized by a total absence of initial off-diagonal coherence. Driven by a Hamiltonian proportional to σ_x , as defined in Eq. (5.30), any coherence generated during the unitary step is subsequently dampended or erased by the channel, establishing the baseline behavior for the macrorealistic metrics. The dephasing channel is implemented as follows [3]:

$$\rho' = \left(1 - \frac{P_{deph}}{2}\right) \rho + \frac{P_{deph}}{2} \sigma_z \rho \sigma_z, \quad (5.34)$$

where $P_{deph} \in [0, 1]$ represents the dephasing strength. When $P_{deph} = 0$, the system undergoes unitary quantum evolution. Conversely, when $P_{deph} = 1$, the system is fully dephased at every step.

5.5.1 Decoherence Results

As a preliminary check, we analyze the behavior of the system described in Eq. (5.33) as the dephasing intensity Eq. (5.34) varies.

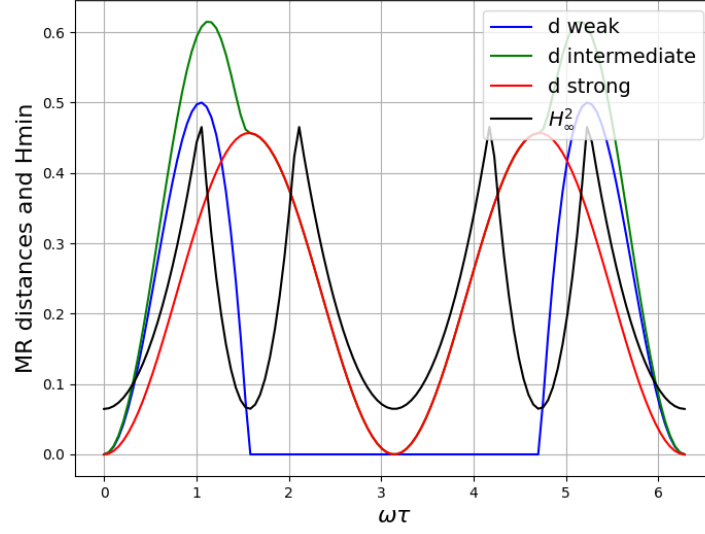


Figure 5.12: $H_{\infty}^{(2)}$ (black line) and Macrorealism distances (d_{weak} Eq. (5.6) in blue, d_{int} Eq. (5.11) in green and d_{strong} Eq. (5.16) in red) as a function of $\omega\tau$ with dephasing parameter set as $P_{\text{deph}} = 0$ (No dephasing), for the initial value of $\theta_0 = 1.36$ in Eq. (5.33).

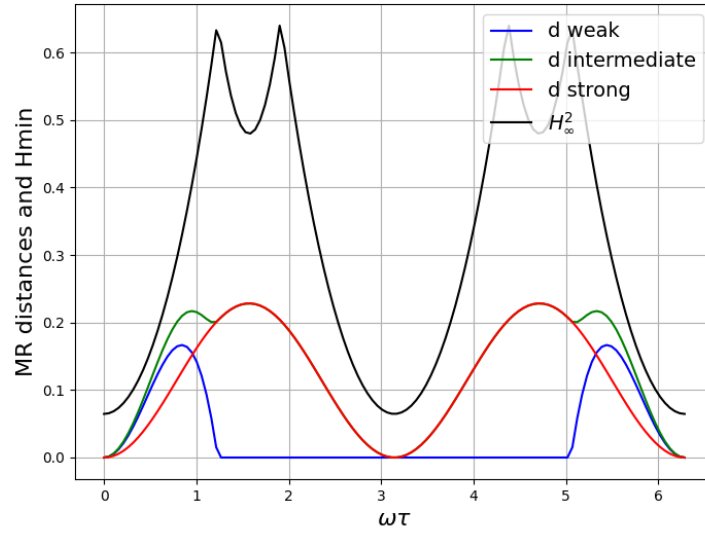


Figure 5.13: $H_{\infty}^{(2)}$ (black line) and Macrorealism distances (d_{weak} Eq. (5.6) in blue, d_{int} Eq. (5.11) in green and d_{strong} Eq. (5.16) in red) as a function of $\omega\tau$ with dephasing parameter set as $P_{\text{deph}} = 0.5$ (Partial dephasing), for the initial value of $\theta_0 = 1.36$ in Eq. (5.33).

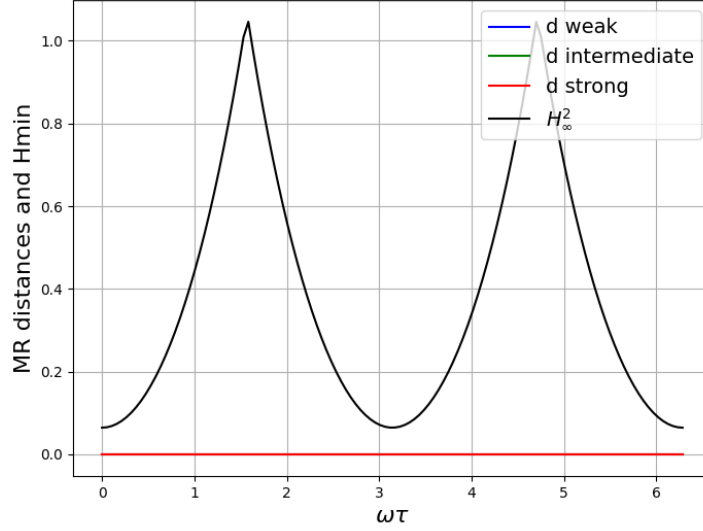


Figure 5.14: $H_{\infty}^{(2)}$ (black line) and Macrorealism distances (d_{weak} Eq. (5.6) in blue, d_{int} Eq. (5.11) in green and d_{strong} Eq. (5.16) in red) as a function of $\omega\tau$ with dephasing parameter set as $P_{\text{deph}} = 1$ (Full dephasing), for the initial value of $\theta_0 = 1.36$ in Eq. (5.33).

Figures 5.12, 5.13, and 5.14 illustrate how the dephasing procedure affects the macrorealism distances defined in Eqs. (5.6), (5.11), and (5.16). In Fig. 5.12, the distances are non-zero, indicating a violation of macrorealism. However, in the fully dephased case shown in Fig. 5.14, all distances vanish, as expected for a classical system.

The minimum bound for the min-entropy is strictly determined by the initial state; e.g., for $\theta_0 = 1.36$, the initial density matrix is:

$$\rho \approx \begin{bmatrix} 0.04 & 0 \\ 0 & 0.96 \end{bmatrix}, \quad (5.35)$$

which leads to a maximum joint probability $P^* = 0.96$ and so a min-entropy $H_{\infty}^{(2)} \approx 0.06$. Consistent with this procedure, we notice that the lower bound for the min-entropy is dictated solely by the initial configuration, spanning the interval $[0, 1]$ depending on the value of θ_0 .

Furthermore, we observe that the range of values assumed by the min-entropy expands as the dephasing parameter increases, indicating a higher sensitivity of the entropic bounds to the dynamical parameters. To quantify this behavior, we performed a simulation across 50 values of $\theta_0 \in (0, 2\pi)$ and analyzed the variation

of the entropic bounds over the evolution domain $\omega\tau \in (0, 2\pi)$.

Specifically, for a given initial state configuration θ_0 , we define the total excursion of the two-time min-entropy, $\Delta H_\infty^{(2)}(\theta_0)$, as:

$$\Delta H_\infty^{(2)}(\theta_0) = \max_{\omega\tau} H_\infty^{(2)}(\theta_0, \omega\tau) - \min_{\omega\tau} H_\infty^{(2)}(\theta_0, \omega\tau). \quad (5.36)$$

Evaluating this quantity highlights how the operational width of the entropic bounds is affected by decoherence. The average values of $\Delta H_\infty^{(2)}$ obtained across the sampled configurations for different dephasing levels are summarized in Table 5.1.

Dephasing Parameter (P_{deph})	Entropy Excursion ($\Delta H_\infty^{(2)}$)
0.0	≈ 0.40
0.5	≈ 0.57
1.0	≈ 0.98

Table 5.1: Average excursion of the two-time min-entropy $\Delta H_\infty^{(2)}$ for different values of the dephasing parameter P_{deph} .

5.5.2 Classical Limit for Maximally Mixed State under σ_y Evolution

We apply our dephasing procedure to the experimental setup described in Ref. [28], which was previously analyzed in Sec. 4. Specifically, we consider the initial density matrix defined in Eq. (5.24), subject to the evolution given by Eq. (5.25). The dephasing protocol, as defined in Eq. (5.34), is applied at each discrete time step of the evolution. We begin by analyzing the system's behavior as a function of the dephasing intensity.

The behavior of the system in the absence of dephasing ($P_{\text{deph}} = 0$) is illustrated in Fig. 5.4. The min-entropy fluctuates between a lower bound of 1 and a maximum of 1.41. The weak and intermediate macrorealism distances (d_{weak} and d_{int}) are non-zero only in regions where the LGI at three times (Eq. (2.23)) are violated, whereas the strong distance (d_{strong}) remains non-zero throughout the evolution.

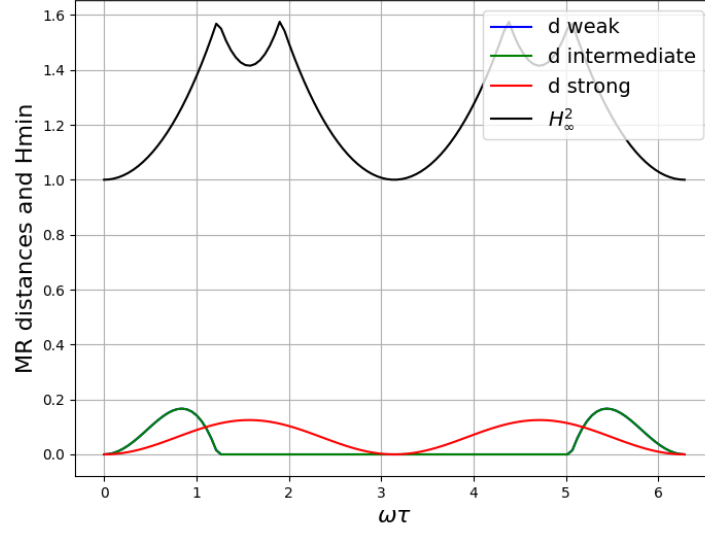


Figure 5.15: Min-entropy (black line) and Macrorealism distances as a function of $\omega\tau$, with the dephasing parameter set to $P_{\text{deph}} = 0.5$.

The impact of intermediate dephasing is shown in Fig. 5.15. We observe an increase in min-entropy, which now reaches a maximum value of 1.57, accompanied by a simultaneous decrease in all macrorealism distances.

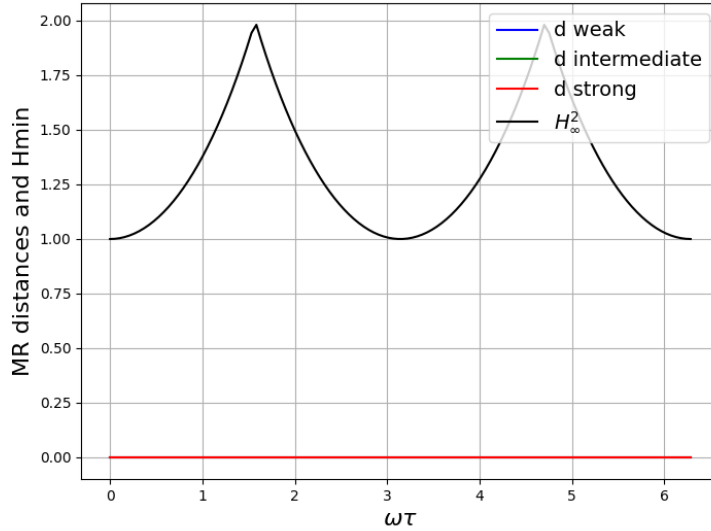


Figure 5.16: Min-entropy (black line) and Macrorealism distances as a function of $\omega\tau$, with full dephasing ($P_{\text{deph}} = 1$).

Under full dephasing ($P_{\text{deph}} = 1$), the system's behavior changes drastically, as illustrated in Fig. 5.16. The min-entropy now reaches a maximum of 2, while

all macrorealism distances vanish ($d = 0$), as expected for a purely classical system. This result can be interpreted as each of the four possible outcomes of the three measurements becoming equally likely, with a probability of 0.25. In Fig. 5.16, it's also important to note that $H_\infty^{(2)}$ is 1 in $0, \pi$ and 2π because of the trivial evolution at these points, which leads to only two possible outcomes 00 or 11, each with a probability of 0.5, this can easily be verified by evaluating a first measurement who give us 0 or 1 with equal probability, then the second measurement it's evolved by $\mathbb{I}$ giving us the same result, 00 or 11.

It is crucial to clarify that the higher entropy in this classical limit does not imply that more randomness can be extracted compared to the quantum case. In the classical regime, the randomness is not certified and could, in principle, be reproduced by a deterministic process. Conversely, in the quantum regime, the randomness is intrinsic and cannot be simulated by local hidden variable models [25].

5.6 Summary of Key Findings

This chapter has provided a detailed numerical and theoretical investigation into the relationship between the violation of macroscopic realism and the certification of quantum randomness. By comparing two different unitary evolutions and various entropic measures, we have identified several fundamental properties that characterize the temporal dynamics of quantum systems. The main findings are summarized below:

- **Hierarchical Structure of Temporal Inequalities:** Our results confirm a clear hierarchy within the Leggett-Garg framework. Specifically, the three-time LGI Eq. (2.23) is consistently more susceptible to violation than its two-time counterparts. This suggests that multi-time correlations capture non-classical effects that are not fully manifest in simpler, pairwise measurement sequences.
- **The Macrorealism Hierarchy and Distance Metrics:** We have demonstrated that the definition of macrorealism itself follows a nested hierarchical structure, as visually represented in Fig. 5.17. In this framework, *Strong Macrorealism* represents the most restrictive condition. Consequently, the associated distance d_{strong} emerges as the most robust metric for non-classicality, capable of certifying quantum behavior in regimes where *Weak* and *Intermediate* metrics vanish.

- **NSIT Condition Sensitivity:** A similar hierarchical trend was observed for the No-Signaling-in-Time conditions. The three-time NSIT formulations (Eq. (2.53) and Eq. (2.54)) are more easily violated than the two-time conditions (Eq. (4.2), (4.3), and (4.4)), further emphasizing the importance of considering the full measurement sequence to detect invasiveness.
- **Decoupling of Non-Classicality and Randomness:** A crucial finding of this work is that the violation of macrorealism does not automatically imply the generation of high-quality randomness. As evidenced in Section 5.4, a system can exhibit strictly non-classical temporal correlations while maintaining a min-entropy below the classical threshold of 1. This "decoupling" highlights the necessity of using both geometric distances and entropic bounds for a reliable QRNG certification.
- **Certified vs. Apparent Randomness:** We observed that while a fully dephased classical system can reach a maximum min-entropy of 2, the quantum systems analyzed (e.g., Sec. 5.3.2) often saturate at lower values, such as 1.41. However, the fundamental distinction lies in the *certifiability*: the randomness in the quantum case is intrinsically linked to the violation of macrorealism and is thus immune to deterministic classical modeling, unlike the apparent randomness of a dephased classical source.

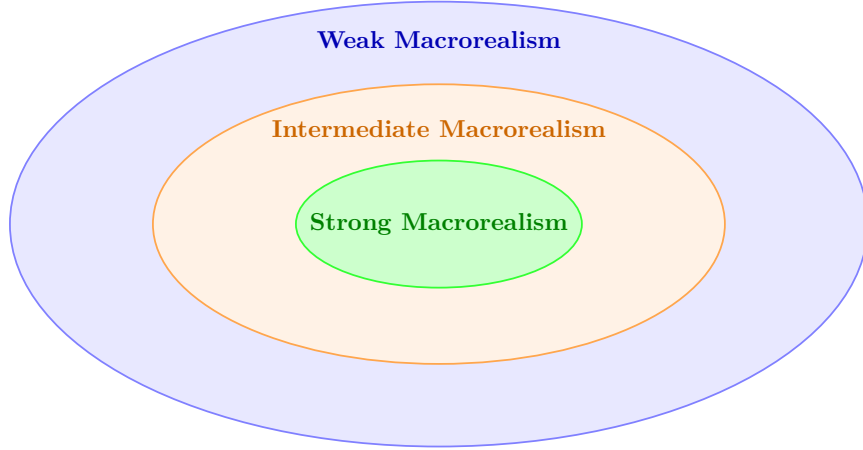


Figure 5.17: Visual representation of the hierarchy of Macrorealism definitions. Strong Macrorealism is a subset of Intermediate Macrorealism, which in turn is a subset of Weak Macrorealism.

Conclusions

The comprehensive research conducted in this thesis has explored the theoretical and experimental foundations of Quantum Random Number Generators (QRNGs) based on temporal correlations, tracing a continuous evolution through each chapter. If Chapter 1 is dedicated to establishing the essential mathematical and conceptual bedrock of this work by reviewing the kinematics of two-level systems (qubits) using both state-vector and density matrix formalisms—including key concepts such as entanglement, the projective measurement process, and the invasive nature of quantum information illustrated through the double-slit experiment—Chapter 2 enters much deeper into the core of purely quantum features and the rigorous methods required to measure them. This second chapter explicitly traces the historical and theoretical transition from Bell’s local realism to Leggett and Garg’s macrorealism. Within this context, we analyze how Leggett-Garg inequalities (LGI) serve as a temporal analogue to Bell inequalities, testing the core postulates of Macrorealism Per Se and Non-Invasive Measurability (NIM), while examining the vital role of Fine’s Theorem in providing the necessary and sufficient conditions for realistic models.

Building upon these fundamental principles, Chapters 3 and 4 shift the focus to how these unique quantum characteristics are intrinsically linked to the concepts of randomness and predictability, and how they can be directly exploited to generate random numbers. These chapters map the profound paradigm shift from classical epistemological unpredictability—which merely arises from incomplete information—to quantum ontological randomness, which is inherently intrinsic to nature. By expounding upon the stringent requirements of modern cryptographic applications and outlining the device-independent paradigm, these sections detail the formal derivation of LGIs directly from the principles of No-Signaling in Time (NSIT) and examine the quantification of randomness via min-entropy formulations.

Crucially, Chapter 5 contains the original results of this thesis, shifting the focus from the introductory literature to our own quantitative and statistical

contributions. This chapter provides a detailed numerical analysis of two distinct physical systems by defining specific violation parameters to evaluate the violation of macrorealism and the min-entropy bounds for randomness certification. By introducing new quantitative indicators to measure the geometric "distance" from the weak, intermediate, and strong levels of macrorealism, this original research maps these distances against min-entropy bounds. Furthermore, we analyze the impact of dephasing and decoherence, demonstrating how these processes drive the system toward the classical limit, and unveiling the intricate trade-offs between quantum non-classicality and secure randomness certification.

The final conclusions drawn from this research allow for a vital conceptual distinction between two properties that are frequently conflated but remain physically distinct: randomness and quantumness. Throughout this work, it has been demonstrated that randomness, in a purely statistical sense, can be generated by both classical and quantum systems. A high-entropy bitstream does not, by itself, reveal the nature of the underlying process, as a classical "Dr. Snoop" or a deterministic algorithm could produce sequences that appear perfectly random. Conversely, the temporal violation of macrorealistic bounds serves as a unique signature of true "quantumness". Unlike simple operational entropy, this property allows us to certify that the bits are truly independent of any pre-existing seed or hidden variable, confirming that while a classical system can simulate randomness, only a quantum system can provide a physical guarantee of it. The Leggett-Garg inequalities and their associated macrorealism distances act as a fundamental certification tool, ensuring that the generated randomness is inherent to the laws of physics, providing a level of security and unpredictability that is unattainable by any classical means.

Looking toward the future, several compelling avenues for research emerge directly from the results presented in this work. The next logical step involves experimental integration, translating these protocols into physical implementations executed on Quantum Computers. Furthermore, future efforts will be directed toward the optimization of min-entropy generation rates. By systematically fine-tuning the measurement timings ($\omega\tau$) for different experimental setups, it will be possible to maximize the certified min-entropy rate while simultaneously maintaining a strong and robust violation of the macrorealism distances, enhancing the practical utility and security of next-generation quantum cryptographic technologies.

Bibliography

- [1] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Vol. III: The New Quantum Mechanics*. Reading, Mass.: Addison-Wesley, 1965. ISBN: 978-0201021189.
- [2] J. J. Sakurai and Jim Napolitano. *Modern Quantum Mechanics*. 3rd. Cambridge University Press, 2017. ISBN: 978-1108473224.
- [3] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge: Cambridge University Press, 2010.
- [4] Kenichi Konishi and Giampiero Paffuti. *Quantum Mechanics: A New Introduction*. Oxford: Oxford University Press, 2009.
- [5] Luigi Picasso. *Lezioni di meccanica quantistica*. Pisa: ETS, 1996.
- [6] David J. Griffiths. *Introduction to Quantum Mechanics*. 2nd ed. Upper Saddle River, NJ: Pearson Prentice Hall, 2004.
- [7] A. Einstein, B. Podolsky, and N. Rosen. “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?” In: *Phys. Rev.* 47 (10 1935), pp. 777–780. DOI: 10.1103/PhysRev.47.777. URL: <https://link.aps.org/doi/10.1103/PhysRev.47.777>.
- [8] J. S. Bell. “On the Einstein Podolsky Rosen paradox”. In: *Physics Physique Fizika* 1 (3 1964), pp. 195–200. DOI: 10.1103/PhysicsPhysiqueFizika.1.195.
- [9] John F. Clauser et al. “Proposed Experiment to Test Local Hidden-Variable Theories”. In: *Phys. Rev. Lett.* 23 (15 1969), pp. 880–884. DOI: 10.1103/PhysRevLett.23.880. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.23.880>.
- [10] Stuart J. Freedman and John F. Clauser. “Experimental Test of Local Hidden-Variable Theories”. In: *Physical Review Letters* 28.14 (1972), pp. 938–941. DOI: 10.1103/PhysRevLett.28.938.

- [11] Alain Aspect, Jean Dalibard, and Gérard Roger. “Experimental Test of Bell’s Inequalities Using Time-Varying Analyzers”. In: *Physical Review Letters* 49.25 (1982), pp. 1804–1807. DOI: 10.1103/PhysRevLett.49.1804.
- [12] B. S. Tsirelson. “Quantum Generalizations of Bell’s Inequality”. In: *Letters in Mathematical Physics* 4.2 (1980), pp. 93–100. DOI: 10.1007/BF00417500.
- [13] A. J. Leggett and Anupam Garg. “Quantum mechanics versus macroscopic realism: Is the flux there when nobody looks?” In: *Phys. Rev. Lett.* 54 (9 1985), pp. 857–860. DOI: 10.1103/PhysRevLett.54.857. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.54.857>.
- [14] Clive Emary, Neill Lambert, and Franco Nori. “Corrigendum: Leggett–Garg inequalities (2014 Rep. Prog. Phys. 77 016001)”. In: *Reports on Progress in Physics* 77 (Feb. 2014), p. 039501. DOI: 10.1088/0034-4885/77/3/039501.
- [15] J. J. Halliwell. “Comparing conditions for macrorealism: Leggett-Garg inequalities versus no-signaling in time”. In: *Physical Review A* 96.1 (July 2017). ISSN: 2469-9934. DOI: 10.1103/physreva.96.012121. URL: <http://dx.doi.org/10.1103/PhysRevA.96.012121>.
- [16] Arthur Fine. “Hidden Variables, Joint Probability, and the Bell Inequalities”. In: *Phys. Rev. Lett.* 48 (5 1982), pp. 291–295. DOI: 10.1103/PhysRevLett.48.291. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.48.291>.
- [17] Lucas Clemente and Johannes Kofler. “Necessary and sufficient conditions for macroscopic realism from quantum mechanics”. In: *Physical Review A* 91.6 (June 2015). ISSN: 1094-1622. DOI: 10.1103/physreva.91.062103. URL: <http://dx.doi.org/10.1103/PhysRevA.91.062103>.
- [18] Augustus De Morgan. *Formal Logic: Or, The Calculus of Inference, Necessary and Probable*. London: Taylor and Walton, 1847.
- [19] Lucas Clemente and Johannes Kofler. “No Fine Theorem for Macrorealism: Limitations of the Leggett-Garg Inequality”. In: *Physical Review Letters* 116.15 (Apr. 2016). ISSN: 1079-7114. DOI: 10.1103/physrevlett.116.150401. URL: <http://dx.doi.org/10.1103/PhysRevLett.116.150401>.
- [20] Roger Eckhardt. “Stan Ulam, John von Neumann, and the Monte Carlo Method”. In: *Los Alamos Science* 15 (1987), pp. 131–137. URL: <https://fas.org/sgp/othergov/doe/lanl/pubs/00326866.pdf>.

- [21] Nicholas Metropolis and Stanislaw Ulam. “The Monte Carlo Method”. In: *Journal of the American Statistical Association* 44.247 (1949), pp. 335–341. DOI: 10.1080/01621459.1949.10483310.
- [22] R. R. Coveyou. “Random number generation is too important to be left to chance”. In: *Studies in Applied Mathematics* 3 (1970), pp. 70–111.
- [23] D. H. Lehmer. “Mathematical methods in large-scale computing units”. In: *Proceedings of the Second Symposium on Large Scale Digital Calculating Machinery*. Cambridge, MA: Harvard University Press, 1951, pp. 141–146.
- [24] Makoto Matsumoto and Takuji Nishimura. “Mersenne Twister: A 623-dimensionally equidistributed uniform pseudorandom number generator”. In: *ACM Transactions on Modeling and Computer Simulation* 8.1 (1998), pp. 3–30. DOI: 10.1145/272991.272995.
- [25] Roger Colbeck. *Quantum And Relativistic Protocols For Secure Multi-Party Computation*. 2011. arXiv: 0911.3814 [quant-ph]. URL: <https://arxiv.org/abs/0911.3814>.
- [26] Alfréd Rényi. “On Measures of Entropy and Information”. In: *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*. Vol. 1. University of California Press, 1961, pp. 547–561.
- [27] Claude E. Shannon. “A Mathematical Theory of Communication”. In: *The Bell System Technical Journal* 27.3 (1948), pp. 379–423.
- [28] Pingal Pratyush Nath et al. “Single-System-Based Generation of Certified Randomness Using Leggett-Garg Inequality”. In: *Physical Review Letters* 133.2 (July 2024). ISSN: 1079-7114. DOI: 10.1103/physrevlett.133.020802. URL: <http://dx.doi.org/10.1103/PhysRevLett.133.020802>.
- [29] Eric G. Cavalcanti and Howard M. Wiseman. “Bell Nonlocality, Signal Locality and Unpredictability (or What Bohr Could Have Told Einstein at Solvay Had He Known About Bell Experiments)”. In: *Foundations of Physics* 42.10 (July 2012), 1329–1338. ISSN: 1572-9516. DOI: 10.1007/s10701-012-9669-1. URL: <http://dx.doi.org/10.1007/s10701-012-9669-1>.
- [30] Shiladitya Mal, Manik Banik, and Sujit Choudhary. “Temporal correlations and device-independent randomness”. In: *Quantum Information Processing* 15 (July 2016). DOI: 10.1007/s11128-016-1321-0.

- [31] Clive Emary, Neill Lambert, and Ramón Aguado. “Leggett–Garg inequalities”. In: *Reports on Progress in Physics* 77.1 (2013), p. 016001. DOI: 10.1088/0034-4885/77/1/016001.
- [32] Costantino Budroni et al. “Bounding Temporal Quantum Correlations”. In: *Physical Review Letters* 111.2 (2013), p. 020403. DOI: 10.1103/PhysRevLett.111.020403.
- [33] J J Halliwell. “Necessary and sufficient conditions for macrorealism using two- and three-time Leggett-Garg inequalities”. In: *Journal of Physics: Conference Series* 1275.1 (Sept. 2019), p. 012008. ISSN: 1742-6596. DOI: 10.1088/1742-6596/1275/1/012008. URL: <http://dx.doi.org/10.1088/1742-6596/1275/1/012008>.
- [34] Charles R. Harris, K. Jarrod Millman, Stéffan J. van der Walt, et al. “Array programming with NumPy”. In: *Nature* 585.7825 (2020), pp. 357–362. DOI: 10.1038/s41586-020-2649-2.
- [35] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python”. In: *Nature Methods* 17 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.
- [36] Wolfram Research, Inc. *Mathematica, Version 14.0*. Champaign, IL: Wolfram Research, Inc., 2024. URL: <https://www.wolfram.com/mathematica>.
- [37] Thomas Bayes. “An Essay towards solving a Problem in the Doctrine of Chances. By the late Rev. Mr. Bayes, F. R. S. communicated by Mr. Price, in a Letter to John Canton, A. M. F. R. S.” In: *Philosophical Transactions of the Royal Society of London* 53 (1763), pp. 370–418.

Appendix A

Analytical Results

This appendix collects the full analytical results obtained via Wolfram Mathematica [36]. These long expressions correspond to the macrorealism distances and the min-entropy plotted and discussed throughout the main text, compiled here to maintain the overall readability of the thesis.

A.1 Case 1: Maximally Mixed State under σ_y Evolution

A.1.1 Leggett-Garg Inequality

The Leggett-Garg inequality violation for the three-time scenario is given by:

$$LGI_{123} = 2 \cos(\theta) - \cos(2\theta) \quad (\text{A.1})$$

A.1.2 Weak Distance

By defining the following macrorealist auxiliary functions for shorthand convenience:

$$A(\theta) = \max \left[0, \frac{1}{4} \sin \left(\frac{\theta}{2} \right) \left(2(1 + 9 \cos(\theta)) \sin \left(\frac{\theta}{2} \right) - \text{Re} \left[\csc \left(\frac{\theta}{2} \right) \right] \sin^2(\theta) \right) \right] \quad (\text{A.2})$$

$$B(\theta) = \max \left[0, \frac{1}{4} \left(2 \cos^2 \left(\frac{\theta}{2} \right) (-5 + \cos(\theta)) + \text{Re} \left[\csc \left(\frac{\theta}{2} \right) \right] \sin \left(\frac{\theta}{2} \right) \sin^2(\theta) \right) \right] \quad (\text{A.3})$$

the macrorealism distance in the weak regime simplifies to:

$$d_{\text{weak}} = \sqrt{A(\theta)^2 + B(\theta)^2} \quad (\text{A.4})$$

A.1.3 Intermediate Distance

The distance evaluated in the intermediate regime yields:

$$d_{\text{intermediate}} = \max \left[0, \frac{1}{4} \sin \left(\frac{\theta}{2} \right) \left(2(1 + 9 \cos(\theta)) \sin \left(\frac{\theta}{2} \right) - \text{Re} \left[\csc \left(\frac{\theta}{2} \right) \right] \sin^2(\theta) \right) \right] \quad (\text{A.5})$$

A.1.4 Strong Distance

The strong macrorealism distance reduces to the following compact expression:

$$d_{\text{strong}} = \frac{\sin^2(\theta)}{4} \quad (\text{A.6})$$

A.1.5 Min-Entropy $H_{\infty}^{(2)}$

The second-order min-entropy is expressed as:

$$H_{\infty}^{(2)} = - \frac{\log \left(\max \left[\frac{1}{2} \cos^2(\theta/2), \frac{1}{2} \cos^2(\theta), \frac{1}{2} \sin^2(\theta/2), \frac{1}{2} \sin^2(\theta) \right] \right)}{\log(2)} \quad (\text{A.7})$$

A.2 Case 2: σ_y Eigenstate under σ_x Evolution

A.2.1 Leggett-Garg Inequality

The Leggett-Garg inequality violation for this configuration is:

$$LGI_{123} = 2 \cos(\theta) - \cos(2\theta) \quad (\text{A.8})$$

A.2.2 Weak Distance

To prevent layout overflows due to the size of the elements under the radical, we introduce the following four components:

$$C_1(\theta) = \max \left[0, 4 \cos(\theta) \sin^2 \left(\frac{\theta}{2} \right) \right] \quad (\text{A.9})$$

$$C_2(\theta) = \max [0, -1 - \cos(\theta) - \sin(\theta)] \quad (\text{A.10})$$

$$C_3(\theta) = \max [0, -1 - \sin(\theta) - \cos(\theta)(1 + 2 \sin(\theta))] \quad (\text{A.11})$$

$$C_4(\theta) = \max [0, -1 - \cos(2\theta) - \sin(2\theta)] \quad (\text{A.12})$$

The weak distance is then computed geometrically as:

$$d_{\text{weak}} = \sqrt{C_1(\theta)^2 + C_2(\theta)^2 + C_3(\theta)^2 + C_4(\theta)^2} \quad (\text{A.13})$$

A.2.3 Intermediate Distance

Utilizing the component $C_1(\theta)$ defined above, the intermediate distance is formulated as:

$$d_{\text{intermediate}} = \frac{1}{8} \sqrt{64 C_1(\theta)^2 + 8 (7 + 5 \cos(2\theta)) \sin^2(\theta)} \quad (\text{A.14})$$

A.2.4 Strong Distance

The strong macrorealism distance for Case 2 evaluates to:

$$d_{\text{strong}} = \frac{|\sin(\theta)| \sqrt{2 + \cos(\theta) + \cos(2\theta)}}{2\sqrt{2}} \quad (\text{A.15})$$

A.2.5 Min-Entropy $H_\infty^{(2)}$

Let $\mathcal{M}(\theta)$ be the maximum argument characterizing the state probabilities:

$$\begin{aligned} \mathcal{M}(\theta) = \max & \left[\frac{1}{2} \cos^2 \left(\frac{\theta}{2} \right), \frac{1}{2} \cos^2(\theta), \frac{1}{2} \sin^2 \left(\frac{\theta}{2} \right), \frac{1}{2} \sin^2(\theta), \right. \\ & \frac{1}{2} \cos^2 \left(\frac{\theta}{2} \right) \left(\cos \left(\frac{\theta}{2} \right) + \sin \left(\frac{\theta}{2} \right) \right)^2, -\frac{1}{2} \cos^2 \left(\frac{\theta}{2} \right) (-1 + \sin(\theta)), \\ & \left. \frac{1}{8} (1 - \cos(\theta) + \sin(\theta))^2, \frac{1}{8} (-1 + \cos(\theta) + \sin(\theta))^2 \right] \end{aligned} \quad (\text{A.16})$$

The corresponding min-entropy is given by:

$$H_\infty^{(2)} = -\frac{\log(\mathcal{M}(\theta))}{\log(2)} \quad (\text{A.17})$$

Appendix B

Predictability

B.1 Experimental Examples of Predictability Evaluation

To provide an empirical grounding for the mathematical quantities introduced above, we examine how the conditional guessing probability behaves when evaluated on experimental bitstreams. For a simplified two-time protocol (t_1, t_2) , the operational metric reduces to the pairwise guessing probability:

$$P_{\text{guess}}(s_2|s_1) = \sum_{s_1} \max_{s_2} p(s_1, s_2). \quad (\text{B.1})$$

We assume that the experimental measurement outcomes are recorded as binary digits $x \in \{0, 1\}$, mapped onto the dichotomous macrorealistic variables $s \in \{+1, -1\}$ via $s = 1 - 2x$.

B.1.1 Case 1: Minimal Predictability (Maximum Stochasticity)

Consider an experimental dataset consisting of $N = 12$ consecutive bit pairs, forming the following stream:

00 10 11 01 10 00 11 01 11 01 10 00

Computing the empirical frequencies $n(x_1, x_2)$ reveals that all four combinations are perfectly balanced:

$$\begin{aligned} n(0, 0) &= 3, & n(0, 1) &= 3, \\ n(1, 0) &= 3, & n(1, 1) &= 3. \end{aligned}$$

Normalizing by the total sample size N yields a uniform empirical joint probability distribution, $p(s_1, s_2) = 0.25 \forall s_1, s_2$. Evaluating the guessing probability under these conditions gives:

$$\begin{aligned} P_{\text{guess}}(s_2|s_1) &= \max[p(0, 0), p(0, 1)] + \max[p(1, 0), p(1, 1)] \\ &= \max(0.25, 0.25) + \max(0.25, 0.25) = 0.5. \end{aligned}$$

Here, $P_{\text{guess}} = 0.5$, hitting the absolute lower bound for a two-level system. The predictability condition is maximally violated; knowing the initial measurement outcome yields zero statistical advantage for predicting the subsequent state, indicating a fully stochastic regime.

B.1.2 Case 2: Maximum Predictability under Identity Map

Conversely, consider an experimental sequence of $N = 10$ pairs representing an identity propagation:

$$00 \ 11 \ 00 \ 11 \ 11 \ 00 \ 11 \ 00 \ 11 \ 00$$

The resulting joint probability distribution is heavily constrained:

$$p(0, 0) = 0.5, \quad p(1, 1) = 0.5, \quad p(0, 1) = p(1, 0) = 0. \quad (\text{B.2})$$

Applying the formal definition of $P_{\text{guess}}(s_2|s_1)$ yields:

$$P_{\text{guess}}(s_2|s_1) = \max(0.5, 0) + \max(0, 0.5) = 1. \quad (\text{B.3})$$

In this case, $P_{\text{guess}} = 1$, and the predictability condition is fully satisfied. Although the global stream fluctuates macroscopically between 00 and 11, the internal dynamics are strictly deterministic: measuring 0 (or 1) at t_1 guarantees the same outcome at t_2 with 100% certainty.

B.1.3 Case 3: Maximum Predictability under a Logical NOT Operation

Finally, to emphasize that predictability is a measure of correlation rather than state invariance, consider a scenario where the system undergoes a deterministic inversion, equivalent to a quantum or classical logical NOT gate. Let the recorded dataset be a sequence of $N = 10$ bit pairs distributed as follows:

01 10 10 01 10 01 10 01 01 10

Counting the occurrences of each pair yields:

$$\begin{aligned} n(0, 1) &= 5, & n(1, 0) &= 5, \\ n(0, 0) &= 0, & n(1, 1) &= 0, \end{aligned}$$

which translates into the empirical joint probabilities:

$$p(0, 1) = 0.5, \quad p(1, 0) = 0.5, \quad p(0, 0) = p(1, 1) = 0. \quad (\text{B.4})$$

Calculating the conditional guessing probability for this inverted distribution gives:

$$\begin{aligned} P_{\text{guess}}(s_2|s_1) &= \max[p(0, 0), p(0, 1)] + \max[p(1, 0), p(1, 1)] \\ &= \max(0, 0.5) + \max(0.5, 0) = 1. \end{aligned}$$

Just as in Case 2, we obtain $P_{\text{guess}} = 1$, demonstrating that the predictability relation is perfectly satisfied. This example confirms that predictability does not require the system to remain static; rather, it requires the evolution to be completely pre-determined. Knowing that the initial bit is 0 allows the observer to predict with absolute certainty (100%) that the second bit will be 1, confirming that a structured, invertible dynamical process preserves perfect classical predictability and excludes genuine randomness.

B.1.4 Single-Experiment Restriction and Contextual Independence

It is crucial to emphasize that the predictability condition can only be meaningfully evaluated within the context of a single, temporally connected physical experiment. Quantifying predictability across distinct, independent experimental realizations is fundamentally flawed due to the absence of causal correlations.

To illustrate this constraint, consider a scenario where two independent observers, Alice and Bob, generate pairs of bits in spatially separated laboratories. Alice records a consecutive pair (s_1, s_2) from her local setup, while Bob independently records another pair (s_3, s_4) from his own apparatus. A third party, Charlie, collects these distinct outputs and concatenates them to form a single four-bit sequence (s_1, s_2, s_3, s_4) .

In this composition, evaluating the conditional predictability of Bob's out-

comes based on the knowledge of Alice’s bits—i.e., computing $P_{\text{guess}}(s_3, s_4|s_1, s_2)$ —is operationally meaningless. Because the two generation processes occur in distinct local contexts, their global joint probability factorizes into the product of their independent marginals:

$$p(s_1, s_2, s_3, s_4) = p_{\text{Alice}}(s_1, s_2) \cdot p_{\text{Bob}}(s_3, s_4). \quad (\text{B.5})$$

Consequently, the conditional probabilities reduce to the unconditioned local ones, $p(s_3, s_4|s_1, s_2) = p_{\text{Bob}}(s_3, s_4)$, reflecting a complete lack of physical correlations between the two laboratories.

While this scenario can be trivially extended to sequences of arbitrary length ($n > 4$), the core implication remains unchanged: predictability is intrinsically a single-experiment property. It serves to quantify the internal, non-local, or temporal correlations driving a specific physical system over time, and it vanishes by definition when evaluating uncorrelated ensembles originating from disconnected experimental frameworks.

Appendix C

Statistical Uncertainties and Error Propagation

In this appendix, we provide a detailed explanation of the error propagation methods used in the analysis of the experimental data, particularly in the context of the certification metrics for quantum randomness.

C.1 Statistical Uncertainties

The primary source of uncertainty in our measurements arises from the finite size of the datasets. Given that we are dealing with binary outcomes (bits), the standard deviation of the mean for a Bernoulli distribution can be expressed as:

$$\sigma = \sqrt{\frac{p(1-p)}{n}} \quad (\text{C.1})$$

where p is the probability of obtaining a "1" (or "0") and n is the total number of samples.

C.2 Propagation of Uncertainties

When calculating derived quantities such as the min-entropy or the macrorealism distances, we must consider how the uncertainties propagate. For a function $f(x_1, x_2, \dots, x_n)$, the uncertainty σ_f is:

$$\sigma_f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \sigma_i \right)^2} \quad (\text{C.2})$$

C.3 Leggett-Garg Inequality Error Propagation

In the context of Leggett-Garg inequalities, the violation is determined by the correlation functions C_{ij} , which are estimated from the joint probabilities $P_{ij}(a, b)$ of obtaining outcomes $a, b \in \{+1, -1\}$ at times t_i and t_j . The correlation function is defined as:

$$C_{ij} = p_{ij}(+1, +1) + p_{ij}(-1, -1) - p_{ij}(+1, -1) - p_{ij}(-1, +1). \quad (\text{C.3})$$

Assuming that the measurements for different outcome pairs are independent and that each probability has an associated statistical uncertainty $\sigma_{p_{ij}}$ (as defined in Eq. C.1), we apply the propagation of uncertainty. Since the derivatives of C_{ij} with respect to each probability are ± 1 , the variance of the correlation function is:

$$\sigma_{C_{ij}}^2 = \sigma_{p_{++}}^2 + \sigma_{p_{--}}^2 + \sigma_{p_{+-}}^2 + \sigma_{p_{-+}}^2. \quad (\text{C.4})$$

In a simplified scenario where we assume a uniform uncertainty σ for all probabilities, the uncertainty for any correlation function reduces to:

$$\sigma_C = \sqrt{4\sigma^2} = 2\sigma. \quad (\text{C.5})$$

The Leggett-Garg Inequality is expressed as sum of these correlation functions:

$$LGI_{123} = C_{12} + C_{23} - C_{13}. \quad (\text{C.6})$$

To determine the uncertainty σ_{LGI} associated with the LGI, we propagate the errors of the individual correlation functions. Since each C_{ij} is obtained from distinct experimental runs (different time intervals), they are treated as independent variables:

$$\sigma_{LGI} = \sqrt{\left(\frac{\partial LGI}{\partial C_{12}} \sigma_{C_{12}}\right)^2 + \left(\frac{\partial LGI}{\partial C_{23}} \sigma_{C_{23}}\right)^2 + \left(\frac{\partial LGI}{\partial C_{13}} \sigma_{C_{13}}\right)^2}. \quad (\text{C.7})$$

Substituting the partial derivatives (which are 1, 1, -1 respectively) and assuming the uniform error σ_C derived above, we obtain:

$$\sigma_{LGI} = \sqrt{\sigma_C^2 + \sigma_C^2 + \sigma_C^2} = \sqrt{3(2\sigma)^2} = 2\sqrt{3}\sigma \approx 3.46\sigma. \quad (\text{C.8})$$

C.4 No Signaling In Time Error Propagation

To evaluate the uncertainty in the NSIT condition, we consider the expression:

$$p(a_i|Q_i) = \sum_{a_j} p(a_i, a_j|Q_i, Q_j). \quad (\text{C.9})$$

Assuming the joint probabilities are independent, its uncertainty is:

$$\sigma_{\text{sum}} = \sqrt{\sigma_{p(+,a_j)}^2 + \sigma_{p(-,a_j)}^2}. \quad (\text{C.10})$$

The total uncertainty on the NSIT condition must also account for the uncertainty of the baseline marginal probability $p(a_i|Q_i)$, which is typically measured in a separate calibration run. Thus:

$$\sigma_{\text{NSIT}} = \sqrt{\sigma_{p(a_i)}^2 + \sigma_{p_{\text{sum}}}^2} = \sqrt{\sigma_{p(a_i)}^2 + \sigma_{p(+,a_j)}^2 + \sigma_{p(-,a_j)}^2}. \quad (\text{C.11})$$

In the specific case where all probabilities are derived from datasets of similar size n and assuming a uniform probability error σ , the uncertainty on the NSIT violation simplifies to:

$$\sigma_{\text{NSIT}} = \sqrt{\sigma^2 + \sigma^2 + \sigma^2} = \sqrt{3}\sigma \approx 1.73\sigma. \quad (\text{C.12})$$

C.5 Macrorealism Error Propagation

C.5.1 Weak Macrorealism Distance

To evaluate the uncertainty on the weak macrorealism distance, we start from its definition:

$$d_{\text{weak}} = \sqrt{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}, \quad (\text{C.13})$$

where α represents the violation of the LGI_{123} Eq. (2.23), and β, γ, δ represent the violations of the three two-time LGIs Eq. (2.26), we employ the general formula:

$$\sigma_{d_{\text{weak}}} = \sqrt{\left(\frac{\partial d_{\text{weak}}}{\partial \alpha} \sigma_{\alpha}\right)^2 + \left(\frac{\partial d_{\text{weak}}}{\partial \beta} \sigma_{\beta}\right)^2 + \left(\frac{\partial d_{\text{weak}}}{\partial \gamma} \sigma_{\gamma}\right)^2 + \left(\frac{\partial d_{\text{weak}}}{\partial \delta} \sigma_{\delta}\right)^2}. \quad (\text{C.14})$$

The partial derivatives are given by:

$$\frac{\partial d_{\text{weak}}}{\partial \alpha} = \frac{\alpha}{\sqrt{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}} = \frac{\alpha}{d_{\text{weak}}}, \quad (\text{C.15})$$

and similarly for the other variables. Substituting these into the propagation formula, we obtain:

$$\sigma_{d_{\text{weak}}} = \frac{1}{d_{\text{weak}}} \sqrt{\alpha^2 \sigma_\alpha^2 + \beta^2 \sigma_\beta^2 + \gamma^2 \sigma_\gamma^2 + \delta^2 \sigma_\delta^2}. \quad (\text{C.16})$$

In the scenario where the uncertainties of the individual LGI violations are approximately equal ($\sigma_\alpha \approx \sigma_\beta \approx \sigma_\gamma \approx \sigma_\delta \approx \sigma_{LGI}$), the expression simplifies significantly:

$$\sigma_{d_{\text{weak}}} = \frac{\sigma_{LGI}}{d_{\text{weak}}} \sqrt{\alpha^2 + \beta^2 + \gamma^2 + \delta^2} = \sigma_{LGI} = 2\sqrt{3}\sigma \approx 3.46\sigma. \quad (\text{C.17})$$

C.6 Intermediate Macroeconomic Distances

To evaluate the uncertainty on the intermediate macroeconomic distance, we start from its definition:

$$d_{\text{int}} = \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2 + \alpha^2}, \quad (\text{C.18})$$

where η_i represents the i -th NSIT condition deviation Eq. (5.8)-(5.10), and α is the Leggett-Garg parameter Eq. (5.2).

The propagation of uncertainty for d_{int} is calculated as:

$$\sigma_{d_{\text{int}}} = \sqrt{\left(\frac{\partial d_{\text{int}}}{\partial \eta_1} \sigma_{\eta_1}\right)^2 + \left(\frac{\partial d_{\text{int}}}{\partial \eta_2} \sigma_{\eta_2}\right)^2 + \left(\frac{\partial d_{\text{int}}}{\partial \eta_3} \sigma_{\eta_3}\right)^2 + \left(\frac{\partial d_{\text{int}}}{\partial \alpha} \sigma_\alpha\right)^2}. \quad (\text{C.19})$$

Evaluating the partial derivatives ($\partial d_{\text{int}} / \partial x_i = x_i / d_{\text{int}}$), we obtain:

$$\sigma_{d_{\text{int}}} = \frac{1}{d_{\text{int}}} \sqrt{\eta_1^2 \sigma_{\eta_1}^2 + \eta_2^2 \sigma_{\eta_2}^2 + \eta_3^2 \sigma_{\eta_3}^2 + \alpha^2 \sigma_\alpha^2}. \quad (\text{C.20})$$

It is important to note that the uncertainties σ_{η_i} and σ_α are not identical, as derived in the previous sections. Specifically, we have:

- $\sigma_{\text{NSIT}} = \sigma_{\eta_i} = \sqrt{3}\sigma$ (assuming uniform probability error σ);
- $\sigma_{\text{LGI}} = \sigma_\alpha = 2\sqrt{3}\sigma$.

Substituting these values into the expression for $\sigma_{d_{\text{int}}}$:

$$\sigma_{d_{\text{int}}} = \frac{1}{d_{\text{int}}} \sqrt{(\eta_1^2 + \eta_2^2 + \eta_3^2)(3\sigma^2) + \alpha^2(12\sigma^2)}. \quad (\text{C.21})$$

Factoring out the common terms, the total uncertainty on the intermediate

distance is:

$$\sigma_{d_{\text{int}}} = \frac{\sigma\sqrt{3}}{d_{\text{int}}} \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2 + 4\alpha^2}. \quad (\text{C.22})$$

In the specific case where the violations are of comparable magnitude ($\eta_1 \approx \eta_2 \approx \eta_3 \approx \alpha$), the expression simplifies to:

$$\sigma_{d_{\text{int}}} = \frac{\sigma\sqrt{3}}{\sqrt{4\alpha^2}} \sqrt{3\alpha^2 + 4\alpha^2} = \sigma\sqrt{\frac{21}{4}} \approx 2.29\sigma. \quad (\text{C.23})$$

C.7 Strong Macrorealism Distance

The strong macrorealism distance d_{strong} is defined as:

$$d_{\text{strong}} = \sqrt{\eta_4^2 + \eta_5^2 + \eta_3^2}, \quad (\text{C.24})$$

where η_4 and η_5 involve the marginalization of three-time joint probabilities $p_{123}(s_1, s_2, s_3)$. To calculate the uncertainty σ_{η_4} (and similarly for σ_{η_5}), we consider the terms:

$$\sigma_{\eta_4} = \sqrt{\sigma_{p_{123}(+,s_2,s_3)}^2 + \sigma_{p_{123}(-,s_2,s_3)}^2 + \sigma_{p_{23}(s_2,s_3)}^2}. \quad (\text{C.25})$$

Assuming a uniform probability error σ , we obtain $\sigma_{\eta_4} = \sigma_{\eta_5} = \sqrt{3}\sigma$. Since we already established in the previous section that $\sigma_{\eta_3} = \sqrt{3}\sigma$, all components of the strong distance share the same uncertainty:

$$\sigma_{\eta_4} \approx \sigma_{\eta_5} \approx \sigma_{\eta_3} \approx \sigma_{\text{NSIT}} = \sqrt{3}\sigma. \quad (\text{C.26})$$

The propagation of uncertainty for d_{strong} is then:

$$\sigma_{d_{\text{strong}}} = \sqrt{\left(\frac{\partial d_{\text{strong}}}{\partial \eta_4} \sigma_{\eta_4}\right)^2 + \left(\frac{\partial d_{\text{strong}}}{\partial \eta_5} \sigma_{\eta_5}\right)^2 + \left(\frac{\partial d_{\text{strong}}}{\partial \eta_3} \sigma_{\eta_3}\right)^2}. \quad (\text{C.27})$$

Substituting the partial derivatives $\partial d_{\text{strong}}/\partial \eta_i = \eta_i/d_{\text{strong}}$ and the uniform uncertainty σ_{Δ} :

$$\sigma_{d_{\text{strong}}} = \frac{\sigma_{\Delta}}{d_{\text{strong}}} \sqrt{\eta_4^2 + \eta_5^2 + \eta_3^2} = \sigma_{\text{NSIT}} \approx 1.73\sigma. \quad (\text{C.28})$$

C.8 Min-Entropy Error Propagation

For a given outcome with probability p , the min-entropy is defined as:

$$H_\infty = -\log_2(p) = -\frac{\ln(p)}{\ln(2)}. \quad (\text{C.29})$$

To determine how the statistical uncertainty σ propagates to the min-entropy, we calculate the derivative of H_∞ with respect to p :

$$\frac{dH_\infty}{dp} = -\frac{1}{\ln(2)} \frac{d}{dp} \ln(p) = -\frac{1}{p \ln(2)}. \quad (\text{C.30})$$

Applying the first-order Taylor expansion for the propagation of uncertainty, the standard deviation σ_{H_∞} is given by:

$$\sigma_{H_\infty} = \left| \frac{dH_\infty}{dp} \right| \sigma = \frac{\sigma}{p \ln(2)}. \quad (\text{C.31})$$

Appendix D

Device-Independent Certification

In this appendix, we present a framework for the certification of quantumness and randomness that does not rely on assumptions regarding the internal workings of the system or the measurement apparatus. This device-independent treatment builds upon the theoretical foundations established in Chapter 5.

D.1 Bit Analysis

To transition from a theoretical framework to a data-driven approach, we analyze binary sequences (0s and 1s) produced by a "black box" system. In this context, we assume knowledge of the generation protocol, which is based on the Leggett-Garg experimental setup described in Chapter 5.

To correctly process the output and extract physical quantities—such as Leggett-Garg parameters, min-entropy, and macrorealism distances—the following technical specifications must be provided:

- The generation protocol of the bits, specifically their temporal ordering (e.g., whether a bit pair corresponds to measurements at times $\{t_1, t_2\}$, $\{t_1, t_3\}$, or $\{t_2, t_3\}$).
- The total number of bits generated for each measurement configuration (n).
- The organization of the data relative to the evolution parameter $\omega\tau$ (i.e., identifying which blocks of bits correspond to specific dynamical steps).

This structured data allows for a rigorous certification of the source, determining whether the underlying process exhibits genuine quantum behavior or can be explained by a classical macrorealistic model.

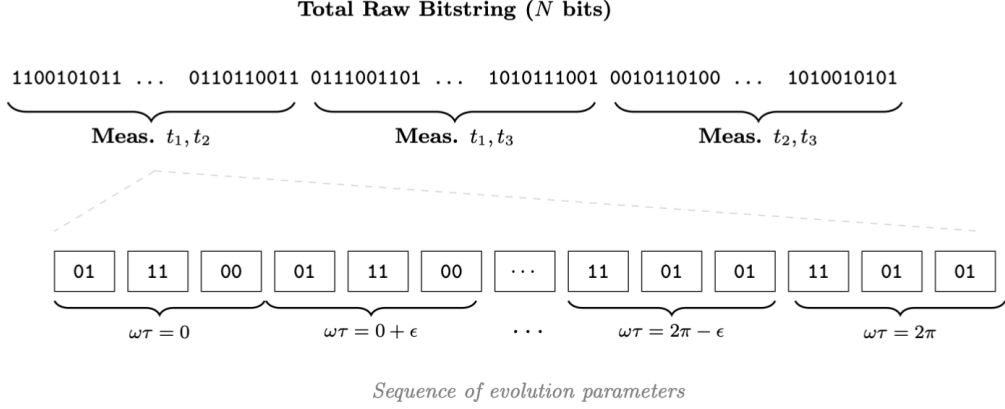


Figure D.1: Hierarchical structure of the experimental dataset. The continuous bitstream is partitioned into three sequences for Leggett-Garg inequality terms, then subdivided by $\omega\tau$ evolution steps, each containing n measurement runs.

The hierarchical organization of the bitstream is visualized in Fig. D.1. In this characterization phase, the raw output of the black box is obtained through a systematic scan over a range of evolution parameters. While an end-user seeking cryptographic randomness would typically require a bitstream generated at a fixed, high-entropy operational point, this scanning procedure is essential for the initial calibration and certification of the device.

Much like the *test print* function in commercial printers—which outputs a comprehensive range of colors and patterns to verify technical integrity and is rarely accessed by the end-user—we require this *black box* to implement a similar diagnostic routine. Specifically, the system should perform an exhaustive scan across the entire parameter space $\omega\tau$, providing a full characterization of the internal state to ensure operational consistency.

By partitioning the total raw bitstring into segments corresponding to specific measurement times and subdividing them by the evolution parameter $\omega\tau$, we can perform a systematic analysis of the system's behavior across its dynamical range. This characterization is fundamental to identify the "optimal operational windows"—specifically, the values of $\omega\tau$ where the violation of macrorealism is most significant and the min-entropy is maximized.

Consequently, once the device is certified, the black box can be **tuned** to one of these specific parameters. This ensures that the final user receives a bitstream whose quantum nature and unpredictability are rigorously guaranteed by the preliminary violation of temporal Bell-like inequalities.

D.1.1 Pairwise Bit Analysis (1-2, 1-3, 2-3)

In this phase of the study, we transition from theoretical predictions to the empirical processing of two distinct datasets, both generated using the physical systems described in detail in Chapter 5:

- **Dataset 1** is constructed using the initial density matrix defined in Eq. (5.29) and the unitary evolution operator specified in Eq. (5.30).
- **Dataset 2** utilizes the initial state defined in Eq. (5.24) and the simplified evolution dynamics expressed in Eq. (5.25).

The primary objective is to verify whether the binary sequences produced by our "black box" exhibit the non-classical signatures expected from the Leggett-Garg framework.

Since the temporal ordering and the specific timing of measurements are critical for evaluating time-correlations, we isolate the outcomes corresponding to the measurement pairs (1-2), (1-3), and (2-3). By processing these bitstrings, we compute the empirical correlation functions— $\langle Q_1 Q_2 \rangle$, $\langle Q_1 Q_3 \rangle$, and $\langle Q_2 Q_3 \rangle$ —which represent the statistical averages of the products of outcomes at different times. These values are then substituted into the three-time LGI LGI_{123} (see Eq. (2.23)).

The results of this computation for Dataset 1 are illustrated in Fig. D.2. It should be noted that Dataset 2 yields qualitatively identical trends in terms of LGI violation; therefore, Dataset 1 is presented as a representative case. The numerical analysis confirms our analytical findings: the LGI is significantly violated in the regions $\omega\tau \in (0, \pi/2)$ and $(3/2\pi, 2\pi)$, providing a necessary but not sufficient certification of the non-macrorealistic nature of the underlying generation process.

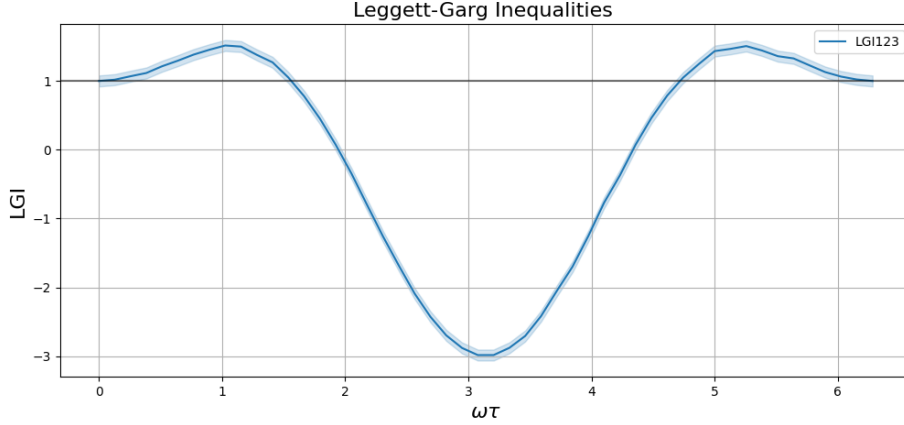


Figure D.2: LGI Eq. (2.23) at three times as a function of $\omega\tau$ for the first dataset. The solid lines represent the continuous trend obtained from the full experimental bitstream analysis, while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

Following the LGI verification, we quantified the "distance" from classicality and the associated degree of randomness. Specifically, we calculated the distance from Intermediate Macrorealism (d_{Int}), defined in Eq. (5.11), and the Min-entropy (H_∞), as defined in Eq. (5.17). The results for Dataset 1 are presented in Fig. D.3.

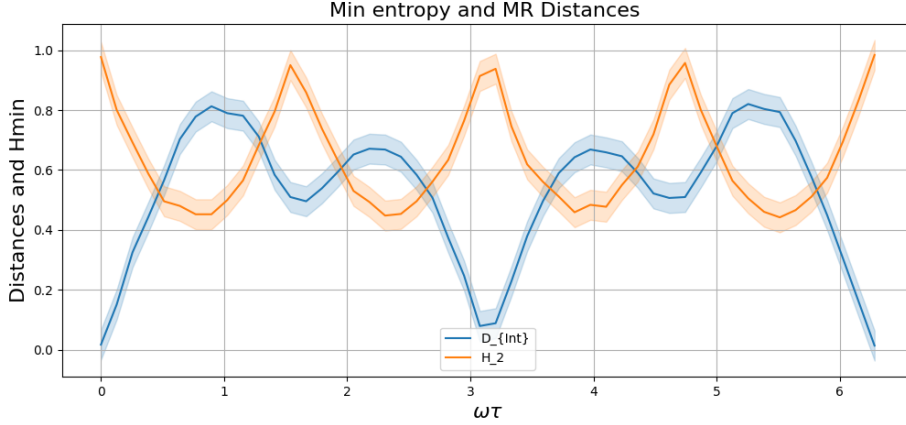


Figure D.3: Comparison between Min-entropy H_∞ (orange) and the distance from Intermediate Macrorealism d_{Int} (blue) for the first dataset. The continuous curves are generated by processing the entire set of $\omega\tau$ evolution steps, while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

In Fig. D.3, we can observe two critical features that validate the theoretical analysis of Chapter 5. First, the framework correctly certifies that the bits are

generated by a non-macrorealistic system, as d_{Int} remains strictly positive across the parameter space. However, we cannot conclude that the system produces high-quality randomness. Although MR_{int} is violated for all values of $\omega\tau$, the min-entropy remains consistently below 1. In information-theoretic terms, this indicates that the outcomes are still highly predictable or biased. Consequently, in this configuration, the black box fails to serve as a secure source of random numbers, as the "quantumness" of the system does not sufficiently translate into "unpredictability."

To investigate how different physical parameters influence these metrics, we performed a secondary analysis on Dataset 2, with the resulting metrics displayed in Fig. D.4.

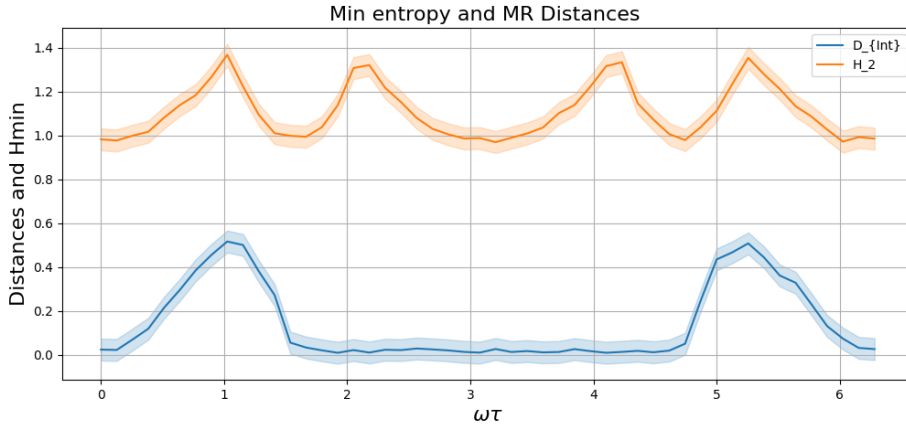


Figure D.4: Min-entropy and d_{Int} as functions of $\omega\tau$ for the second dataset, while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

Fig. D.4 reveals a trend similar to Dataset 1 regarding the violation of Intermediate Macrorealism, which persists across the entire range of $\omega\tau$. However, a significant quantitative difference arises in the entropy profile: the min-entropy is substantially higher in this dataset, reaching values of approximately 1.4. In this scenario, we can provide a much stronger certification: the black box is not only non-macrorealistic but also produces bitstrings with sufficient entropy to be considered genuinely random. Under these conditions, the device-independent protocol validates the black box as a viable source for secure cryptographic applications.

Discussion and Interpretation of Results

Based on the comparison of Figures D.3 and D.4, several key observations can be summarized:

- **Certification Limits:** Pairwise bit analysis is inherently limited to certifying the violation of *intermediate* macrorealism and the *two-time* min-entropy. This limitation exists because these quantities are calculated solely from the marginal probabilities of measurement pairs (e.g., $p(Q_i, Q_j)$). Without access to the full three-time joint probability distribution $p(Q_1, Q_2, Q_3)$ which would require more complex measurement protocols or different experimental assumptions—it is impossible to evaluate stronger forms of macrorealism or higher-order entropy bounds.
- **Decoupling of Quantumness and Randomness:** As analytically derived in Chapter 5, the violation of MR_{int} is not sufficient condition for maximum randomness. Even when the system is non-classical ($d_{\text{Int}} > 0$), the min-entropy can remain low. This highlights the importance of our dual-certification approach: we must verify both the nature of the source (non-macrorealism) and the quality of its output (entropy) separately to ensure cryptographic security.

To establish a robust validation of our certification framework, it is essential to consider the behavior of a purely classical system. This benchmarking process allows us to demonstrate that the framework is not only capable of detecting quantumness but, equally importantly, is immune to "false positives" generated by classical processes that mimic randomness.

For the sake of illustration, let us consider a hypothetical scenario involving an adversary, "Dr. Snoop." Suppose Dr. Snoop provides us with a "black box" supercomputer, claiming it to be a state-of-the-art quantum bit generator capable of producing randomness of a quality unparalleled on the market.

To an unwary user, the output might appear perfectly random. However, without a device-independent certification, there is no way to verify if these bits are truly unpredictable or merely the result of a complex, but deterministic, classical algorithm. To simulate this, we analyzed a dataset generated by a classical stochastic process. The results of this analysis are presented in Fig. D.5.

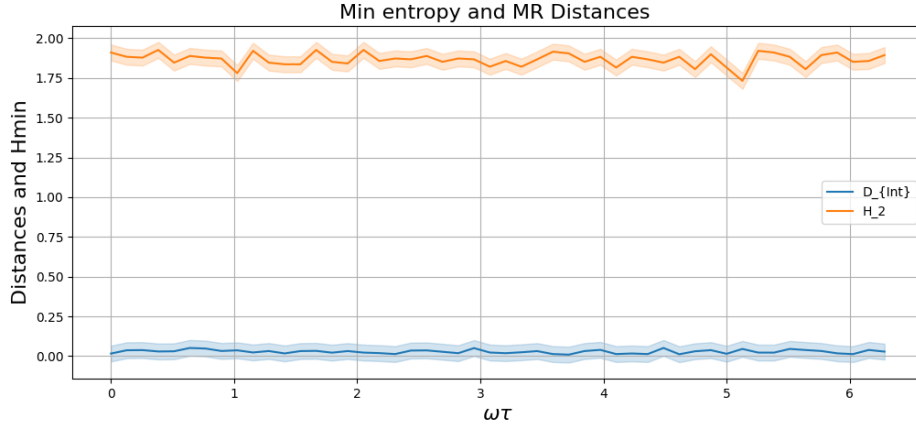


Figure D.5: Certification metrics for the classical benchmark: Min-entropy H_∞ from Eq. (5.17) (orange) and the distance from Intermediate Macrorealism d_{Int} from Eq. (5.11) (blue). While shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

Interestingly, as can be seen in Fig. D.5, the bits generated by this classical "supercomputer" appear to be of superior quality in terms of raw entropy, with a min-entropy H_∞ consistently near the maximum value of 2. At first glance, such a high entropy value might suggest a high-quality source. However, our framework reveals the true nature of the device: the distance from intermediate macrorealism, d_{Int} , remains zero across the entire range of the evolution parameter $\omega\tau$. This signifies that the system consistently satisfies the macrorealistic condition.

This result highlights a fundamental vulnerability in randomness generation. Dr. Snoop's supercomputer is not a source of secure, device-independent random numbers. Because it obeys macrorealism, the bits it generates are intrinsically classical. This implies that the sequence is likely the product of a PRNG as seen in Chapter 3; while the output may pass standard statistical tests for randomness [25], it remains fully predictable to anyone who possesses knowledge of the underlying algorithm and the initial seed.

Our framework thus correctly identifies the classical nature of this system by the absence of any violation of macrorealism. These empirical results align perfectly with the theoretical observations made by Colbeck [25], who remarked that a classical system can produce bits that appear "more random" (i.e., having higher entropy) than a quantum system when analyzed through certain restricted metrics. Indeed, while the classical case in Fig. D.5 shows $H_\infty \approx 2$, the certified quantum cases in Fig. D.3 and Fig. D.4 exhibit lower entropy values (below 1.5).

D.1.2 Single and Couple Bits Analysis (1-2, 1-3, 2-3, 1, 2, 3)

The subsequent stage of our device-independent certification involves the simultaneous analysis of datasets containing both pairwise correlations ("couple bits") and single-measurement statistics ("single bits").

The necessity for this expanded approach arises when we attempt to evaluate the *weak distance* (d_{Weak}), defined in Eq. (5.6). Unlike the intermediate macrorealism condition, the weak distance cannot be determined solely through pairwise correlation data (the couple bits). This is because d_{Weak} requires knowledge of the individual expectation values $\langle Q_i \rangle$ at each time t_i . In an experimental context, while $\langle Q_1 Q_2 \rangle$ is obtained from joint measurements, the individual averages $\langle Q_i \rangle$ must be collected from independent, single-measurement runs to ensure they remain unaffected by any potential invasiveness from previous measurement steps.

Following this protocol, we analyzed the same two physical configurations described previously, now incorporating both "couple" and "single" bitstrings. For clarity, we refer to these as $\mathcal{D}_i^{Couple}$ and $\mathcal{D}_i^{single}$ ($i = 1, 2$). This multi-dataset approach allows us to evaluate not only the three-time LGI, Eq. (2.23), but also the three two-time LGIs, defined in Eq. (2.26).

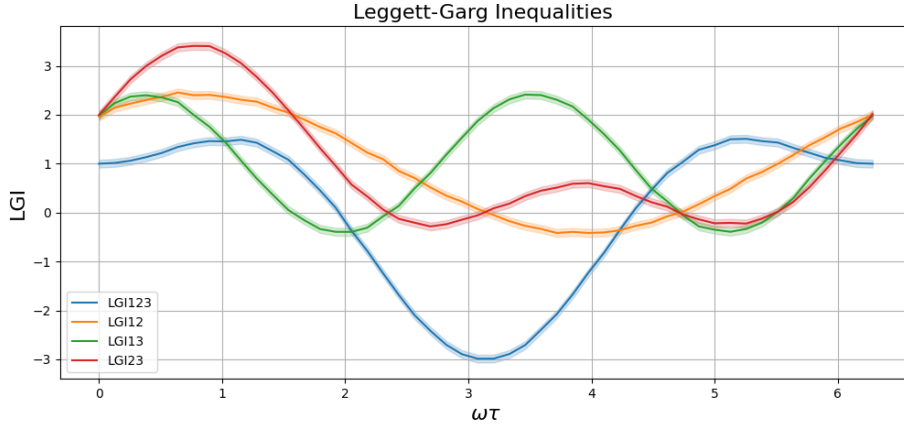


Figure D.6: Three-time LGI (LGI_{123} , blue) and two-time LGIs (LGI_{12} orange, LGI_{13} green, LGI_{23} red) as functions of $\omega\tau$ for $\mathcal{D}_1^{single}$ and $\mathcal{D}_1^{Couple}$. The solid lines represent the continuous experimental trend, while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

As illustrated in Fig. D.6, both the two-time and three-time LGIs exhibit violations across the parameter space. This consistent violation across multiple

inequalities provides a certification of the system’s non-classicality.

We then proceeded to calculate the weak distance (d_{Weak}) and the corresponding min-entropies for both datasets. The results for the first configuration are presented in Fig. D.7, while the results for the second configuration ($\mathcal{D}_2^{\text{single}}$ and $\mathcal{D}_2^{\text{Couple}}$) are shown in Fig. D.8.

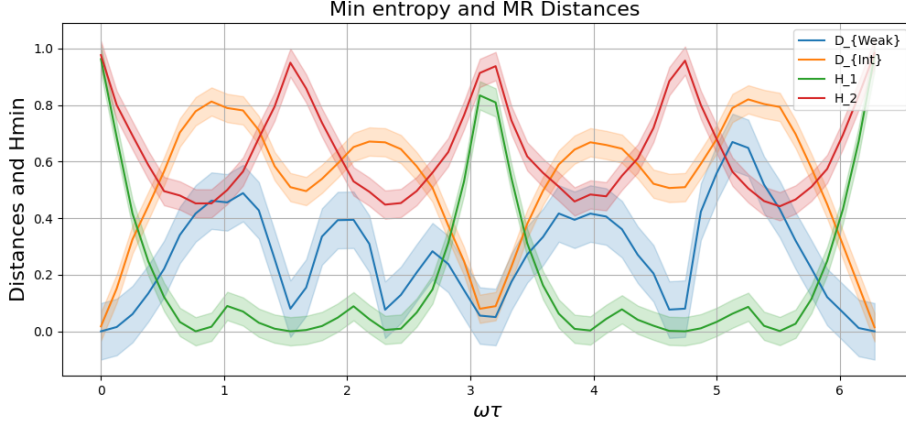


Figure D.7: Evolution of $H_\infty^{(1)}$ (green), $H_\infty^{(2)}$ (red), d_{Weak} (blue), and d_{Int} (orange) as functions of $\omega\tau$ for Dataset 1. The curves reflect the full experimental bitstream, with shadowed areas representing the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

The analysis of Fig. D.7 confirms that while the system is strictly quantum—evidenced by the fact that the weak distance d_{Weak} and the intermediate distance d_{Int} are positive across almost the entire range of $\omega\tau$ —it remains a poor source of randomness. Specifically, the single-time min-entropy $H_\infty^{(1)}$ remains well below 1 and approaches 0 for a significant portion of the domain, indicating highly predictable outcomes.

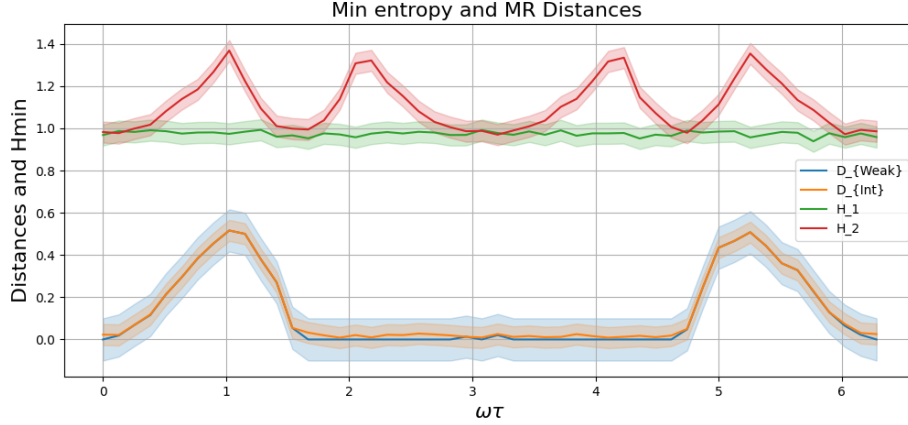


Figure D.8: $H_{\infty}^{(1)}$ (green), $H_{\infty}^{(2)}$ (red), d_{Weak} (blue), and d_{Int} (orange) for $\mathcal{D}_2^{\text{single}}$ and $\mathcal{D}_2^{\text{Couple}}$. while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

In contrast, Dataset 2 (Fig. D.8) demonstrates a much more favorable regime for randomness generation. Here, the weak distance d_{Weak} is consistently lower than the intermediate distance d_{Int} throughout the range of $\omega\tau$. Crucially, in the regions $\omega\tau \in (0, \pi/2)$ and $\omega\tau \in (3/2\pi, 2\pi)$, both distances are positive, certifying the quantum nature of the source. Simultaneously, the min-entropy $H_{\infty}^{(1)}$ reaches a value of 1, and the two-time min-entropy $H_{\infty}^{(2)}$ exceeds 1, indicating that the system is an excellent source of certified, secure randomness in these specific regions.

In conclusion, this combined analysis confirms that Dataset 1, while exhibiting non-classical features, is unsuitable for randomness-based applications. Dataset 2, however, provides a robust operational window for $\omega\tau \in (0, \pi/2) \cup (3/2\pi, 2\pi)$ where both quantumness and high-quality randomness are simultaneously certified.

D.1.3 Triple Bits Analysis (1-2-3)

The final and most rigorous stage of our device-independent certification involves the analysis of "triple bits" datasets. To evaluate the *strong distance* from macro-realism (d_{Strong}), as defined in Eq. (5.16), the pairwise and single-measurement statistics utilized in previous sections are no longer sufficient.

The certification of strong macrorealism requires access to the full joint probability distribution $p(Q_1, Q_2, Q_3)$. Experimentally, this necessitates a specific measurement protocol where the system is measured sequentially at three dis-

tinct time steps (t_1 , t_2 , and t_3), with the prescribed unitary evolution occurring between each intervention. Unlike the earlier datasets, these "triple-bit" sequences capture the correlations of the entire three-time evolution in a single experimental run. By integrating these new datasets with the previous single and couple bits, we can provide a complete characterization of the system, incorporating the strong distance measure and the three-time min-entropy $H_\infty^{(3)}$.

The comprehensive results, which include all min-entropy values and distance measures across the three configurations, are presented in Fig. D.9 for Dataset 1 and in Fig. D.10 for Dataset 2.

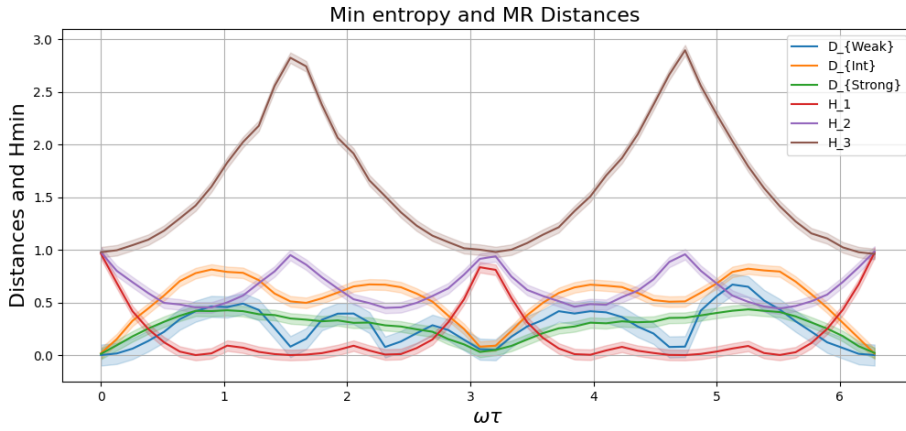


Figure D.9: Complete certification metrics for Dataset 1: $H_\infty^{(1)}$ (red), $H_\infty^{(2)}$ (purple), $H_\infty^{(3)}$ (brown), d_{Weak} (blue), d_{Int} (orange), and d_{Strong} (green) as functions of $\omega\tau$. while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

As observed in Fig. D.9 (Dataset 1), the strong distance d_{Strong} remains strictly positive across the entire range of the evolution parameter $\omega\tau$. This provides the most robust validation of the system's non-classicality to date, confirming that the generation process cannot be described by any macrorealistic model. Despite this strong certification of quantumness, the entropy analysis remains consistent with our previous findings: the system's inherent correlations significantly limit its unpredictability, as evidenced by the low min-entropy values.

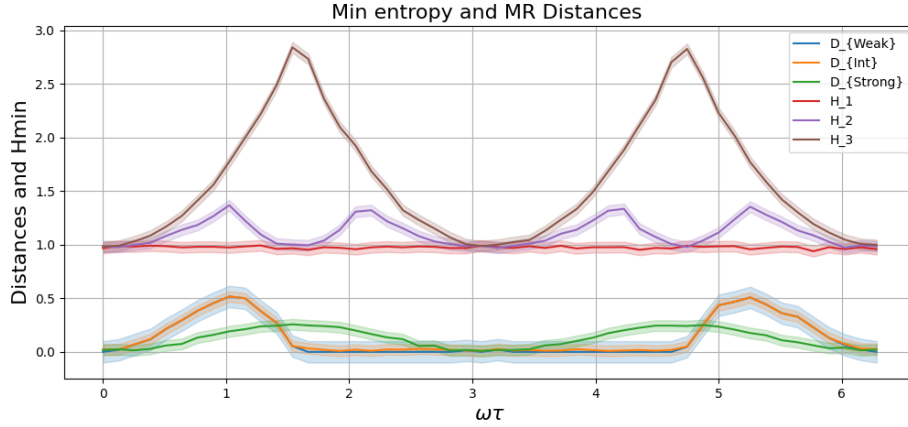


Figure D.10: Complete certification metrics for Dataset 2: $H_\infty^{(1)}$ (red), $H_\infty^{(2)}$ (purple), $H_\infty^{(3)}$ (brown), d_{Weak} (blue), d_{Int} (orange), and d_{Strong} (green) as functions of $\omega\tau$. while shadowed areas represent the error bounds, obtained through propagation of uncertainties $\propto \sqrt{n}$, with n being the number of samples, for more details see Appendix C.

In contrast, the analysis of Dataset 2 (Fig. D.10) reveals a particularly interesting feature. While the weak (d_{Weak}) and intermediate (d_{Int}) distances vanish at specific points in the $\omega\tau$ domain, the strong distance d_{Strong} remains positive throughout the entire range. This demonstrates the power of the three-time protocol: d_{Strong} is capable of certifying the quantum nature of the black box even in regimes where simpler pairwise tests might fail to detect non-classical behavior. This allows for a much broader operational range for the device, ensuring that its quantum origin is guaranteed for all values of $\omega\tau$, even if the quality of the randomness (captured by the entropy lines) varies.

Consistent with the visual standards established in this chapter, the curves in Figures D.9 and D.10 are derived from the high-resolution scan of the experimental bitstream.

This multi-layered analysis concludes our data-driven certification. By progressively moving from single measurements to triple-time correlations, we have successfully mapped the transition from classical to quantum behavior, providing a rigorous hierarchy of tests that certify both the physical nature of the source and the security of its output.

D.2 Conclusion

In this chapter, we have transitioned from the theoretical framework of device-independent certification to a rigorous, data-driven analysis. By treating the

experimental source as a "black box," we have demonstrated that non-classicality and genuine randomness can be certified solely through the statistical properties of the generated bitstreams, provided that the underlying measurement protocol is known.

To summarize our methodology and the hierarchy of tests performed, we clarify the relationship between the experimental data sources and the computed physical quantities in Table D.1.

Data Source	Evaluated Min-entropy	Evaluated Distance
Couple bits only	$H_{\min}^2$	d_{Int}
Couple + Single bits	$H_{\min}^1, H_{\min}^2$	d_{Weak}, d_{Int}
Triple bits (Full Joint)	$H_{\min}^3$	d_{Strong}

Table D.1: Summary of experimental data sources and the corresponding evaluated physical quantities.

As summarized in Table D.1, the analysis of the triple-bit files provides the most comprehensive characterization of the system's properties. While single and couple measurement configurations allow for the detection of non-classicality through the violation of intermediate or weak macrorealism, only the triple-bit data—capturing the full joint probability distribution $p(Q_1, Q_2, Q_3)$ —enables the evaluation of the strong distance d_{Strong} . This hierarchical approach confirms that the system maintains its quantum nature even in regimes where simpler pairwise correlations might fail to provide a definitive certification.

Furthermore, our comparison between quantum datasets and classical benchmarks (as seen in the "Dr. Snoop" scenario) underscores the fundamental distinction between apparent stochasticity and certified quantum randomness. While a classical system can produce high-entropy bitstreams, it consistently fails to violate macrorealistic bounds, revealing its deterministic and predictable nature.

It is important to emphasize that, in a device-independent context, prior knowledge of the bit generation process remains essential. Without a clear specification of the experimental sequence and the temporal measurement protocols, it would be impossible to correctly interpret the bitstreams. By anchoring our analysis to the Leggett-Garg framework, we have successfully achieved our primary objectives: providing a robust, data-centric methodology for certifying the quantum properties of a physical system and ensuring the presence of secure, genuine randomness for information-theoretic applications.

Appendix E

Predictability and Non-Signaling in Three-Time Operational Frameworks

In this chapter, we extend the foundational concepts of temporal correlation frameworks from two-time operational sequences to higher-order multi-time scenarios. Specifically, we establish a mathematical relationship between three-time predictability and its two-time counterpart. Leveraging this baseline property, we derive a generalized version of the Lemma originally introduced by Mal et al. [30], establishing an analytical bridge between multi-time predictability and macroscopic non-signaling bounds.

Throughout this chapter, we consider sequential projective measurements performed on a single physical system at consecutive time instants $t_A < t_B < t_C$. The measurement settings (or contexts) selected at each time are denoted by $Q_A, Q_B, Q_C \in \mathcal{M}$, and their corresponding operational outcomes are defined as $A, B, C \in \mathcal{O}$, respectively, where $\mathcal{M}$ is the set of measurement settings and $\mathcal{O}$ is the set of operational outcomes.

E.1 Review of Two-Time Operational Bounds

To anchor our multi-time extension, we briefly recall the mathematical structures established for two-time operational sequences in literature [30] (and derived in Chapter 4). Within a chronological bilateral framework, a system satisfies two-time predictability (Pred_2) if the joint probability distribution over the measure-

ment outcomes collapses strictly to a deterministic, binary range:

$$\text{Pred}_2 : P(AB|Q_A Q_B) \in \{0, 1\} \quad \forall A, B \in \mathcal{O}, \forall Q_A, Q_B \in \mathcal{M}. \quad (\text{E.1})$$

Concurrently, the standard two-time No-Signaling-in-Time (NSIT_{2T}) defined in Eq. (4.2) - (4.4) constraints enforce the requirement that a subsequent measurement choice cannot non-invasively update the statistics of a past event, and that a past action does not alter the marginal probability of a future event under unselective operations. These conditions are explicitly, in the formulation introduced in this chapter, written as:

$$P(B|Q_A Q_B) = P(B|Q_B), \quad (\text{E.2})$$

$$P(C|Q_B Q_C) = P(C|Q_C), \quad (\text{E.3})$$

$$P(C|Q_A Q_C) = P(C|Q_C). \quad (\text{E.4})$$

Within this structural backdrop, Mal et al. [30] demonstrated a Lemma showing that the intersection of non-invasive macrorealism and local predictability forces the joint distribution to factorize:

$$\text{NSIT}_{2T} \wedge \text{Pred}_2 \implies P(AB|Q_A Q_B) = P(A|Q_A)P(B|Q_B). \quad (\text{E.5})$$

A direct operational consequence of this factorization is captured by the Corollary [30], which guarantees that any statistically significant violation of the baseline Leggett-Garg inequality (LGI₁₂₃) under fulfilled NSIT_{2T} conditions implies a non-classical breakdown of underlying system predictability:

$$\text{NSIT}_{2T} \wedge \neg \text{LGI}_{123} \implies \neg \text{Pred}_2. \quad (\text{E.6})$$

E.2 Formal Definitions in the Three-Time Scenario

Before formulating the generalized multi-time framework, we explicitly define the operational conditions for three-time measurement sequences.

Definition E.2.1 (Three-Time Predictability). *A sequential measurement framework satisfies three-time predictability (Pred₃) if the joint probability distribution over the complete profile of three consecutive measurement outcomes is determin-*

istic:

$$Pred_3 : P(ABC|Q_A Q_B Q_C) \in \{0, 1\} \quad \forall A, B, C \in \mathcal{O}, \forall Q_A, Q_B, Q_C \in \mathcal{M}. \quad (E.7)$$

While we previously introduced the No-Signaling-in-Time conditions in Chapter 2 (specifically in Eq. 2.52, Eq. 2.53, and Eq. 2.54), we now reformulate them following the proof and definitions established by [30]. Accordingly, we formalize the set of conditions governing the three-time operational sequence as follows:

- NSIT₂₃: The marginal probability of a measurement outcome at t_C is statistically independent of the past measurement setting selected at t_B :

$$NSIT_{23} : P(C|Q_B Q_C) = P(C|Q_C). \quad (E.8)$$

- NSIT₍₁₎₂₃: The joint probability distribution of the outcomes at t_B and t_C is independent of whether an unselective or unperformed measurement configuration occurred at the initial time t_A :

$$NSIT_{(1)23} : P(BC|Q_A Q_B Q_C) = P(BC|Q_B Q_C). \quad (E.9)$$

- NSIT₁₍₂₎₃: The joint probability distribution of the outcomes at t_A and t_C is independent of the choice of measurement setting at the intermediate time t_B :

$$NSIT_{1(2)3} : P(AC|Q_A Q_B Q_C) = P(AC|Q_A Q_C). \quad (E.10)$$

E.3 Hierarchical Implication: From Three-Time to Two-Time Predictability

Before evaluating the implications of the multi-time Lemma, it is foundational to establish the mathematical relationship between predictability conditions at different temporal orders. Specifically, we prove that three-time predictability ($Pred_3$) strictly implies two-time predictability ($Pred_2$).

Proposition E.3.1. *In a sequential measurement framework, if the condition of three-time predictability ($Pred_3$) is satisfied, then the two-time predictability condition ($Pred_2$) must necessarily hold:*

$$Pred_3 \implies Pred_2. \quad (E.11)$$

Proof. Let us choose an arbitrary configuration of measurement outcomes and settings for the first two time steps, denoted by $\bar{A}, \bar{B} \in \mathcal{O}$ and $\bar{Q}_A, \bar{Q}_B \in \mathcal{M}$. To evaluate the two-time probability $P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B)$, we invoke the consistency property of marginalization by considering an arbitrary future measurement setting $\bar{Q}_C \in \mathcal{M}$ at time t_C :

$$P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) = \sum_{C \in \mathcal{O}} P(\bar{A}\bar{B}C|\bar{Q}_A\bar{Q}_B\bar{Q}_C). \quad (\text{E.12})$$

By the core hypothesis of three-time predictability [Eq. (E.7)], every single joint probability distribution term inside the summation is strictly restricted to a binary deterministic range:

$$P(\bar{A}\bar{B}C|\bar{Q}_A\bar{Q}_B\bar{Q}_C) \in \{0, 1\} \quad \forall C \in \mathcal{O}. \quad (\text{E.13})$$

Since $P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B)$ is a well-defined probability, it is bounded by definition within the unit interval:

$$0 \leq P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) \leq 1. \quad (\text{E.14})$$

We can now exhaustively analyze the restricted non-negative integer values that the summation in Eq. (E.12) can take:

1. **Case 1:** If all individual elements in the set $\{P(\bar{A}\bar{B}C|\bar{Q}_A\bar{Q}_B\bar{Q}_C)\}_C$ are identically equal to 0, the summation collapses to zero, yielding:

$$P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) = 0. \quad (\text{E.15})$$

2. **Case 2:** If exactly one outcome $\bar{C} \in \mathcal{O}$ exists such that $P(\bar{A}\bar{B}\bar{C}|\bar{Q}_A\bar{Q}_B\bar{Q}_C) = 1$, while the remaining outcomes yield zero, the summation collapses to unity, yielding:

$$P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) = 1. \quad (\text{E.16})$$

3. **Case 3:** If two or more distinct outcomes in the sample space of C were equal to 1, the summation would satisfy $\sum_C P(\bar{A}\bar{B}C|\bar{Q}_A\bar{Q}_B\bar{Q}_C) \geq 2$. This scenario is strictly impossible as it violates the Kolmogorov probability bound $P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) \leq 1$.

Because Cases 1 and 2 represent the only mathematically admissible configura-

tions, the joint probability for the chosen arbitrary sequence must satisfy:

$$P(\bar{A}\bar{B}|\bar{Q}_A\bar{Q}_B) \in \{0, 1\}. \quad (\text{E.17})$$

Since the initial choice of settings and outcomes $(\bar{A}, \bar{B}, \bar{Q}_A, \bar{Q}_B)$ was arbitrary, this binary constraint holds universally $\forall A, B \in \mathcal{O}$ and $\forall Q_A, Q_B \in \mathcal{M}$. Therefore, the two-time predictability condition (Pred_2) is fulfilled. $\square$

E.4 Lemma Extension to Three-Time Frameworks

Lemma E.4.1. *In a three-time sequential measurement scenario, if the condition of three-time predictability (Pred_3) holds, the conjunction of the past-directed non-signaling bounds NSIT_{23} and $\text{NSIT}_{(1)23}$ logically implies the non-signaling condition $\text{NSIT}_{1(2)3}$:*

$$\text{Pred}_3 \wedge \text{NSIT}_{23} \wedge \text{NSIT}_{(1)23} \implies \text{NSIT}_{1(2)3}. \quad (\text{E.18})$$

Proof. Let us consider the joint probability distribution of outcomes A and C conditioned on the complete measurement array Q_A, Q_B, Q_C . By applying the definition of conditional probability, we can write:

$$P(AC|Q_AQ_BQ_C) = P(C|AQ_AQ_BQ_C)P(A|Q_AQ_BQ_C). \quad (\text{E.19})$$

By invoking the arrow of time principle (or chronological induction), future measurement configurations cannot causally affect the statistics of past unselective outcomes. Therefore, the marginal probability of A depends solely on its local setting Q_A :

$$P(A|Q_AQ_BQ_C) = P(A|Q_A), \quad (\text{E.20})$$

yielding:

$$P(AC|Q_AQ_BQ_C) = P(C|AQ_AQ_BQ_C)P(A|Q_A). \quad (\text{E.21})$$

We now enforce the three-time predictability condition (Pred_3). Under Pred_3 , the probability of obtaining outcome C is fully pre-determined to be either 0 or 1 prior to any assessment of A . Consequently, gaining operational knowledge of the outcome A cannot update or alter the statistical inference of C . This establishes the property:

$$P(C|AQ_AQ_BQ_C) = P(C|Q_AQ_BQ_C). \quad (\text{E.22})$$

Substituting this back into the joint distribution expression gives:

$$P(AC|Q_A Q_B Q_C) = P(C|Q_A Q_B Q_C)P(A|Q_A). \quad (\text{E.23})$$

To expand the right-hand side, we insert the completeness identity over the intermediate outcome B , namely $1 = \sum_{B \in \mathcal{O}} P(B|Q_A Q_B Q_C)$:

$$P(AC|Q_A Q_B Q_C) = P(C|Q_A Q_B Q_C) \sum_{B \in \mathcal{O}} P(B|Q_A Q_B Q_C)P(A|Q_A). \quad (\text{E.24})$$

Distributing the terms inside the summation, we obtain:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A) \sum_{B \in \mathcal{O}} [P(C|Q_A Q_B Q_C)P(B|Q_A Q_B Q_C)]. \quad (\text{E.25})$$

Applying Bayes' theorem to the terms within the summation allows us to reconstruct the joint three-time distribution:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A) \sum_{B \in \mathcal{O}} P(BC|Q_A Q_B Q_C). \quad (\text{E.26})$$

At this stage, we assume that the non-signaling condition $\text{NSIT}_{(1)23}$ is satisfied, which states that $P(BC|Q_A Q_B Q_C) = P(BC|Q_B Q_C)$. The expression simplifies to:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A) \sum_{B \in \mathcal{O}} P(BC|Q_B Q_C). \quad (\text{E.27})$$

By applying Bayes' theorem [37] again inside the sum and leveraging the predictability condition Pred_2 (see Proposition E.3.1) to isolate the marginal distribution of C , we can write:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A) \sum_{B \in \mathcal{O}} [P(C|Q_B Q_C)P(B|Q_B Q_C)] \quad (\text{E.28})$$

$$= P(A|Q_A)P(C|Q_B Q_C) \sum_{B \in \mathcal{O}} P(B|Q_B Q_C). \quad (\text{E.29})$$

Since the summation over all possible intermediate outcomes yields unity ($\sum_B P(B|Q_B Q_C) = 1$), the expression reduces to:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A)P(C|Q_B Q_C). \quad (\text{E.30})$$

Next, we require the fulfillment of the NSIT_{23} condition, meaning $P(C|Q_B Q_C) =$

$P(C|Q_C)$. This yields a complete factorization of the joint distribution:

$$P(AC|Q_A Q_B Q_C) = P(A|Q_A)P(C|Q_C). \quad (\text{E.31})$$

Finally, by applying the arrow of time principle in reverse to the product of independent marginals, we note that $P(A|Q_A)P(C|Q_C) = P(AC|Q_A Q_C)$. Therefore:

$$P(AC|Q_A Q_B Q_C) = P(AC|Q_A Q_C), \quad (\text{E.32})$$

which is precisely the definition of $\text{NSIT}_{1(2)3}$. This completes the proof:

$$\text{Pred}_3 \wedge \text{NSIT}_{23} \wedge \text{NSIT}_{(1)23} \implies \text{NSIT}_{1(2)3}. \quad (\text{E.33})$$

□

E.5 Logical Analysis: Strong Macrorealism and Predictability

To map the foundational implications of Lemma E.4.1 onto the broader definitions of Macrorealism, we introduce a structured propositional logic framework. Let us define the following core propositions:

$$\mathcal{A} \equiv \text{NSIT}_{23}, \quad (\text{E.34})$$

$$\mathcal{B} \equiv \text{NSIT}_{(1)23}, \quad (\text{E.35})$$

$$\mathcal{C} \equiv \text{NSIT}_{1(2)3}. \quad (\text{E.36})$$

Based on these definitions, Lemma E.4.1 and the standard axiomatic definition of Strong Macrorealism ($\text{MR}_{\text{strong}}$) can be formulated as:

$$(\text{Pred}_3 \wedge \mathcal{A} \wedge \mathcal{B}) \implies \mathcal{C}, \quad (\text{E.37})$$

$$\text{MR}_{\text{strong}} \iff (\mathcal{A} \wedge \mathcal{B} \wedge \mathcal{C}). \quad (\text{E.38})$$

E.5.1 Sufficiency for Strong Macrorealism

By linking the implications of Eq. (E.37) and Eq. (E.38), it follows that three-time predictability combined with the first two past-directed non-signaling constraints

constitutes a sufficient operational condition to guarantee Strong Macrorealism:

$$(\text{Pred}_3 \wedge \text{NSIT}_{23} \wedge \text{NSIT}_{(1)23}) \implies \text{MR}_{\text{strong}}. \quad (\text{E.39})$$

Proof. Assume the premises Pred_3 , $\mathcal{A}$, and $\mathcal{B}$ hold true simultaneously. According to the implication in Eq. (E.37), the proposition $\mathcal{C}$ must logically be true. Given that $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are concurrently true, the equivalence condition in Eq. (E.38) is satisfied, proving that $\text{MR}_{\text{strong}}$ is true. $\square$

E.5.2 Operational Contrapositives and Experimental Relevance

By applying logical transposition rules to the framework established above, we can derive two distinct contrapositive relations that offer deep insights into quantum macrorealism violations.

The first contrapositive relation isolates the intermediate non-signaling condition:

$$\text{NSIT}_{23} \wedge \text{NSIT}_{(1)23} \wedge \neg \text{NSIT}_{1(2)3} \implies \neg \text{Pred}_3. \quad (\text{E.40})$$

Equation (E.40) represents a direct higher-order analogue to the original lemma introduced by Mal et al. [30]. From an experimental standpoint, this relation is highly valuable: it establishes that if a physical system satisfies the baseline past-directed non-signaling bounds (NSIT_{23} and $\text{NSIT}_{(1)23}$) but exhibits a statistically significant violation of the intermediate context constraint ($\text{NSIT}_{1(2)3}$), one can definitively infer a non-classical breakdown of underlying three-time predictability ($\neg \text{Pred}_3$) in a device-independent manner.

Alternatively, transposition can be applied directly to the Strong Macrorealism implication, yielding:

$$\text{NSIT}_{23} \wedge \text{NSIT}_{(1)23} \wedge \neg \text{MR}_{\text{strong}} \implies \neg \text{Pred}_3. \quad (\text{E.41})$$

While logically sound, Equation (E.41) connects predictability directly to the abstract global definition of Strong Macrorealism. Operationally, this second relation is less useful in a laboratory setting, as testing $\neg \text{MR}_{\text{strong}}$ requires verifying all non-signaling configurations simultaneously, whereas Equation (E.40) provides a targeted, fine-grained witness of non-classicality via a single localized NSIT violation.

E.6 Results and Physical Discussion

We apply the generalized multi-time framework to the physical system introduced in Section 5.3.2. Our analytical and numerical evaluation reveals that while NSIT_{23} remains satisfied—in agreement with the findings of Nath et al. [28]—the past-directed non-signaling condition $\text{NSIT}_{(1)23}$ is also strictly fulfilled. Conversely, the intermediate context condition $\text{NSIT}_{1(2)3}$ is significantly violated.

By virtue of the contrapositive relation derived in Eq. (E.40), this specific pattern of signaling validation and violation provides a definitive, device-independent certification that the system breaks the three-time predictability condition Pred_3 [Eq. (E.7)]. Concurrently, this implies a non-classical breakdown of the global strong macrorealism condition [previously defined in Eq. (5.12)].

To visualize these behaviors, Fig. E.1 illustrates the profile of the three NSIT conditions mapped onto the operational distance parameters η_3 , η_4 , and η_5 (as detailed in Section 5.1.3) evaluated as a function of the parameter $\omega\tau$.

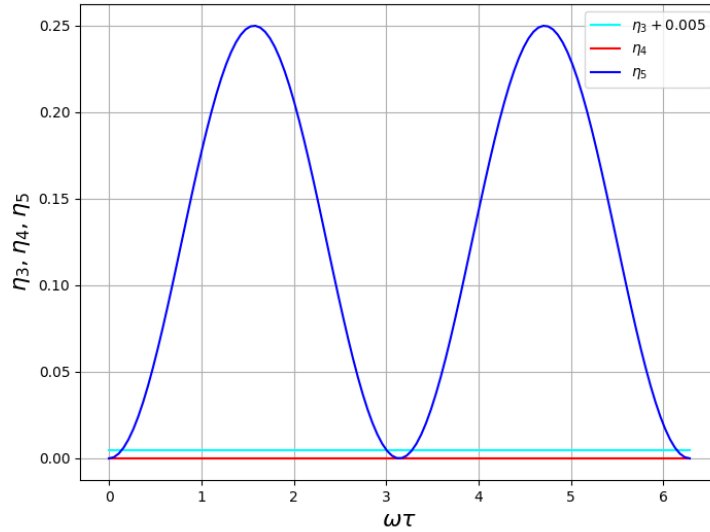


Figure E.1: Operational distance parameters η_3 (cyan line, slightly shifted for visual clarity), η_4 (red line), and η_5 (blue line) as a function of $\omega\tau$ for the sequential measurement scenario defined in Section 5.3.2. The system is initialized in the maximally mixed state (identity density matrix) and evolves under a Hamiltonian proportional to σ_y .

As evidenced by the plot, both η_3 and η_4 remain identically zero across the entire spectrum of $\omega\tau$, confirming that NSIT_{23} and $\text{NSIT}_{(1)23}$ are robustly satisfied. In contrast, η_5 exhibits non-zero values for all configurations of $\omega\tau$, highlighting a persistent violation of $\text{NSIT}_{1(2)3}$. Comparing this outcome with the macrore-

alist violations shown in Fig. 5.3, we can uniquely conclude that the breakdown of strong macrorealism in this regime is entirely driven by the failure of the intermediate-time non-signaling constraint $\text{NSIT}_{1(2)3}$.

E.6.1 Comparison with Previous Literature and Entropy Enhancement

To frame the operational advantages of our multi-time extension, we benchmark our results against the established two-time protocol introduced by Nath et al. [28].

In their framework, the evaluation of the two-time min-entropy $H_\infty^{(2)}$ (represented by the blue curve in Fig. E.2) yields a maximum certified value of approximately 1.41 bits out of a theoretical upper bound of 2.0. Crucially, this quantum-backed randomness certification can only be guaranteed within restricted parameter windows, specifically for $\omega\tau \in (0, \pi/2) \cup (3\pi/2, 2\pi)$.

By upgrading to the three-time sequential framework, we evaluate the higher-order min-entropy $H_\infty^{(3)}$ (represented by the red curve in Fig. E.2). Our protocol achieves a peak min-entropy of 2.85 bits out of a maximum bound of 3.0. Remarkably, this certified randomness is sustained and accessible across the entire range of the parameter $\omega\tau$.

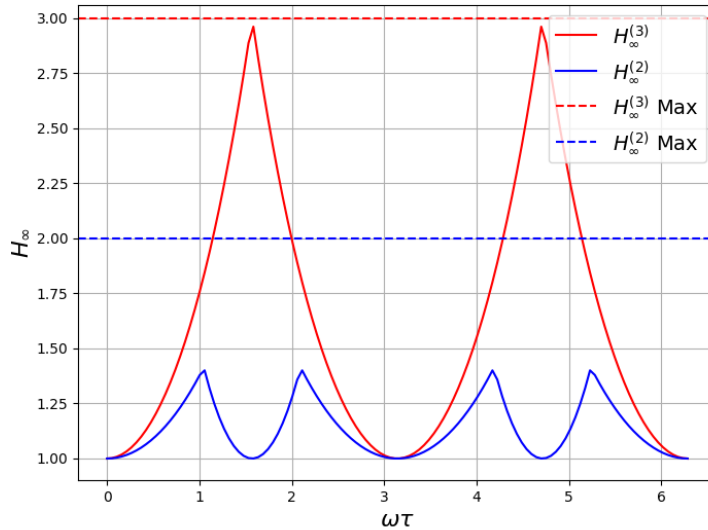


Figure E.2: Two-time min-entropy $H_\infty^{(2)}$ (blue line) and three-time min-entropy $H_\infty^{(3)}$ (red line) plotted against $\omega\tau$. The system configurations match those of Fig. E.1. The horizontal dotted lines denote the respective theoretical maximum bounds for each framework.

A comparative quantitative analysis shows that while the two-time protocol [28] reaches roughly 70.5% of its ultimate information capacity, our three-time approach unlocks over 95.0% of the maximum available min-entropy.

This significant enhancement in entropic output can be understood through the underlying quantum measurement dynamics. When the same physical system undergoes consecutive, selective projective measurements at times t_A , t_B , and t_C , each operational intervention actively back-reacts on the quantum state. The state collapse induced at t_A drastically disrupts the system's memory before it reaches t_B , and the subsequent measurement at t_B further randomizes the state prior to t_C .

This back-action cascade effectively acts as a dynamical noise injector or a sequential randomizing filter. Consequently, the 3-bit string generated within a single temporal shot loses the underlying quantum coherent correlations typically responsible for Leggett-Garg inequality violations, causing the setup to behave as an almost ideal physical source of random numbers.

In conclusion, our three-time framework proves to be highly superior for quantum information tasks, as it is capable of generating output bits with significantly higher entropy (near-perfect randomness) while simultaneously removing any restrictive parameter tuning on $\omega\tau$.

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"Quellu che nu sôn, a nu sôn"

— Nonna Ese o Socrate

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